

Block 4: Calculus

Short explanations, worked examples and self-check questions for every unit, with every idea also explained in Malayalam.

ഓരോ ആശയവും ലളിതമായ ഇംഗ്ലീഷിലും മലയാളത്തിലും, ഉദാഹരണങ്ങളോടെ.

UNIT 11

Limit and Continuity

x ഒരു value-ലേക്ക് അടുത്തടുത്ത് വരുമ്പോൾ $f(x)$ എന്ത് value-ലേക്കാണ് നീങ്ങുന്നത് എന്നതാണ് limit. ഈ unit-ൽ limit കണ്ടുപിടിക്കാനുള്ള methods (direct substitution, factorisation, LCM, rationalisation, standard results), infinite limit, L.H.L./R.H.L., അവസാനം continuity എന്നിവ പഠിക്കുന്നു.

Concept of limit

Direct substitution method and algebra of limits

When direct substitution fails: indeterminate forms

Factorisation method

L.C.M. method

Rationalisation method

Standard results

Infinite limits ($x \rightarrow \infty$)

Left hand and right hand limits

Continuity at a point

Continuous functions

Concept of limit

Definition: $\lim_{x \rightarrow a} f(x) = l$ means that $f(x)$ approaches the number l as x approaches a ($x \neq a$) from both the left and the right.

x , a -യോട് ($x \neq a$) ഇടത്തുനിന്നും വലത്തുനിന്നും അടുക്കുമ്പോൾ $f(x)$ ഒരു number l -നോട് അടുക്കുന്നു എന്നാണ് $\lim_{x \rightarrow a} f(x) = l$ എന്നതിന്റെ അർത്ഥം.

The value $f(a)$ tells us what the function gives at $x = a$. The **limit** tells us what $f(x)$ gets closer and closer to as x comes closer and closer to a , without x ever needing to equal a . We can approach a from the left ($x < a$) or from the right ($x > a$). If $f(x)$ approaches the same number l from both sides, we write $\lim_{x \rightarrow a} f(x) = l$.

മലയാളത്തിൽ

ബസ് stop-ലേക്ക് ബസ് അടുക്കുന്നത് ഓർക്കുക: stop-ൽ എത്തിയോ ഇല്ലയോ എന്നത് നമ്മൾ നോക്കുന്നില്ല, ബസ് എങ്ങോട്ടാണ് നീങ്ങുന്നത് എന്ന് മാത്രം. അതുപോലെ x , a -യോട് വളരെ അടുത്ത് വരുമ്പോൾ $f(x)$ ഏത് number-ലേക്ക് അടുക്കുന്നു എന്നതാണ് limit. ഇടത് വശത്തുനിന്നും ($x < a$) വലത് വശത്തുനിന്നും ($x > a$) വരുമ്പോൾ ഒരേ number കിട്ടിയാൽ ആ number ആണ് limit.

$\lim_{x \rightarrow a} f(x) = l$ means: as $x \rightarrow a$ ($x \neq a$), $f(x) \rightarrow l$

" $x \rightarrow 2$ " is read "x tends to 2" or "x approaches 2"

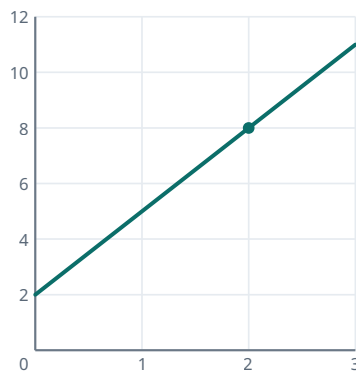
EXAMPLE (FROM THE UNIT)

Find what $y = f(x) = 3x + 2$ approaches as $x \rightarrow 2$, using tables.

x (from left)	1.9	1.98	1.998	1.9998	...	1.99999998
f(x)	7.7	7.94	7.994	7.9994	...	7.99999994

x (from right)	2.1	2.01	2.001	2.0001	...	2.00000001
f(x)	8.3	8.03	8.003	8.0003	...	8.00000003

From both sides $f(x)$ gets closer to 8. So $\lim_{x \rightarrow 2} (3x + 2) = 8$.



$y = 3x + 2$: from the left and from the right, the curve heads to the point (2, 8).

Remember: the limit is about the values near a , not at a . A function may have a limit at a point where it is not even defined (see the factorisation method).

Direct substitution method and algebra of limits

Direct Substitution Method (D.S.M.): Evaluating $\lim_{x \rightarrow a} f(x)$ by simply putting $x = a$; valid when no indeterminate form appears (always for polynomials).

$x = a$ നേരിട്ട് substitute ചെയ്ത് $\lim_{x \rightarrow a} f(x)$ കണ്ടുപിടിക്കുന്ന രീതി; indeterminate form വരാത്തപ്പോൾ ഇത് ശരിയാണ് (polynomials-ന് എപ്പോഴും).

Algebra of limits: If $\lim f(x)$ and $\lim g(x)$ exist, the limit of their sum, difference, product or quotient ($\lim g \neq 0$) is the sum, difference, product or quotient of the limits.

$\lim f(x)$ -ഉം $\lim g(x)$ -ഉം exist ചെയ്യുന്നുണ്ടെങ്കിൽ, അവയുടെ sum, difference, product, quotient ($\lim g \neq 0$) എന്നിവയുടെ limit, limits-ന്റെ sum, difference, product, quotient തന്നെയാണ്.

If $f(x)$ approaches the same value from both sides and there is no trouble at $x = a$, simply put $x = a$. This is the **Direct Substitution Method (D.S.M.)**. It always works for polynomials. It is justified by the algebra of limits below, which holds whenever $\lim f(x)$ and $\lim g(x)$ both exist.

മലയാളത്തിൽ

ഏറ്റവും എളുപ്പമുള്ള വഴി: x -ന്റെ സ്ഥാനത്ത് a നേരിട്ട് substitute ചെയ്യുക. ഉത്തരം ഒരു സാധാരണ number ആയി കിട്ടിയാൽ അതാണ് limit. Polynomial-ൽ ഇത് എപ്പോഴും work ചെയ്യും. Sum-ന്റെ limit = limits-ന്റെ sum, product-ന്റെ limit = limits-ന്റെ product എന്നിങ്ങനെയുള്ള rules ആണ് ഇതിന് അടിസ്ഥാനം.

1. $\lim (f + g) = \lim f + \lim g$ 2. $\lim (f - g) = \lim f - \lim g$
 3. $\lim (f \cdot g) = (\lim f)(\lim g)$ 4. $\lim (f/g) = (\lim f)/(\lim g)$, provided $\lim g \neq 0$
 5. $\lim \sqrt[n]{f(x)} = \sqrt[n]{\lim f(x)}$ 6. $\lim 1/g(x) = 1/\lim g(x)$, $g(x) \neq 0$ near a
 7. $\lim \log f(x) = \log (\lim f(x))$ 8. $\lim e^{f(x)} = e^{\lim f(x)}$
 9. $\lim f(x)^{g(x)} = (\lim f(x))^{\lim g(x)}$
 10. $\lim k \cdot f(x) = k \cdot \lim f(x)$ 11. $\lim k = k$ (k constant)
- (all limits as $x \rightarrow a$)

EXAMPLE 1 (FROM THE UNIT)

(i) $\lim_{x \rightarrow 3} (x^2 - 2x + 4) = 3^2 - 2 \cdot 3 + 4 = 9 - 6 + 4 = 7$.

(ii) $\lim_{x \rightarrow 2} x(x^2 - 4) = 2(4 - 4) = 0$.

(iii) $\lim_{x \rightarrow -1} (1 + x + x^2 + \dots + x^{100}) = 1 + (-1) + 1 + (-1) + \dots + 1$. After the first 1 there are 100 terms: 50 equal to +1 and 50 equal to -1, which cancel. Answer **1**.

(iv) $\lim_{x \rightarrow \sqrt{2}} (x^3 + 3)/(3 - x^4) = ((\sqrt{2})^3 + 3)/(3 - (\sqrt{2})^4) = (2\sqrt{2} + 3)/(3 - 4) = -(2\sqrt{2} + 3)$. (Denominator limit is $-1 \neq 0$, so rule 4 applies.)

(v) $\lim_{x \rightarrow 3} \sqrt{25 - x^2} = \sqrt{25 - 9} = \sqrt{16} = 4$.

(vi) $f(x) = 2 - x^2$ for $x \neq 0$ and $f(0) = 2$. As $x \rightarrow 0$ we have $x \neq 0$, so we use $2 - x^2$: $\lim = 2 - 0 = 2$.

EXAMPLE 2 (CHECK YOUR PROGRESS 1, Q1)

(i) $\lim_{x \rightarrow 2} (x^2 - 2x + 3)^{x^2 + 1} = (4 - 4 + 3)^{4 + 1} = 3^5 = \mathbf{243}$ (rule 9).

(ii) $\lim_{x \rightarrow 1} \log(x^4 + x^2 + 1) = \log(1 + 1 + 1) = \mathbf{\log 3}$ (rule 7).

(iii) $\lim_{x \rightarrow 5} 3 = 3$ (constant function, rule 11).

(iv) $\lim_{x \rightarrow 3} 4f(x)$, $f(x) = (x - 5)^2$: $4 \cdot (3 - 5)^2 = 4 \cdot 4 = \mathbf{16}$.

Exam tip: in Example 1(iv) the unit's question line prints $x^2 + 3$ but its solution uses $x^3 + 3$; the answer $-(2\sqrt{2} + 3)$ belongs to $x^3 + 3$. (With $x^2 + 3$ the answer would be $5/(-1) = -5$.) Also, IGNOU calls the method "D.S.M." in solutions; you may write "[By D.S.M.]" as a reason.

When direct substitution fails: indeterminate forms

Definition: An indeterminate form is a form obtained on substitution that has no definite meaning: $0/0$, ∞/∞ , $0 \times \infty$, $\infty - \infty$, 0^0 , 1^∞ or ∞^0 .

Substitute ചെയ്യുമ്പോൾ കിട്ടുന്ന, നിശ്ചിതമായ അർത്ഥമില്ലാത്ത form ആണ് indeterminate form: $0/0$, ∞/∞ , $0 \times \infty$, $\infty - \infty$, 0^0 , 1^∞ അല്ലെങ്കിൽ ∞^0 .

If substitution produces one of the seven **indeterminate forms**, the expression has no meaning as it stands and D.S.M. fails. We must first rewrite the expression, using one of the methods that follow, and then substitute.

മലയാളത്തിൽ

$x = a$ ഇടപോൾ $0/0$ പോലുള്ള ഒരു "അർത്ഥമില്ലാത്ത" form കിട്ടിയാൽ, limit ഇല്ല എന്നല്ല അർത്ഥം; D.S.M. പറ്റില്ല എന്ന് മാത്രം. അപ്പോൾ expression-നെ factorise ചെയ്തോ, LCM എടുത്തോ, rationalise ചെയ്തോ, standard results ഉപയോഗിച്ചോ മാറ്റി എഴുതിയ ശേഷം വീണ്ടും substitute ചെയ്യണം.

Indeterminate forms: $0/0$, ∞/∞ , $0 \times \infty$, $\infty - \infty$, 0^0 , 1^∞ , ∞^0

Methods: I. Factorisation II. L.C.M. III. Rationalisation IV. Standard results

What you get on substituting	Try
$0/0$ with polynomials	Factorisation: cancel $(x - a)$
$\infty - \infty$ with fractions	L.C.M., which usually turns it into $0/0$
$0/0$ with square roots	Rationalisation (multiply by the conjugate)
$0/0$ with $\sin$, $\tan$, e^x , a^x , $\log$, or $x^n - a^n$	Standard results
∞/∞ as $x \rightarrow \infty$	Divide by a power of x

Exam tip: always write the form you got, e.g. "[$0/0$ form, so D.S.M. fails]", before switching method. IGNOU solutions do this every time and it earns a step mark.

Factorisation method

Definition: For a $0/0$ form $\lim_{x \rightarrow a} f(x)/g(x)$ with $f(a) = g(a) = 0$, factorise out the common factor $(x - a)$, cancel it, and then substitute $x = a$.

$f(a) = g(a) = 0$ ആയ $0/0$ form $\lim_{x \rightarrow a} f(x)/g(x)$ -ൽ, പൊതുവായ factor $(x - a)$ factorise ചെയ്ത് cancel ചെയ്തശേഷം $x = a$ substitute ചെയ്യുന്നു.

Use this for $\lim f(x)/g(x)$ when both $f(a) = 0$ and $g(a) = 0$. Then $(x - a)$ is a factor of both (factor theorem). Write $f(x) = (x - a) \cdot p(x)$, $g(x) = (x - a) \cdot q(x)$ (by factoring or long division), cancel $(x - a)$, and try D.S.M. again. If you still get $0/0$, repeat.

മലയാളത്തിൽ

0/0 കിട്ടിയാൽ numerator-ലും denominator-ലും $(x - a)$ എന്ന common factor ഒളിഞ്ഞിരിപ്പുണ്ട്. അത് പുറത്തെടുത്ത് cancel ചെയ്യുക. "0-നെ 0 കൊണ്ട് cancel ചെയ്യാമോ?" എന്ന് സംശയിക്കേണ്ട: x , a -യോട് അടുക്കുന്നതേയുള്ളൂ, a -ക്ക് തുല്യമല്ല, അതുകൊണ്ട് $x - a \neq 0$ ആണ്. വീണ്ടും 0/0 വന്നാൽ ഒരിക്കൽ കൂടി factorise ചെയ്യുക.

$f(a) = 0$ and $g(a) = 0 \Rightarrow (x - a)$ divides both

$$\lim_{x \rightarrow a} f(x)/g(x) = \lim_{x \rightarrow a} p(x)/q(x)$$

Useful: $a^2 - b^2 = (a - b)(a + b)$, $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$, $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

EXAMPLE 1 (FROM THE UNIT)

Evaluate $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2)$.

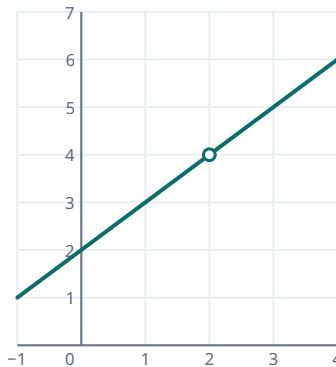
Substituting gives 0/0, so D.S.M. fails.

$$\lim_{x \rightarrow 2} (x - 2)(x + 2)/(x - 2) = \lim_{x \rightarrow 2} (x + 2) \quad [x \rightarrow 2 \Rightarrow x - 2 \neq 0, \text{ so cancel}]$$

$$= 2 + 2 = 4.$$

x	1.9	1.99	1.999	2	2.001	2.01	2.1
$(x^2 - 4)/(x - 2)$	3.9	3.99	3.999	undefined	4.001	4.01	4.1

The table confirms it: the function is undefined at 2, yet the values close in on 4.



$f(x) = (x^2 - 4)/(x - 2)$ is the line $y = x + 2$ with a hole at $(2, 4)$: $f(2)$ is undefined, but the limit is 4.

EXAMPLE 2 (FROM THE UNIT)

$$(i) \lim_{x \rightarrow -1} (x^3 + 1)/(x + 1) \quad [0/0]$$

$$= \lim_{x \rightarrow -1} (x + 1)(x^2 - x + 1)/(x + 1) = \lim_{x \rightarrow -1} (x^2 - x + 1) = 1 + 1 + 1 = 3.$$

$$(ii) \lim_{x \rightarrow 1} (x^2 + x - 2)/(x^2 - 5x + 4) \quad [0/0]$$

$$= \lim_{x \rightarrow 1} (x + 2)(x - 1)/((x - 4)(x - 1)) = \lim_{x \rightarrow 1} (x + 2)/(x - 4) = 3/(-3) = -1.$$

EXAMPLE 3 (FROM THE UNIT, LONG DIVISION)

Evaluate $\lim_{x \rightarrow 3} (x^3 - 3x^2 + x - 3)/(x^3 - 10x^2 + 27x - 18)$.

At $x = 3$: numerator $27 - 27 + 3 - 3 = 0$, denominator $27 - 90 + 81 - 18 = 0$. [0/0]

Divide each by $x - 3$: $x^3 - 3x^2 + x - 3 = (x - 3)(x^2 + 1)$ and $x^3 - 10x^2 + 27x - 18 = (x - 3)(x^2 - 7x + 6)$.

Limit = $\lim_{x \rightarrow 3} (x^2 + 1)/(x^2 - 7x + 6) = (9 + 1)/(9 - 21 + 6) = 10/(-6) = -5/3$.

EXAMPLE 4 (CHECK YOUR PROGRESS 1, Q2: FACTORISE TWICE)

Evaluate $\lim_{x \rightarrow 2} (x^3 - 7x^2 + 16x - 12)/(x^4 - 6x^3 - 3x^2 + 52x - 60)$.

At $x = 2$: $8 - 28 + 32 - 12 = 0$ and $16 - 48 - 12 + 104 - 60 = 0$. [0/0]

Step 1: = $\lim (x - 2)(x^2 - 5x + 6)/((x - 2)(x^3 - 4x^2 - 11x + 30)) = \lim (x^2 - 5x + 6)/(x^3 - 4x^2 - 11x + 30)$.

At $x = 2$ this is again $(4 - 10 + 6)/(8 - 16 - 22 + 30) = 0/0$.

Step 2: = $\lim (x - 2)(x - 3)/((x - 2)(x^2 - 2x - 15)) = \lim (x - 3)/(x^2 - 2x - 15) = (-1)/(4 - 4 - 15) = 1/15$.

Similarly $\lim_{x \rightarrow 2} (x^3 - 4x^2 + 5x - 2)/(x^3 - 2x - 4) = \lim (x^2 - 2x + 1)/(x^2 + 2x + 2) = 1/10$.

Exam tip: in the unit's Example 2(iv) the numerator is printed as $x^3 - 3x^2 + x$, but that is 3 (not 0) at $x = 3$, and it is not $(x - 3)(x^2 + 1)$. The intended numerator is $x^3 - 3x^2 + x - 3 = (x - 3)(x^2 + 1)$, and with it the answer $-5/3$ is correct.

L.C.M. method

Definition: For an $\infty - \infty$ form, combine the fractions over their L.C.M. and simplify; this usually gives a 0/0 form, which is then finished by factorisation.

$\infty - \infty$ form-ൽ fractions-നെ L.C.M. എടുത്ത് ഒന്നാക്കി simplify ചെയ്യുന്നു; സാധാരണ ഇത് ഒരു 0/0 form ആകും, അത് factorisation വഴി പൂർത്തിയാക്കുന്നു.

Use this for the $\infty - \infty$ form, which appears when two fractions both blow up at $x = a$. Combine them over their L.C.M. and simplify. The result is usually a 0/0 form, which you then finish by factorisation.

മലയാളത്തിൽ

രണ്ട് fractions തമ്മിൽ കറയ്ക്കുമ്പോൾ രണ്ടിന്റെയും denominator 0 ആയാൽ $\infty - \infty$ form കിട്ടും. സ്കൂളിൽ ചെയ്തതുപോലെ LCM എടുത്ത് ഒരു fraction ആക്കുക. മിക്കപ്പോഴും അത് 0/0 ആയി മാറും, പിന്നെ factorisation കൊണ്ട് തീർക്കാം.

$\infty - \infty$ form $\rightarrow$ take L.C.M. $\rightarrow$ (usually) $0/0 \rightarrow$ factorise and cancel

EXAMPLE 1 (FROM THE UNIT)

Evaluate $\lim_{x \rightarrow 3} [1/(x-3) - 3/(x^2-3x)]$.

$[\infty - \infty$ form, so D.S.M. fails.] Note $x^2 - 3x = x(x-3)$.

$1/(x-3) - 3/(x(x-3)) = (x-3)/(x(x-3)) = 1/x$ ($x \neq 3$).

Limit = $1/3$.

EXAMPLE 2 (CHECK YOUR PROGRESS 1, Q3)

Evaluate $\lim_{x \rightarrow -2} [1/(x+2) - 6/(x^3+x^2-2x)]$.

Factor: $x^3 + x^2 - 2x = x(x^2 + x - 2) = x(x+2)(x-1)$.

L.C.M.: $[x(x-1) - 6]/(x(x+2)(x-1)) = (x^2 - x - 6)/(x(x+2)(x-1)) = (x-3)(x+2)/(x(x+2)(x-1))$.

Cancel $x+2$: $\lim_{x \rightarrow -2} (x-3)/(x(x-1)) = (-5)/((-2)(-3)) = -5/6$.

Rationalisation method

Definition: When a square root causes a $0/0$ form, multiply numerator and denominator by the conjugate of the root expression ($\sqrt{A} - B \rightarrow \sqrt{A} + B$), then cancel the common factor.

ഒരു square root കാരണം $0/0$ form വരുമ്പോൾ, root expression-ന്റെ conjugate ($\sqrt{A} - B \rightarrow \sqrt{A} + B$) കൊണ്ട് numerator-നെയും denominator-നെയും ഗുണിച്ച്, പൊതുവായ factor cancel ചെയ്യുന്നു.

When a square root causes the $0/0$ form, multiply numerator and denominator by the **conjugate** of the root expression ($\sqrt{A} - B$ becomes $\sqrt{A} + B$). The identity $(a-b)(a+b) = a^2 - b^2$ removes the root, and the troublesome factor $(x-a)$ then appears and cancels.

മലയാളത്തിൽ

Square root ഉള്ളിടത്ത് $0/0$ വന്നാൽ conjugate കൊണ്ട് മുകളിലും താഴെയും

ഗുണിക്കുക: $\sqrt{(4+x)} - 2$ ആണെങ്കിൽ $\sqrt{(4+x)} + 2$ കൊണ്ട്. $(a-b)(a+b) = a^2 - b^2$

ഉപയോഗിച്ച് root മാറ്റം, അപ്പോൾ cancel ചെയ്യാൻ പറ്റുന്ന factor പുറത്ത് വരും.

$$(\sqrt{A} - \sqrt{B})(\sqrt{A} + \sqrt{B}) = A - B$$

$$(\sqrt{A} - b)(\sqrt{A} + b) = A - b^2$$

EXAMPLE 1 (FROM THE UNIT)

Evaluate $\lim_{x \rightarrow 0} (\sqrt{4+x} - 2)/x$.

[0/0] Multiply by $(\sqrt{4+x} + 2)/(\sqrt{4+x} + 2)$:

$$\begin{aligned} &= \lim ((4+x) - 4)/(x(\sqrt{4+x} + 2)) = \lim x/(x(\sqrt{4+x} + 2)) = \lim 1/(\sqrt{4+x} + 2) \\ &= 1/(2+2) = \mathbf{1/4}. \end{aligned}$$

EXAMPLE 2 (FROM THE UNIT)

Evaluate $\lim_{x \rightarrow 3} (\sqrt{5x-6} - \sqrt{x+6})/(x^2-9)$.

[0/0, since $\sqrt{9} - \sqrt{9} = 0$] Multiply by $\sqrt{5x-6} + \sqrt{x+6}$:

Numerator becomes $(5x-6) - (x+6) = 4x-12 = 4(x-3)$.

$$\begin{aligned} &= \lim 4(x-3)/((x-3)(x+3)(\sqrt{5x-6} + \sqrt{x+6})) = \lim 4/((x+3)(\sqrt{5x-6} + \sqrt{x+6})) \\ &= 4/(6 \cdot (3+3)) = 4/36 = \mathbf{1/9}. \end{aligned}$$

EXAMPLE 3 (CHECK YOUR PROGRESS 1, Q4)

Evaluate $\lim_{x \rightarrow 2} (\sqrt{3+x} - \sqrt{5})/(x-2)$.

$$= \lim ((3+x) - 5)/((x-2)(\sqrt{3+x} + \sqrt{5})) = \lim 1/(\sqrt{3+x} + \sqrt{5}) = 1/(2\sqrt{5}) = \mathbf{\sqrt{5}/10}.$$

Standard results

Definition: Standard results are known limits used without proof, e.g. $\lim_{\theta \rightarrow 0} (\sin \theta)/\theta = 1$, $\lim_{x \rightarrow 0} (a^x - 1)/x = \log_e a$ and $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$.

Proof ഇല്ലാതെ നേരിട്ട് ഉപയോഗിക്കുന്ന അറിയപ്പെടുന്ന limits ആണ് standard results; ഉദാ.

$$\lim_{\theta \rightarrow 0} (\sin \theta)/\theta = 1, \lim_{x \rightarrow 0} (a^x - 1)/x = \log_e a, \lim_{x \rightarrow 0} (1+x)^{1/x} = e.$$

Some 0/0 and 1^∞ forms cannot be factorised. For these we use standard limits (quoted without proof). The trick is to make the expression look exactly like a standard form: whatever is inside sin, tan, $e^{(\dots)}$ or $\log(1 + \dots)$ must also appear, unchanged, in the denominator. Multiply and divide by a suitable number to achieve this. Angles are in radians.

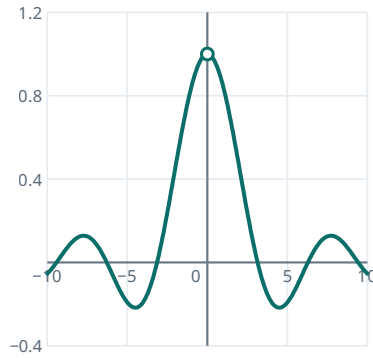
മലയാളത്തിൽ

ഇവ മനഃപാഠമാക്കേണ്ട formulas ആണ്. $\sin 4x/3x$ പോലുള്ളതിൽ $\sin$ -ന്റെ ഉള്ളിലുള്ള $4x$ തന്നെ താഴെയും വേണം; അതിനായി $4x$ കൊണ്ട് ഗുണിച്ചും ഹരിച്ചും ക്രമീകരിക്കുക. കടയിൽ ഒരു താക്കോലിന് duplicate ഉണ്ടാക്കുന്നത് പോലെ, "shape" കൃത്യമായി match ചെയ്താലേ formula തുറക്കൂ. $x \rightarrow 0$ ആണെങ്കിൽ $4x \rightarrow 0$ -ഉം ആണ്.

1. $\lim_{x \rightarrow a} (x^n - a^n)/(x - a) = n \cdot a^{n-1}$
2. $\lim_{\theta \rightarrow 0} (\sin \theta)/\theta = \lim_{\theta \rightarrow 0} \theta/(\sin \theta) = 1$
3. $\lim_{\theta \rightarrow 0} \cos \theta = 1$
4. $\lim_{\theta \rightarrow 0} (\tan \theta)/\theta = \lim_{\theta \rightarrow 0} \theta/(\tan \theta) = 1$
5. $\lim_{x \rightarrow 0} (a^x - 1)/x = \log_e a$ ($a > 0, a \neq 1$); in particular $\lim_{x \rightarrow 0} (e^x - 1)/x = 1$
6. $\lim_{x \rightarrow 0} \log(1 + x)/x = 1$
7. $\lim_{x \rightarrow 0} (1 + x)^{1/x} = e$
8. $\lim_{x \rightarrow \infty} (1 + 1/x)^x = e$

Value tables make these believable. Notice that the function is undefined at $x = 0$ in every row, yet both sides close in on the same number.

x	-0.5	-0.1	-0.01	0	0.01	0.1	0.5
$(\sin x)/x$	0.958851	0.998334	0.999983	$\rightarrow 1$	0.999983	0.998334	0.958851
$(e^x - 1)/x$	0.786939	0.951626	0.995017	$\rightarrow 1$	1.005017	1.051709	1.297443
$\log(1 + x)/x$	1.386294	1.053605	1.005034	$\rightarrow 1$	0.995033	0.953102	0.810930
$(2^x - 1)/x$	0.585786	0.669670	0.690750	$\rightarrow \log 2$ $\approx$ 0.6931	0.695555	0.717735	0.828427
$(1 + x)^{1/x}$	4	2.867972	2.731999	$\rightarrow e \approx$ 2.7183	2.704814	2.593742	2.25



$y = (\sin x)/x$ (x in radians). It is undefined at $x = 0$, yet the values close in on 1 from both sides.

EXAMPLE 1: $(X^n - A^n)/(X - A)$ (FROM THE UNIT)

(i) $\lim_{x \rightarrow 3} (x^5 - 243)/(x^4 - 81) = \lim (x^5 - 3^5)/(x^4 - 3^4)$. Divide top and bottom by $x - 3$:

$$= [\lim (x^5 - 3^5)/(x - 3)] / [\lim (x^4 - 3^4)/(x - 3)] = 5 \cdot 3^4 / (4 \cdot 3^3) = \mathbf{15/4}.$$

(ii) $\lim_{x \rightarrow 2} (x^{10/3} - 2^{10/3})/(x^{4/3} - 2^{4/3}) = [(10/3) \cdot 2^{7/3}] / [(4/3) \cdot 2^{1/3}] = (10/4) \cdot 2^2 = \mathbf{10}.$

(iii) (CYP 2) $\lim_{x \rightarrow \sqrt{2}} (x^{10} - 32)/(x - \sqrt{2})$: since $32 = (\sqrt{2})^{10}$, this is $10(\sqrt{2})^9 = 10 \cdot 16\sqrt{2} = \mathbf{160\sqrt{2}}.$

EXAMPLE 2: TRIGONOMETRIC (FROM THE UNIT)

(i) $\lim_{x \rightarrow 0} (\sin 4x)/(3x) = \lim (\sin 4x)/(4x) \cdot (4/3) = 1 \cdot 4/3 = \mathbf{4/3}$ [$x \rightarrow 0 \Rightarrow 4x \rightarrow 0$]

(ii) $\lim_{x \rightarrow 0} \cos 5x = \mathbf{1}$ [$5x \rightarrow 0$ and $\lim \cos \theta = 1$]

(iii) $\lim_{x \rightarrow 0} (\tan 3x)/(\sin 2x) = \lim (\tan 3x)/(3x) \cdot (2x)/(\sin 2x) \cdot (3/2) = 1 \cdot 1 \cdot 3/2 = \mathbf{3/2}.$

EXAMPLE 3: EXPONENTIAL AND LOG (FROM THE UNIT)

(i) $\lim_{x \rightarrow 0} (2^{5x} - 1)/x = 5 \cdot \lim (2^{5x} - 1)/(5x) = \mathbf{5 \log_e 2}.$

(ii) $\lim_{x \rightarrow 0} (e^{ax} - 1)/x = a \cdot \lim (e^{ax} - 1)/(ax) = a \cdot 1 = \mathbf{a}.$

(iii) $\lim_{x \rightarrow 0} \log(1 + 5x)/x = 5 \cdot \lim \log(1 + 5x)/(5x) = \mathbf{5}.$

(iv) $\lim_{x \rightarrow 0} \log(1 + 3x)/(e^{2x} - 1) = (3/2) \cdot [\log(1 + 3x)/(3x)] \cdot [(2x)/(e^{2x} - 1)] \rightarrow (3/2) \cdot 1 \cdot 1 = \mathbf{3/2}.$

(v) $\lim_{x \rightarrow 0} (1 + 8x)^{1/x} = \lim [(1 + 8x)^{1/(8x)}]^8 = \mathbf{e^8}$ [1^∞ form; $8x \rightarrow 0$]

EXAMPLE 4 (CHECK YOUR PROGRESS 2, Q1)

(i) $\lim_{x \rightarrow 0} ((ab)^{3x} - 1)/x = 3 \cdot \lim_{x \rightarrow 0} ((ab)^{3x} - 1)/(3x) = 3 \log(ab)$.

(ii) $\lim_{x \rightarrow 0} (e^{\sin x} - 1)/\tan x = [(e^{\sin x} - 1)/\sin x] \cdot [(\sin x)/x] \cdot [x/\tan x] \rightarrow 1 \cdot 1 \cdot 1 = 1$ (as $x \rightarrow 0$, $\sin x \rightarrow 0$).

(iii) $\lim_{x \rightarrow 0} \log(1 + 8x^2)/(e^{x^2} - 1) = 8 \cdot [\log(1 + 8x^2)/(8x^2)] \cdot [x^2/(e^{x^2} - 1)] \rightarrow 8$.

(iv) $\lim_{x \rightarrow 0} (a^x - e^x)/x = \lim [(a^x - 1)/x - (e^x - 1)/x] = \log a - 1 = \log(a/e)$.

(v) $\lim_{x \rightarrow 0} (e^x - (1 + 2x))/(2^x - 1)$. Divide top and bottom by x : $[(e^x - 1)/x - 2]/[(2^x - 1)/x] \rightarrow (1 - 2)/\log 2 = -1/\log 2$.

(vi) $\lim_{x \rightarrow 0} [x(1 + 2x)^{1/x} - (e^{2x} - 1)]/x = \lim (1 + 2x)^{1/x} - \lim (e^{2x} - 1)/x = [(1 + 2x)^{1/(2x)}]^2 - 2 \cdot 1 \rightarrow e^2 - 2$.

Exam tip: $(\sin x)/x \rightarrow 1$ only when x is in radians and $x \rightarrow 0$. Do not "cancel" $\sin$: $\sin 4x/\sin 2x$ is not $4x/2x$ by algebra, it only has the same limit because each $\sin \theta$ behaves like θ near 0. In the unit's Example 5(ii) the final value is not clearly printed; it is 10, as worked above.

Infinite limits ($x \rightarrow \infty$)

Definition: $\lim_{x \rightarrow \infty} f(x) = l$ means $f(x)$ approaches l as x grows beyond every real number; the basic result is $\lim_{x \rightarrow \infty} 1/x^n = 0$ for $n > 0$.

x ഏതൊരു real number-നെക്കാളും വലുതായി വളരുമ്പോൾ $f(x)$, l -നോട് അടുക്കുന്നു എന്നാണ് $\lim_{x \rightarrow \infty} f(x) = l$ -ന്റെ അർത്ഥം; $n > 0$ ആണെങ്കിൽ $\lim_{x \rightarrow \infty} 1/x^n = 0$ എന്നതാണ് അടിസ്ഥാന result.

" $x \rightarrow \infty$ " means x grows beyond every real number K . As the denominator of $1/x$ grows (10, 100, 1000, ...), the value 0.1, 0.01, 0.001, ... shrinks towards 0. So $1/x^n \rightarrow 0$ for every $n > 0$. For a rational function as $x \rightarrow \infty$, divide numerator and denominator by a suitable power of x (the unit uses the smaller of the two degrees), then use $1/x^n \rightarrow 0$.

മലയാളത്തിൽ

ഒരു കേക്ക് 10 പേർക്ക്, 100 പേർക്ക്, 1000 പേർക്ക് വീതിച്ചാൽ ഓരോരുത്തർക്കും കിട്ടുന്ന പങ്ക് 0-ലേക്ക് അടുക്കും: അതാണ് $1/x \rightarrow 0$. Rational function-ൽ x -ന്റെ ഒരു power കൊണ്ട് മുകളിലും താഴെയും ഹരിച്ചാൽ $1/x$, $1/x^2$ പോലുള്ള terms എല്ലാം 0 ആകും; ബാക്കിയുള്ളതാണ് ഉത്തരം. Degree-കൾ compare ചെയ്താൽ ഉത്തരം മുൻകൂട്ടി അറിയാം.

$$\lim_{x \rightarrow \infty} 1/x^n = 0, \quad n > 0$$

$$\lim_{x \rightarrow \infty} (a_0 x^m + a_1 x^{m-1} + \dots + a_m) / (b_0 x^n + b_1 x^{n-1} + \dots + b_n) =$$

$$a_0/b_0 \text{ if } m = n; \quad 0 \text{ if } m < n; \quad \infty \text{ if } m > n \quad (-\infty \text{ if } a_0/b_0 < 0)$$

EXAMPLE 1 (FROM THE UNIT)

(i) $\lim_{x \rightarrow \infty} (3x^2 + 5x - 1)/(4x^2 - 3x + 9)$. Degrees equal (2). Divide by x^2 :

$$= \lim (3 + 5/x - 1/x^2)/(4 - 3/x + 9/x^2) = (3 + 0 - 0)/(4 - 0 + 0) = 3/4.$$

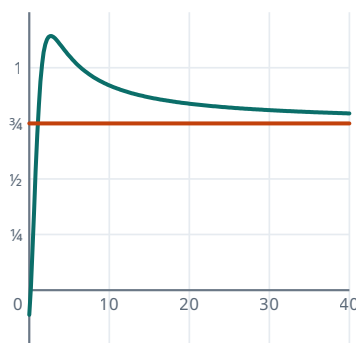
x	10	100	1000	10000
f(x)	0.9208	0.7681	0.7518	0.7502

(ii) $\lim_{x \rightarrow \infty} (5x^5 + x + 1)/(x^3 + 5)$. Numerator degree larger. Divide by x^3 :

$$= \lim (5x^2 + 1/x^2 + 1/x^3)/(1 + 5/x^3) = (\infty + 0 + 0)/(1 + 0) = \infty.$$

(iii) $\lim_{x \rightarrow \infty} (x^5 + 1)/(4x^7 + 3x^2 + 7)$. Denominator degree larger. Divide by x^5 :

$$= \lim (1 + 1/x^5)/(4x^2 + 3/x^3 + 7/x^5) = 1/\infty = 0.$$



$y = (3x^2 + 5x - 1)/(4x^2 - 3x + 9)$ settles down to the line $y = 3/4$ as $x \rightarrow \infty$.

EXAMPLE 2 (CHECK YOUR PROGRESS 2, Q2: ROOTS)

Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3} + \sqrt[4]{x^4 + 2})/(\sqrt[7]{x^7 + x^2} - \sqrt[3]{x^2 + 5})$.

Each of $\sqrt{x^2 + 3}$, $\sqrt[4]{x^4 + 2}$, $\sqrt[7]{x^7 + x^2}$ behaves like x , while $\sqrt[3]{x^2 + 5}$ grows only like $x^{2/3}$. Divide top and bottom by x :

$$= \lim [\sqrt{1 + 3/x^2} + \sqrt[4]{1 + 2/x^4}] / [\sqrt[7]{1 + 1/x^5} - \sqrt[3]{1/x + 5/x^3}]$$

$$= (1 + 1)/(1 - 0) = 2.$$

Exam tip: dividing by the highest power of the denominator also works and is often cleaner. Whichever you choose, state " $1/x^n \rightarrow 0$ as $x \rightarrow \infty$ for $n > 0$ " as your reason.

Left hand and right hand limits

Left hand limit (L.H.L.): The value $f(x)$ approaches as $x \rightarrow a$ through values less than a : $\lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0^+} f(a - h)$.

a -യെക്കാൾ ചെറിയ values വഴി $x \rightarrow a$ ആകുമ്പോൾ $f(x)$ അടുക്കുന്ന value: $\lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0^+} f(a - h)$.

Right hand limit (R.H.L.): The value $f(x)$ approaches as $x \rightarrow a$ through values greater than a : $\lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0^+} f(a + h)$.

a -യെക്കാൾ വലിയ values വഴി $x \rightarrow a$ ആകുമ്പോൾ $f(x)$ അടുക്കുന്ന value: $\lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0^+} f(a + h)$.

Existence of limit: $\lim_{x \rightarrow a} f(x)$ exists only when L.H.L. and R.H.L. both exist and are equal.

L.H.L.-ഉം R.H.L.-ഉം exist ചെയ്യുകയും തുല്യമാവുകയും ചെയ്യുമ്പോൾ മാത്രമേ $\lim_{x \rightarrow a} f(x)$ exist ചെയ്യൂ.

Approaching a from values slightly less than a gives the **left hand limit**, written $\lim_{x \rightarrow a^-} f(x)$. Approaching from values slightly greater than a gives the **right hand limit**, $\lim_{x \rightarrow a^+} f(x)$. To compute them put $x = a - h$ (for L.H.L.) or $x = a + h$ (for R.H.L.), where $h > 0$ is very small, and let $h \rightarrow 0^+$. We need these for modulus functions and for piecewise functions that change formula at a . **The limit exists only when L.H.L. and R.H.L. both exist and are equal.**

മലയാളത്തിൽ

ഒരു പാലത്തിന്റെ നടുവിലേക്ക് രണ്ട് കരകളിൽ നിന്നും പണി തുടങ്ങി എന്ന് കരുതുക. രണ്ട് ഭാഗവും ഒരേ ഉയരത്തിൽ കൂടിച്ചേർന്നാലേ പാലം ശരിയാകൂ; അതുപോലെ L.H.L. = R.H.L. ആയാലേ limit നിലവിലുള്ളൂ. $|x|$ അല്ലെങ്കിൽ piecewise function കണ്ടാൽ ഉടനെ L.H.L., R.H.L. വെച്ചേറെ കണ്ടുപിടിക്കുക. L.H.L.-ന് $x = a - h$, R.H.L.-ന് $x = a + h$ എന്ന് ഇടുക (h ഒരു ചെറിയ positive number).

$$\text{L.H.L.} = \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0^+} f(a - h)$$

$$\text{R.H.L.} = \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0^+} f(a + h)$$

$$\lim_{x \rightarrow a} f(x) \text{ exists} \Leftrightarrow \text{L.H.L.} = \text{R.H.L.}$$

$$|x| = x \text{ if } x \geq 0, |x| = -x \text{ if } x < 0; \text{ so } |-h| = h \text{ for } h > 0$$

EXAMPLE 1 (FROM THE UNIT)

(i) $\lim_{x \rightarrow 0} |x|$. L.H.L.: $x = 0 - h$ gives $|-h| = h \rightarrow 0$. R.H.L.: $x = 0 + h$ gives $|h| = h \rightarrow 0$. Equal, so the limit exists and is **0**.

(ii) $\lim_{x \rightarrow 3} |x - 3|$. L.H.L.: $|3 - h - 3| = |-h| = h \rightarrow 0$. R.H.L.: $|3 + h - 3| = h \rightarrow 0$. Limit = **0**.

EXAMPLE 2 (FROM THE UNIT)

$f(x) = x^2 + 1$ for $x < 1$ and $f(x) = 1 - x^2$ for $x \geq 1$. Find $\lim_{x \rightarrow 1} f(x)$.

L.H.L.: x slightly less than 1, so $f(x) = x^2 + 1 \rightarrow 1 + 1 = 2$.

R.H.L.: x slightly greater than 1, so $f(x) = 1 - x^2 \rightarrow 1 - 1 = 0$.

x	0.9	0.99	0.999	1	1.001	1.01	1.1
$f(x)$	1.81	1.9801	1.998001	0	-0.002001	-0.0201	-0.21

L.H.L. = $2 \neq 0$ = R.H.L., so $\lim_{x \rightarrow 1} f(x)$ **does not exist**.



Example 2 above: left piece heads to 2, right piece starts at 0. L.H.L. = $2 \neq 0$ = R.H.L., so no limit at $x = 1$.

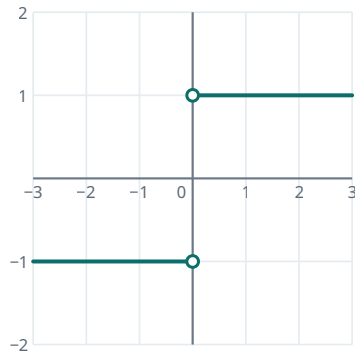
EXAMPLE 3 (FROM THE UNIT)

$f(x) = |x - 4|/(x - 4)$ for $x \neq 4$ and $f(4) = 0$. Find $\lim_{x \rightarrow 4} f(x)$.

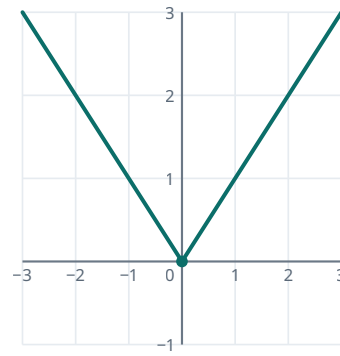
L.H.L.: $x = 4 - h$: $|-h|/(-h) = h/(-h) = -1$.

R.H.L.: $x = 4 + h$: $|h|/h = h/h = 1$.

$-1 \neq 1$, so the limit **does not exist**. (The value $f(4) = 0$ plays no part in the limit.) The same reasoning shows $\lim_{x \rightarrow 0} |x|/x$ does not exist: L.H.L. = -1 , R.H.L. = 1 .



$y = |x|/x$: L.H.L. = -1, R.H.L. = 1
at $x = 0$, so the limit does not exist.



$y = |x|$: both sides reach 0 at $x = 0$, so the limit is 0 (and $|x|$ is continuous there).

EXAMPLE 4 (FROM THE UNIT: FIND K)

$f(x) = x - |x|$ for $x < 0$ and $f(x) = k$ for $x \geq 0$. If $\lim_{x \rightarrow 0} f(x)$ exists, find k .

L.H.L.: $x = -h$: $(-h) - |-h| = -h - h = -2h \rightarrow 0$.

R.H.L.: $f(x) = k$, so R.H.L. = k .

The limit exists $\Rightarrow$ L.H.L. = R.H.L. $\Rightarrow k = 0$.

EXAMPLE 5 (CHECK YOUR PROGRESS 2, Q3 AND Q4)

(i) $\lim_{x \rightarrow 0^-} (5x + |x|)/(3|x| - 7x)$. Put $x = -h$: $(-5h + h)/(3h + 7h) = -4h/10h = -2/5$.

(ii) $\lim_{x \rightarrow 5^+} (3 - |x|)$. Put $x = 5 + h$: $3 - (5 + h) = -2 - h \rightarrow -2$.

(iii) $f(x) = ax + 3$ for $x < 3$ and $2(x + 1)$ for $x > 3$; $\lim_{x \rightarrow 3} f(x)$ exists. L.H.L. = $3a + 3$, R.H.L. = $2(4) = 8$. So $3a + 3 = 8$, $a = 5/3$.

Exam tip: when you see $|...|$ or a function defined in pieces, never use D.S.M. directly at the break point. Write L.H.L. and R.H.L. separately, label them (1) and (2), then compare. Final line: "L.H.L. $\neq$ R.H.L., hence the limit does not exist."

Continuity at a point

Definition: A function f is continuous at $x = a$ if $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$, i.e. L.H.L. = R.H.L. = value of f at a .

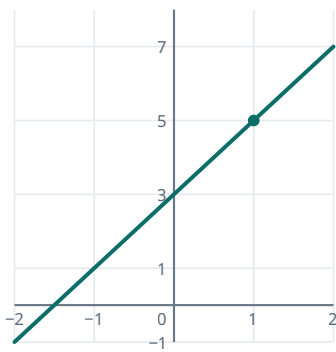
$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$ ആണെങ്കിൽ, അതായത് L.H.L. = R.H.L. = a -യിലെ f -ന്റെ value ആണെങ്കിൽ, f എന്ന function $x = a$ -യിൽ continuous ആണ്.

A function f is **continuous at $x = a$** if $L.H.L. = R.H.L. = f(a)$. This needs three things: $f(a)$ is defined, the limit at a exists, and the limit equals $f(a)$. In a picture: you can draw the graph through $x = a$ without lifting the pencil. If any of the three conditions fails, f is discontinuous at a .

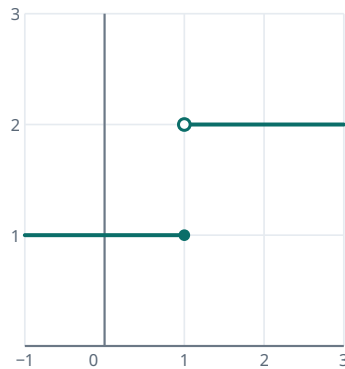
മലയാളത്തിൽ

Continuity എന്നാൽ graph-ൽ "പൊട്ടൽ" ഇല്ല എന്നർത്ഥം: pencil പേപ്പറിൽ നിന്ന് ഉയർത്താതെ വരയ്ക്കാൻ പറ്റണം. ഇതിന് മൂന്ന് കാര്യങ്ങൾ ഒരേ സമയം ശരിയാകണം: L.H.L., R.H.L., $f(a)$ മൂന്നും തുല്യം. Limit ഉണ്ടെങ്കിലും $f(a)$ വേറെ value ആണെങ്കിൽ (ഒരു "കുഴി" മാത്രം) അതും discontinuity തന്നെ.

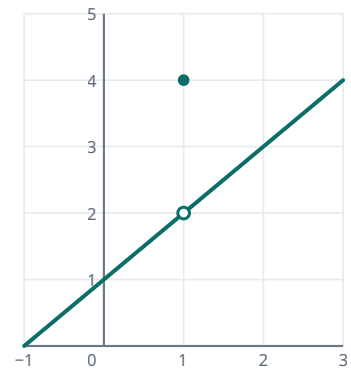
f is continuous at $x = a \Leftrightarrow \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$
i.e. L.H.L. at a = R.H.L. at a = value of f at a



Continuous: $y = 2x + 3$ (Fig. 11.6). No gap anywhere; the pencil never leaves the paper.



Jump discontinuity (Fig. 11.7): L.H.L. = 1, R.H.L. = 2 at $x = 1$.



Removable discontinuity: the limit at $x = 1$ is 2 but $f(1) = 4$. Redefining $f(1) = 2$ fixes it.

Picture	L.H.L.	R.H.L.	$f(a)$	Continuous?
$y = 2x + 3$ at $x = 1$	5	5	5	Yes
Step: 1 ($x \leq 1$), 2 ($x > 1$) at $x = 1$	1	2	1	No (jump)
$(x^2 - 1)/(x - 1)$, $f(1) = 4$, at $x = 1$	2	2	4	No (removable)

EXAMPLE 1 (FROM THE UNIT)

Discuss continuity.

(i) $f(x) = |x|$ at $x = 0$: L.H.L. = R.H.L. = 0 and $f(0) = |0| = 0$. **Continuous.**

(ii) $f(x) = |x - 3|$ at $x = 3$: L.H.L. = R.H.L. = 0 and $f(3) = |0| = 0$. **Continuous.**

(iii) $f(x) = x^2 + 1$ ($x < 1$), $1 - x^2$ ($x \geq 1$) at $x = 1$: L.H.L. = 2 $\neq$ R.H.L. = 0. **Not continuous.**

(iv) $f(x) = |x - 4|/(x - 4)$ ($x \neq 4$), $f(4) = 0$ at $x = 4$: L.H.L. = -1 $\neq$ R.H.L. = 1. **Not continuous.**

(v) $f(x) = |x|/x$ at $x = 0$: L.H.L. = -1 $\neq$ R.H.L. = 1 (and $f(0)$ is not even defined). **Not continuous.**

EXAMPLE 2 (FROM THE UNIT: FIND A AND B)

$f(x) = 7$ for $x < 2$, $f(x) = ax + b$ for $x \geq 2$, $f(2) = a + 5$. f is continuous at $x = 2$. Find a , b .

L.H.L. = 7; R.H.L. = $2a + b$; $f(2) = a + 5$.

Continuity: $7 = 2a + b = a + 5$.

From $7 = a + 5$: $a = 2$. From $7 = 2a + b$: $b = 7 - 4 = 3$. So **$a = 2$, $b = 3$** .

EXAMPLE 3 (CHECK YOUR PROGRESS 2, Q5)

$f(x) = 2x - a$ for $x \geq 0$ and $f(x) = ax + b + 3$ for $x < 0$ is continuous at $x = 0$. Find the relation between a and b .

L.H.L.: $x = -h$: $a(-h) + b + 3 \rightarrow b + 3$.

R.H.L. = $\lim (2x - a) = -a$, and $f(0) = 2(0) - a = -a$.

Continuity: $b + 3 = -a$, i.e. **$a + b + 3 = 0$** .

Exam tip: for "find k / find a and b " questions, write the three quantities L.H.L., R.H.L., $f(a)$, set them equal, and solve. In the unit's Example 9(ii) the line " $= f(0)$ " is a misprint for $f(3)$.

Continuous functions

Definition: A function f is a continuous function if it is continuous at every point of its domain.

Domain-ലെ എല്ലാ point-ലും continuous ആയ function-നെ continuous function എന്ന് പറയുന്നു.

A function is called a **continuous function** if it is continuous at every point of its domain. For example $f(x) = 2x + 3$ on $\mathbb{R}$ has no gap anywhere, so it is a continuous function. The step function $f(x) = 1$ ($x \leq 1$), 2 ($x > 1$) is continuous at every point except $x = 1$, so it is not a continuous function on $\mathbb{R}$. Sums, differences, products and (where the denominator is non-zero) quotients of continuous functions are continuous.

മലയാളത്തിൽ

Domain-ലെ ഓരോ point-ലും continuous ആണെങ്കിൽ ആ function-നെ continuous function എന്ന് വിളിക്കാം. Polynomials ($2x + 3$, $x^2 - 5x + 6$...) എല്ലായിടത്തും continuous ആണ്. രണ്ട് continuous functions കൂട്ടിയാലും കുറച്ചാലും ഗുണിച്ചാലും continuous തന്നെ; ഹരിക്കുമ്പോൾ മാത്രം denominator 0 ആകാത്ത points-ൽ.

f continuous on its domain $\Leftrightarrow f$ continuous at every point of the domain

If f, g are continuous on a common domain: $f + g$, $f - g$, $f \cdot g$ continuous;

f/g continuous provided $g(x) \neq 0$ for all x in the domain

EXAMPLE

Where is $h(x) = (x^2 + 1)/(x - 2)$ continuous?

$x^2 + 1$ and $x - 2$ are polynomials, hence continuous everywhere. Their quotient is continuous wherever $x - 2 \neq 0$.

So h is continuous at every $x \neq 2$, i.e. on $\mathbb{R} - \{2\}$. (At $x = 2$ it is not even defined.)

EXAMPLE

Is $f(x) = |x| + 3x^2$ a continuous function on $\mathbb{R}$?

$|x|$ is continuous everywhere (at 0 we checked L.H.L. = R.H.L. = $f(0) = 0$; elsewhere it is a line). $3x^2$ is a polynomial. Their sum is continuous, so **yes**.

Quick recap

- $\lim_{x \rightarrow a} f(x) = l$ means $f(x)$ gets close to l as x gets close to a ($x \neq a$), from both sides.
- D.S.M.: put $x = a$. Always works for polynomials, and whenever no indeterminate form appears.
- Indeterminate forms: $0/0$, ∞/∞ , $0 \times \infty$, $\infty - \infty$, 0^0 , 1^∞ , ∞^0 . Getting one means "change method", not "no limit".
- $0/0$ with polynomials: factorise, cancel $(x - a)$ (allowed since $x \neq a$), substitute; repeat if needed.
- $\infty - \infty$ with fractions: take L.C.M., then usually factorise.
- $0/0$ with square roots: multiply by the conjugate.
- $(x^n - a^n)/(x - a) \rightarrow na^{n-1}$; $(\sin \theta)/\theta$, $(\tan \theta)/\theta \rightarrow 1$; $\cos \theta \rightarrow 1$ ($\theta \rightarrow 0$, radians).
- $(a^x - 1)/x \rightarrow \log a$; $(e^x - 1)/x \rightarrow 1$; $\log(1 + x)/x \rightarrow 1$; $(1 + x)^{1/x} \rightarrow e$ (all $x \rightarrow 0$).
- Match the inside: $\sin 4x$ needs $4x$ underneath; multiply and divide to create it.
- $x \rightarrow \infty$: $1/x^n \rightarrow 0$ ($n > 0$). Rational function: a_0/b_0 if degrees equal, 0 if top degree smaller, ∞ if larger.
- L.H.L.: put $x = a - h$; R.H.L.: put $x = a + h$; $h \rightarrow 0^+$. Use for $|x|$ and piecewise functions.

- Limit exists $\Leftrightarrow$ L.H.L. = R.H.L.
- Continuous at $a \Leftrightarrow$ L.H.L. = R.H.L. = $f(a)$. Picture: no lifting of the pencil.
- Continuous function = continuous at every point of its domain; $f \pm g$, fg , f/g ($g \neq 0$) stay continuous.

Check yourself

Q. 1. Evaluate $\lim_{x \rightarrow 2} (x^3 + 2x^2 - 1)/(x + 3)$.

D.S.M. works (denominator $5 \neq 0$): $(8 + 8 - 1)/5 = 15/5 = 3$.

Q. 2. Evaluate $\lim_{x \rightarrow 3} (x^2 - 5x + 6)/(x^2 - 9)$.

0/0 form. $(x - 2)(x - 3)/((x - 3)(x + 3)) = (x - 2)/(x + 3) \rightarrow 1/6$.

Q. 3. Evaluate $\lim_{x \rightarrow 2} (x^3 - 4x^2 + 5x - 2)/(x^3 - 2x - 4)$.

0/0 form. Numerator = $(x - 2)(x^2 - 2x + 1)$, denominator = $(x - 2)(x^2 + 2x + 2)$.

Limit = $(4 - 4 + 1)/(4 + 4 + 2) = 1/10$.

Q. 4. Evaluate $\lim_{x \rightarrow 1} [1/(x - 1) - 2/(x^2 - 1)]$.

$\infty - \infty$ form. L.C.M.: $(x + 1 - 2)/((x - 1)(x + 1)) = (x - 1)/((x - 1)(x + 1)) = 1/(x + 1) \rightarrow 1/2$.

Q. 5. Evaluate $\lim_{x \rightarrow 0} (\sqrt{1 + x} - \sqrt{1 - x})/x$.

0/0 form. Multiply by $\sqrt{1 + x} + \sqrt{1 - x}$: numerator becomes $(1 + x) - (1 - x) = 2x$.

$= \lim 2/(\sqrt{1 + x} + \sqrt{1 - x}) = 2/2 = 1$.

Q. 6. Evaluate $\lim_{x \rightarrow 0} (\sin 5x)/(\tan 3x)$ and $\lim_{x \rightarrow 0} (3^x - 1)/(2x)$.

$(\sin 5x)/(5x) \cdot (3x)/(\tan 3x) \cdot 5/3 \rightarrow 5/3$.

$(1/2) \cdot (3^x - 1)/x \rightarrow (1/2) \log 3$.

Q. 7. Evaluate $\lim_{x \rightarrow 0} (1 + 3x)^{2/x}$.

1^∞ form. $(1 + 3x)^{2/x} = [(1 + 3x)^{1/(3x)}]^6 \rightarrow e^6$ ($3x \rightarrow 0$).

Q. 8. Evaluate $\lim_{x \rightarrow \infty} (2x^3 - x + 4)/(5x^3 + 7x^2)$ and $\lim_{x \rightarrow \infty} (x^2 + 1)/(x^4 - 3)$.

Equal degrees: $2/5$. Second: degree of numerator (2) < degree of denominator (4), so the limit is **0**.

Q. 9. Does $\lim_{x \rightarrow 2} |x - 2|/(x - 2)$ exist?

L.H.L.: $x = 2 - h$: $|-h|/(-h) = -1$. R.H.L.: $x = 2 + h$: $h/h = 1$.

L.H.L. $\neq$ R.H.L., so the limit **does not exist**.

Q. 10. $f(x) = kx + 1$ for $x \leq 2$ and $f(x) = 3x - 1$ for $x > 2$. Find k so that f is continuous at $x = 2$.

L.H.L. = $f(2) = 2k + 1$. R.H.L. = $3(2) - 1 = 5$.

Continuity: $2k + 1 = 5 \Rightarrow k = 2$.

Differentiation

ഒരു function എത്ര വേഗത്തിൽ മാറുന്നു എന്ന് അളക്കുന്നതാണ് derivative; graph-ൽ അത് tangent-ന്റെ slope ആണ്. First principle, product/quotient/chain rule, exponential, logarithmic, parametric, implicit functions, higher order derivatives, maxima-minima എന്നിവയെല്ലാം ഈ unit-ൽ step by step പഠിക്കുന്നു.

Derivative: definition, meaning and geometry

Derivative at a point

Derivative by first principle

Table of standard derivatives and basic rules

Product rule

Quotient rule

Chain rule

Exponential and logarithmic functions

Parametric functions

Implicit functions

Derivatives of higher orders

Maxima and minima: the idea and the tests

Maxima and minima: worked problems

Derivative: definition, meaning and geometry

Definition: The derivative of f at $x = a$ is $f'(a) = \lim_{h \rightarrow 0} (f(a + h) - f(a))/h$, provided the limit exists; geometrically, it is the slope of the tangent to $y = f(x)$ at $x = a$.

Limit exist ചെയ്യുന്നുണ്ടെങ്കിൽ, $x = a$ -യിൽ f -ന്റെ derivative $f'(a) = \lim_{h \rightarrow 0} (f(a + h) - f(a))/h$ ആണ്; geometrically ഇത് $x = a$ -യിൽ $y = f(x)$ -ന്റെ tangent-ന്റെ slope ആണ്.

Let $f : D \rightarrow \mathbb{R}$ with $D \subseteq \mathbb{R}$ and $a \in D$. The derivative of f at $x = a$, written $f'(a)$, is the limit below, provided it exists. It measures the rate at which $f(x)$ changes at $x = a$. Geometrically, $f'(a)$ is the **slope of the tangent** to the curve $y = f(x)$ at the point $x = a$, i.e. $\tan \psi$, where ψ is the angle the tangent makes with the positive x -axis.

മലയാളത്തിൽ

Curve-ൽ P എന്ന point-ഉം അതിനടുത്ത് Q എന്ന point-ഉം എടുക്കുക. PQ ചേർക്കുന്ന line (chord) ന്റെ slope $= RQ/PR = (f(a + h) - f(a))/h$. ഇനി h ചെറുതാക്കി Q -നെ P -ലേക്ക് അടുപ്പിച്ചാൽ ആ chord, P -യിലെ tangent ആയി മാറും. അങ്ങനെ കിട്ടുന്ന limit ആണ് derivative. ബൈക്കിന്റെ speedometer കാണിക്കുന്ന ആ നിമിഷത്തെ speed പോലെയാണ് ഇത്: ആ ഒരു point-ലെ rate of change.

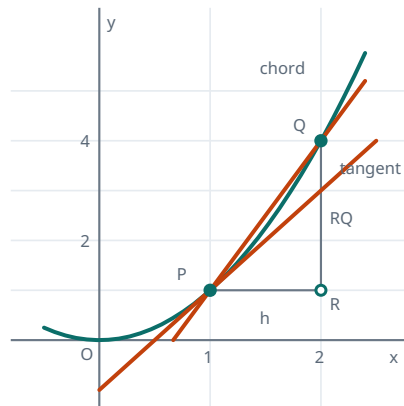
$f'(a) = \lim_{h \rightarrow 0} (f(a+h) - f(a))/h$, provided the limit exists

$f'(a) = \tan \psi$ = slope of the tangent at $x = a$

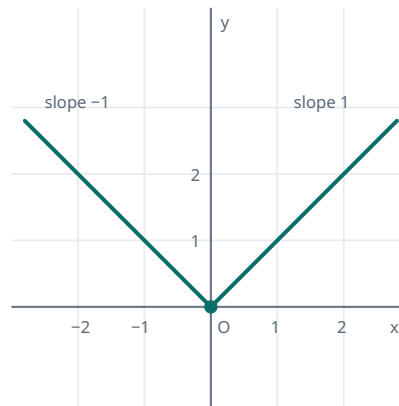
Left hand derivative: $L f'(a) = \lim_{h \rightarrow 0^-} (f(a+h) - f(a))/h$

Right hand derivative: $R f'(a) = \lim_{h \rightarrow 0^+} (f(a+h) - f(a))/h$

$f'(a)$ exists $\Leftrightarrow$ L.H.D. and R.H.D. both exist and L.H.D. = R.H.D.



$y = x^2$. Chord PQ has slope $RQ/PR = 3$. As Q slides down to P ($h \rightarrow 0$) the chord turns into the tangent at $P(1, 1)$, whose slope is $f'(1) = 2$.



$f(x) = |x|$. At the corner $x = 0$ the left slope is -1 and the right slope is 1 , so $f'(0)$ does not exist.

If $x = a$ is a **corner point** of the graph, the slope of the tangent just to the left differs from the slope just to the right, so the limit does not exist and f is not differentiable there. Example: $f(x) = |x|$ is differentiable everywhere except $x = 0$.

മലയാളത്തിൽ

Graph-ൽ ഒരു മുർച്ചയുള്ള മൂല (corner) ഉണ്ടെങ്കിൽ അവിടെ ഒരു tangent മാത്രം വരയ്ക്കാൻ പറ്റില്ല. ഇടത്തുനിന്ന് വരുമ്പോൾ ഒരു slope, വലത്തുനിന്ന് വരുമ്പോൾ വേറൊരു slope. അതുകൊണ്ട് corner point-ൽ derivative ഇല്ല. $|x|$ ന്റെ V shape ആണ് ഏറ്റവും നല്ല ഉദാഹരണം.

Exam tip: The unit also says the derivative "measures the rate at which the tangent is changing". Read that as: $f'(a)$ is the slope of the tangent, i.e. the rate at which $f(x)$ changes at a . The rate at which the slope itself changes is the second derivative $f''(a)$.

Derivative at a point

Left hand derivative (L.H.D.): $L f'(a) = \lim_{h \rightarrow 0^-} (f(a+h) - f(a))/h$.

$L f'(a) = \lim_{h \rightarrow 0^-} (f(a+h) - f(a))/h$ ആണ് left hand derivative.

Right hand derivative (R.H.D.): $R f'(a) = \lim_{h \rightarrow 0^+} (f(a+h) - f(a))/h$.

$R f'(a) = \lim_{h \rightarrow 0^+} (f(a+h) - f(a))/h$ ആണ് right hand derivative.

Existence of $f'(a)$: $f'(a)$ exists only when L.H.D. and R.H.D. both exist and are equal.

L.H.D.-ഉം R.H.D.-ഉം exist ചെയ്യുകയും തുല്യമാവുകയും ചെയ്യുമ്പോൾ മാത്രമേ $f'(a)$ exist ചെയ്യൂ.

To find $f'(a)$ from the definition: write $f(a+h)$ and $f(a)$, subtract, simplify, cancel h , then put $h = 0$. If the function is defined differently on the two sides of a (a modulus or a piecewise function), find L.H.D. ($h \rightarrow 0^-$) and R.H.D. ($h \rightarrow 0^+$) separately and compare.

മലയാളത്തിൽ

Recipe ലളിതമാണ്: $f(a+h) - f(a)$ കണ്ടുപിടിക്കുക, h കൊണ്ട് ഹരിക്കുക, h cancel ചെയ്യുക, പിന്നെ $h = 0$ ഇടുക. Function രണ്ടു വശത്തും വേറെ വേറെ formula ആണെങ്കിൽ, ഇടതുവശത്തെ derivative-ഉം വലതുവശത്തെ derivative-ഉം തനിയെ കണ്ടുപിടിച്ച് തുല്യമാണോ എന്ന് നോക്കണം; തുല്യമല്ലെങ്കിൽ derivative ഇല്ല.

$$f'(a) = \lim_{h \rightarrow 0} (f(a+h) - f(a))/h$$

For $|h|$: when $h < 0$, $|h| = -h$; when $h > 0$, $|h| = h$

EXAMPLE 1 (FROM THE UNIT)

Find (i) $f'(5)$ for $f(x) = a$ (constant), (ii) $f'(2)$ for $f(x) = ax + b$, (iii) $f'(3)$ for $f(x) = ax^2 + bx + c$, (iv) $f'(1)$ for $f(x) = 1/x$.

$$(i) f'(5) = \lim_{h \rightarrow 0} (a - a)/h = \lim_{h \rightarrow 0} 0 = 0.$$

$$(ii) f'(2) = \lim_{h \rightarrow 0} [a(2+h) + b - (2a + b)]/h = \lim_{h \rightarrow 0} ah/h = a.$$

$$(iii) f(3+h) - f(3) = a(9 + 6h + h^2) + b(3+h) + c - (9a + 3b + c) = ah^2 + 6ah + bh.$$

$$f'(3) = \lim_{h \rightarrow 0} (ah + 6a + b) = 6a + b.$$

$$(iv) f(1+h) - f(1) = 1/(1+h) - 1 = (1 - (1+h))/(1+h) = -h/(1+h).$$

$$f'(1) = \lim_{h \rightarrow 0} -1/(1+h) = -1.$$

EXAMPLE 2 (FROM THE UNIT)

Does the derivative exist? (i) $f(x) = |x|$ at $x = 0$; (ii) $f(x) = 5 + 2x$ for $x \geq 1$, $9 - 2x$ for $x < 1$, at $x = 1$.

(i) $(f(0 + h) - f(0))/h = |h|/h$.

L.H.D.: $h < 0$, so $|h|/h = -h/h = -1$. L.H.D. = -1 .

R.H.D.: $h > 0$, so $|h|/h = h/h = 1$. R.H.D. = 1 .

L.H.D. $\neq$ R.H.D., so $f'(0)$ does not exist.

(ii) $f(1) = 5 + 2 = 7$.

L.H.D. ($h < 0$, $1 + h < 1$): $[9 - 2(1 + h) - 7]/h = -2h/h = -2$.

R.H.D. ($h > 0$): $[5 + 2(1 + h) - 7]/h = 2h/h = 2$.

$-2 \neq 2$, so $f'(1)$ does not exist. $x = 1$ is a corner of this V-shaped graph.

EXAMPLE (CHECK YOUR PROGRESS 1, Q1)

(i) $f(x) = x^3 + x + 1$ at $x = -1$. (ii) $f(x) = 2 - 3x^2$ at $x = 1/2$.

(i) $f(-1) = -1$. $f(-1 + h) = (-1 + 3h - 3h^2 + h^3) + (-1 + h) + 1 = -1 + 4h - 3h^2 + h^3$.

$f(-1 + h) - f(-1) = h^3 - 3h^2 + 4h$. Divide by h : $h^2 - 3h + 4 \rightarrow 4$. So $f'(-1) = 4$.

(ii) $f(1/2) = 2 - 3/4$. $f(1/2 + h) = 2 - 3(1/4 + h + h^2) = 2 - 3/4 - 3h - 3h^2$.

Difference = $-3h - 3h^2$. Divide by h : $-3 - 3h \rightarrow -3$. So $f'(1/2) = -3$.

EXAMPLE (CHECK YOUR PROGRESS 1, Q2)

Find a if $f'(-2) = 3$, where $f(x) = 2x^2 - 3ax + 5$.

$f(-2 + h) - f(-2) = 2(4 - 4h + h^2) - 3a(-2 + h) + 5 - (8 + 6a + 5) = 2h^2 - 8h - 3ah$.

$f'(-2) = \lim_{h \rightarrow 0} (2h - 8 - 3a) = -8 - 3a$.

$-8 - 3a = 3 \Rightarrow a = -11/3$.

Exam tip: With $|x|$ or a piecewise function, always write L.H.D. and R.H.D. separately; writing just " $\lim |h|/h$ " and stopping earns no marks.

Derivative by first principle

Definition: The first principle finds the derivative from its definition: $dy/dx = \lim_{\delta x \rightarrow 0} (f(x + \delta x) - f(x))/\delta x$; it is also called the delta method or ab-initio method.

Definition ഉപയോഗിച്ച് derivative കണ്ടുപിടിക്കുന്ന രീതിയാണ് first principle: $dy/dx =$

$\lim_{\delta x \rightarrow 0} (f(x + \delta x) - f(x))/\delta x$; ഇതിനെ delta method അല്ലെങ്കിൽ ab-initio method എന്നും പറയുന്നു.

Finding $f(a)$ separately for many points is slow, so we find the derivative at a general point x once and then substitute. This is the **first principle** (also called the definition, delta method or ab-initio method). Give x a small increment δx ; y changes by δy . Then dy/dx is the limit of $\delta y/\delta x$ as $\delta x \rightarrow 0$.

മലയാളത്തിൽ

ഓരോ point-നും വേറെ വേറെ limit എടുക്കുന്നതിന് പകരം, ഒരു general x -ൽ ഒറ്റത്തവണ derivative കണ്ടുപിടിച്ചാൽ പിന്നെ ഏത് value-ഉം substitute ചെയ്യാം. ഒരു recipe ഒരിക്കൽ പഠിച്ചാൽ എത്ര പേർക്കും ചായ ഉണ്ടാക്കാം എന്നപോലെ. അഞ്ച് steps ഓർത്താൽ മതി: increment, subtract, divide, limit, answer.

Step I: $y + \delta y = f(x + \delta x)$

Step II: $\delta y = f(x + \delta x) - f(x)$

Step III: $\delta y/\delta x = (f(x + \delta x) - f(x))/\delta x$ (simplify first)

Step IV: take $\lim_{\delta x \rightarrow 0}$ on both sides

Step V: $dy/dx = \lim_{\delta x \rightarrow 0} (f(x + \delta x) - f(x))/\delta x$

Useful limits: $\lim_{x \rightarrow a} (x^n - a^n)/(x - a) = na^{n-1}$; $\lim_{x \rightarrow 0} (e^x - 1)/x = 1$; $\lim_{x \rightarrow 0} (1 + x)^{1/x} = e$

RESULT 1: $F(X) = K$ (CONSTANT)

$y = k$; $y + \delta y = k \Rightarrow \delta y = 0 \Rightarrow \delta y/\delta x = 0 \Rightarrow dy/dx = 0$. So $d/dx(k) = 0$.

RESULT 2: $F(X) = X^2$ AND X^3

x^2 : $y + \delta y = (x + \delta x)^2 = x^2 + 2x \cdot \delta x + (\delta x)^2$.

$\delta y = 2x \cdot \delta x + (\delta x)^2 \Rightarrow \delta y/\delta x = 2x + \delta x \Rightarrow dy/dx = \lim_{\delta x \rightarrow 0} (2x + \delta x) = 2x$.

x^3 : $\delta y = (x + \delta x)^3 - x^3 = 3x^2 \cdot \delta x + 3x(\delta x)^2 + (\delta x)^3$.

$\delta y/\delta x = 3x^2 + 3x \cdot \delta x + (\delta x)^2 \rightarrow 3x^2$. So $d/dx(x^3) = 3x^2$.

Second method: $\delta y/\delta x = ((x + \delta x)^n - x^n)/((x + \delta x) - x)$. As $\delta x \rightarrow 0$, $x + \delta x \rightarrow x$, so by the standard limit this tends to nx^{n-1} . Hence $d/dx(x^n) = nx^{n-1}$.

RESULT 3: $F(X) = (AX + B)^2$

$\delta y = (ax + a\delta x + b)^2 - (ax + b)^2$. Multiply and divide by a :

$\delta y/\delta x = a \cdot [(ax + a\delta x + b)^2 - (ax + b)^2]/[(ax + a\delta x + b) - (ax + b)]$.

As $\delta x \rightarrow 0$, $ax + a\delta x + b \rightarrow ax + b$, so the bracket $\rightarrow 2(ax + b)$.

$d/dx(ax + b)^2 = 2a(ax + b)$. In general $d/dx(ax + b)^n = na(ax + b)^{n-1}$.

RESULT 4: $F(X) = E^{AX}$

$$\delta y = e^{a(x + \delta x)} - e^{ax} = e^{ax}(e^{a\delta x} - 1).$$

$$\delta y / \delta x = a e^{ax} \cdot (e^{a\delta x} - 1) / (a\delta x). \text{ As } \delta x \rightarrow 0, a\delta x \rightarrow 0 \text{ and } (e^{a\delta x} - 1) / (a\delta x) \rightarrow 1.$$

$$d/dx(e^{ax}) = a e^{ax}.$$

RESULT 5: $F(X) = \log_a X$ ($A > 0, A \neq 1, X > 0$)

$$\delta y = \log_a(x + \delta x) - \log_a x = \log_a(1 + \delta x/x).$$

$$\delta y / \delta x = (1/x) \cdot (x/\delta x) \log_a(1 + \delta x/x) = (1/x) \log_a(1 + \delta x/x)^{x/\delta x}.$$

As $\delta x \rightarrow 0$, $\delta x/x \rightarrow 0$ and $(1 + \delta x/x)^{x/\delta x} \rightarrow e$ (log is continuous, so the limit goes inside).

$$d/dx(\log_a x) = (1/x) \log_a e. \text{ With } a = e: d/dx(\log x) = 1/x.$$

EXAMPLE (EXTRA): $F(X) = \sqrt{X}$

$$\delta y = \sqrt{x + \delta x} - \sqrt{x}. \text{ Multiply top and bottom by } \sqrt{x + \delta x} + \sqrt{x}:$$

$$\delta y = \delta x / (\sqrt{x + \delta x} + \sqrt{x}) \Rightarrow \delta y / \delta x = 1 / (\sqrt{x + \delta x} + \sqrt{x}).$$

$$dy/dx = 1 / (\sqrt{x} + \sqrt{x}) = 1 / (2\sqrt{x}).$$

EXAMPLE (EXTRA): $F(X) = 1/X$

$$\delta y = 1/(x + \delta x) - 1/x = (x - x - \delta x)/(x(x + \delta x)) = -\delta x/(x(x + \delta x)).$$

$$\delta y / \delta x = -1/(x(x + \delta x)) \rightarrow -1/x^2. \text{ So } d/dx(1/x) = -1/x^2.$$

Exam tip: "Using first principle" means the marks are for the five steps. Jumping straight to the formula nx^{n-1} scores nothing, even if the answer is right.

Table of standard derivatives and basic rules

Power rule: $d/dx(x^n) = n x^{n-1}$; the derivative of a constant is 0.

$d/dx(x^n) = n x^{n-1}$; ഒരു constant-ന്റെ derivative 0 ആണ്.

Constant multiple rule: $d/dx(cu) = c du/dx$.

$d/dx(cu) = c du/dx$; constant പുറത്തേക്ക് വരുന്നു.

Sum/difference rule: $d/dx(u \pm v) = du/dx \pm dv/dx$.

$d/dx(u \pm v) = du/dx \pm dv/dx$; sum-ന്റെ അല്ലെങ്കിൽ difference-ന്റെ derivative, derivatives-ന്റെ sum അല്ലെങ്കിൽ difference ആണ്.

The unit's aim is to use these results, not to derive every one. Learn the table below; u and v are functions of x and c, k, a, b, n are constants. A constant multiple comes outside, and the derivative of a sum or difference is the sum or difference of the derivatives.

മലയാളത്തിൽ

ഇത് ഗുണനപ്പട്ടിക പോലെ മനഃപാഠമാക്കേണ്ട table ആണ്. Exam-ൽ ഓരോ problem-ഉം ഈ formulas ചേർത്ത് വെച്ചാണ് solve ചെയ്യുന്നത്. Constant-ന്റെ derivative പൂജ്യം; constant multiple പുറത്തേക്ക് വരും; sum ആണെങ്കിൽ ഓരോ term-ഉം വേറെ differentiate ചെയ്ത് കൂട്ടിയാൽ മതി.

No.	Function	Derivative
1	k (constant)	0
2	x^n	nx^{n-1}
3	$(ax + b)^n$	$na(ax + b)^{n-1}$
4	$a^x; a^{bx}$	$a^x \log a; b a^{bx} \log a$
5	$e^x; e^{bx}$	$e^x; b e^{bx}$
6	$\log_a x$	$(1/x) \log_a e$
7	$\log x (= \log_e x)$	$1/x$
8	cu	$c \, du/dx$
9	$u \pm v$	$du/dx \pm dv/dx$
10	uv (product rule)	$u \, dv/dx + v \, du/dx$
11	u/v (quotient rule)	$(v \, du/dx - u \, dv/dx)/v^2$
12	$[f(x)]^n$, n any real	$n[f(x)]^{n-1} f'(x)$
13	$1/f(x)$	$-f'(x)/[f(x)]^2$
14	$e^{f(x)}$	$e^{f(x)} f'(x)$
15	$\log f(x)$	$f'(x)/f(x)$
16	$y = f(u)$, $u = g(w)$, $w = h(x)$ (chain rule)	$dy/dx = (dy/du)(du/dw)(dw/dx)$
17	$x = f(t)$, $y = g(t)$ (parametric)	$dy/dx = (dy/dt)/(dx/dt)$
18	$a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n$	$na_0 x^{n-1} + (n-1)a_1 x^{n-2} + \dots + a_{n-1}$
19	$\sqrt{x}; 1/x; 1/x^n$	$1/(2\sqrt{x}); -1/x^2; -n/x^{n+1}$

EXAMPLE 3 (FROM THE UNIT): CONSTANTS AND POWERS

(i) $y = 5 \Rightarrow dy/dx = 0$. (ii)–(iv) Any constant such as $\alpha + \beta$, $1/11$ or $-\pi/17$ also gives 0.

(v) $y = x^{11} \Rightarrow 11x^{10}$. (vi) $y = x^{5/2} \Rightarrow (5/2)x^{3/2}$.

(vii) $y = 1/\sqrt[3]{x} = x^{-1/3} \Rightarrow -(1/3)x^{-4/3}$. (viii) $y = 1/x^7 = x^{-7} \Rightarrow -7x^{-8} = -7/x^8$.

(ix) $y = 1/x = x^{-1} \Rightarrow -1/x^2$. (x) $y = \sqrt{x} = x^{1/2} \Rightarrow (1/2)x^{-1/2} = 1/(2\sqrt{x})$.

EXAMPLE 3 (FROM THE UNIT): $(AX + B)^N$

$$(xi) y = (2x + 5)^3 \Rightarrow 3(2x + 5)^2 \cdot 2 = 6(2x + 5)^2.$$

$$(xii) y = (4 - 3x)^8 \Rightarrow 8(4 - 3x)^7 \cdot (-3) = -24(4 - 3x)^7.$$

$$(xiii) y = (5 - 3x/2)^{4/9} \Rightarrow (4/9)(5 - 3x/2)^{-5/9} \cdot (-3/2) = -(2/3)(5 - 3x/2)^{-5/9}.$$

$$(xiv) y = \sqrt{(2 + 3x)} = (2 + 3x)^{1/2} \Rightarrow (1/2)(2 + 3x)^{-1/2} \cdot 3 = 3/(2\sqrt{(2 + 3x)}).$$

EXAMPLE 3 (FROM THE UNIT): POLYNOMIALS AND SUMS

$$(xv) y = x^5 + x^2 + 1 \Rightarrow 5x^4 + 2x.$$

$$(xvi) y = (x^3 + x^2 + 1)/x = x^2 + x + 1/x \Rightarrow 2x + 1 - 1/x^2.$$

$$(xx) y = x^{-3}(1 + x^2 + x^5 + x^8) = x^{-3} + x^{-1} + x^2 + x^5 \Rightarrow -3x^{-4} - x^{-2} + 2x + 5x^4.$$

EXAMPLE (CHECK YOUR PROGRESS 1, Q3 AND Q4)

$$(i) y = e \text{ (a constant)} \Rightarrow 0. (ii) y = 1/x^7 \Rightarrow -7/x^8. (iii) y = \sqrt{x} \Rightarrow 1/(2\sqrt{x}). (iv) y = (4 - 3x)^8 \Rightarrow -24(4 - 3x)^7.$$

$$Q4(i) y = (\sqrt{x} + 1/\sqrt{x})(x + 1/x) = x^{3/2} + x^{1/2} + x^{-1/2} + x^{-3/2}.$$

$$dy/dx = (3/2)x^{1/2} + (1/2)x^{-1/2} - (1/2)x^{-3/2} - (3/2)x^{-5/2} = (3/2)\sqrt{x} + 1/(2\sqrt{x}) - 1/(2x^{3/2}) - 3/(2x^{5/2}).$$

$$Q4(ii) y = (x^3 + 1/x^3)(x + 1/x) = x^4 + x^2 + x^{-2} + x^{-4} \Rightarrow 4x^3 + 2x - 2/x^3 - 4/x^5.$$

$$Q4(iv) y = (x^2 + x)/a, a \text{ a constant. No quotient rule needed: } dy/dx = (1/a)(2x + 1).$$

Exam tip: Before differentiating, rewrite roots and fractions as powers ($1/\sqrt[3]{x} = x^{-1/3}$) and multiply out simple products. It turns a product or quotient problem into the power rule.

Product rule

Definition: Product rule: $d/dx(uv) = u dv/dx + v du/dx$, i.e. first $\times$ derivative of second + second $\times$ derivative of first.

Product rule: $d/dx(uv) = u dv/dx + v du/dx$, അതായത് first $\times$ second-ന്റെ derivative + second $\times$ first-ന്റെ derivative.

For a product of two functions: first $\times$ derivative of second + second $\times$ derivative of first. For three functions each term differentiates one factor and keeps the other two. After differentiating, take out common factors to simplify.

മലയാളത്തിൽ

രണ്ട് functions ഗുണിച്ചിരിക്കുമ്പോൾ, ഒരു സമയം ഒന്നിനെ മാത്രം differentiate ചെയ്യുക, മറ്റേതിനെ അതുപോലെ വെക്കുക, പിന്നെ തിരിച്ചും; രണ്ടും കൂട്ടുക. മൂന്നെണ്ണമാണെങ്കിൽ ഓരോരുത്തർക്കും ഒരു turn കിട്ടും, ഒരു കുടുംബത്തിൽ ഓരോരുത്തരും ഊഴമിട്ട് പാത്രം കഴുകുന്നതുപോലെ.

$$d/dx(uv) = u \, dv/dx + v \, du/dx$$

$$d/dx(uvw) = uv \, dw/dx + uw \, dv/dx + vw \, du/dx$$

EXAMPLE 3(XIX) (FROM THE UNIT)

$$y = (x^2 + 1)(x - 1).$$

$$\text{Method 1: } y = x^3 - x^2 + x - 1 \Rightarrow dy/dx = 3x^2 - 2x + 1.$$

$$\text{Method 2 (product rule): } (x^2 + 1)(1) + (x - 1)(2x) = x^2 + 1 + 2x^2 - 2x = 3x^2 - 2x + 1. \text{ Same answer.}$$

EXAMPLE 3(XXI) (FROM THE UNIT)

$$y = (x^2 + 1)(x^3 + x^2 + 1).$$

$$\begin{aligned} dy/dx &= (x^2 + 1)(3x^2 + 2x) + (x^3 + x^2 + 1)(2x) \\ &= 3x^4 + 2x^3 + 3x^2 + 2x + 2x^4 + 2x^3 + 2x \\ &= 5x^4 + 4x^3 + 3x^2 + 4x. \end{aligned}$$

EXAMPLE 3(XXII) (FROM THE UNIT)

$$y = (4x + 1)^3(7x + 1)^4.$$

$$\begin{aligned} dy/dx &= (4x + 1)^3 \cdot 4(7x + 1)^3 \cdot 7 + (7x + 1)^4 \cdot 3(4x + 1)^2 \cdot 4 \\ &= 4(4x + 1)^2(7x + 1)^3[7(4x + 1) + 3(7x + 1)] \\ &= 4(4x + 1)^2(7x + 1)^3(28x + 7 + 21x + 3) = 4(4x + 1)^2(7x + 1)^3(49x + 10). \end{aligned}$$

EXAMPLE 3(XXIII) (FROM THE UNIT, THREE FACTORS)

$$y = (x + 2)^2(x + 3)^4(x + 1)^5.$$

$$dy/dx = (x + 2)^2(x + 3)^4 \cdot 5(x + 1)^4 + (x + 2)^2(x + 1)^5 \cdot 4(x + 3)^3 + (x + 3)^4(x + 1)^5 \cdot 2(x + 2).$$

$$\begin{aligned} \text{Common factor } (x + 2)(x + 3)^3(x + 1)^4: \\ &= (x + 2)(x + 3)^3(x + 1)^4[5(x + 2)(x + 3) + 4(x + 2)(x + 1) + 2(x + 3)(x + 1)] \\ &= (x + 2)(x + 3)^3(x + 1)^4[5(x^2 + 5x + 6) + 4(x^2 + 3x + 2) + 2(x^2 + 4x + 3)] \\ &= (x + 2)(x + 3)^3(x + 1)^4(11x^2 + 45x + 44). \end{aligned}$$

Exam tip: In Example 3(xxiii) the unit's last line writes $(x + 3)^2$ in front. The correct factor is $(x + 3)^3$: the common factor taken out is $(x + 3)^3$, and nothing changes it in the final step.

Quotient rule

Definition: Quotient rule: $d/dx(u/v) = (v du/dx - u dv/dx)/v^2, v \neq 0$.

Quotient rule: $d/dx(u/v) = (v du/dx - u dv/dx)/v^2, v \neq 0$.

For u/v : (denominator $\times$ derivative of numerator $-$ numerator $\times$ derivative of denominator) $\div$ (denominator) 2 .
The order in the numerator matters because of the minus sign. If the numerator is a constant, write the function as $c \cdot [v]^{-1}$ and use the power rule instead.

മലയാളത്തിൽ

"താഴെ $\times$ മുകളിലത്തെ derivative $-$ മുകളിൽ $\times$ താഴത്തെ derivative, എല്ലാം താഴത്തെതിന്റെ square കൊണ്ട് ഹരിക്കുക" എന്ന് ഓർത്തുവെക്കുക. Minus ഉള്ളതുകൊണ്ട് order തെറ്റിയാൽ sign മാറും. മുകളിൽ വെറും constant ആണെങ്കിൽ quotient rule വേണ്ട; power -1 ആക്കി എഴുതിയാൽ മതി.

$$d/dx(u/v) = (v du/dx - u dv/dx)/v^2$$

$$d/dx(c/v) = -c (dv/dx)/v^2$$

EXAMPLE 4 (FROM THE UNIT)

(i) $y = (x + 1)/(x - 1)$.

$$dy/dx = [(x - 1)(1) - (x + 1)(1)]/(x - 1)^2 = (x - 1 - x - 1)/(x - 1)^2 = -2/(x - 1)^2.$$

(ii) $y = (8x + 3)/(6 - 5x)$.

$$dy/dx = [(6 - 5x)(8) - (8x + 3)(-5)]/(6 - 5x)^2 = (48 - 40x + 40x + 15)/(6 - 5x)^2 = 63/(6 - 5x)^2.$$

(iii) $y = a^2/(x^2 + a^2) = a^2(x^2 + a^2)^{-1}$ (numerator has no x , so no quotient rule).

$$dy/dx = a^2(-1)(x^2 + a^2)^{-2} \cdot 2x = -2a^2x/(x^2 + a^2)^2.$$

EXAMPLE 4(IV) (FROM THE UNIT, QUOTIENT INSIDE A ROOT)

$$y = \sqrt{(x^2 + 1)/(x + 1)}.$$

Power rule with $n = 1/2$: $dy/dx = (1/2) \cdot [(x^2 + 1)/(x + 1)]^{-1/2} \cdot d/dx[(x^2 + 1)/(x + 1)]$.

Quotient rule: $d/dx[(x^2 + 1)/(x + 1)] = [(x + 1)(2x) - (x^2 + 1)(1)]/(x + 1)^2 = (x^2 + 2x - 1)/(x + 1)^2$.

$$\text{So } dy/dx = (1/2) \cdot \sqrt{(x + 1)/(x^2 + 1)} \cdot (x^2 + 2x - 1)/(x + 1)^2 = (x^2 + 2x - 1)/(2(x + 1)^{3/2} \sqrt{(x^2 + 1)}).$$

EXAMPLE (CHECK YOUR PROGRESS 1, Q4(III))

$$y = x^2/(x^3 + 1).$$

$$dy/dx = [(x^3 + 1)(2x) - x^2(3x^2)]/(x^3 + 1)^2 = (2x^4 + 2x - 3x^4)/(x^3 + 1)^2 = (2x - x^4)/(x^3 + 1)^2 = x(2 - x^3)/(x^3 + 1)^2.$$

Exam tip: In Example 4(iv) the unit ends with $(x - 1)^2$ in the numerator. That is wrong: $(x - 1)^2 = x^2 - 2x + 1$, but the numerator is $x^2 + 2x - 1$, which does not factor nicely. Check with $x = 0$: dy/dx should be $-1/2$; the unit's form gives $+1/2$.

Chain rule

Definition: Chain rule: if $y = f(u)$ and $u = g(x)$, then $dy/dx = (dy/du)(du/dx)$; e.g. $d/dx[f(x)]^n = n[f(x)]^{n-1} f'(x)$.

Chain rule: $y = f(u)$, $u = g(x)$ ആണെങ്കിൽ $dy/dx = (dy/du)(du/dx)$; ഉദാ. $d/dx[f(x)]^n = n[f(x)]^{n-1} f'(x)$.

When y depends on x through a chain $y = f(u)$, $u = g(w)$, $w = h(x)$, multiply the derivatives link by link. The power rule $[f(x)]^n \rightarrow n[f(x)]^{n-1} f'(x)$ is the chain rule in its most common form: differentiate the outside, keep the inside, then multiply by the derivative of the inside.

മലയാളത്തിൽ

ഒരു ചങ്ങലയിലെ കണ്ണികൾ പോലെ: y , u -നെ ആശ്രയിക്കുന്നു, u , w -നെ, w , x -നെ.

ഓരോ കണ്ണിയുടെയും derivative എടുത്ത് ഗുണിക്കുക. സവാള

തൊലിയുരിക്കുന്നതുപോലെ പുറത്തെ layer ആദ്യം, പിന്നെ അകത്തേത്. അവസാനം u , v എല്ലാം x -ൽ തിരിച്ചെഴുതാൻ മറക്കരുത്.

$$dy/dx = (dy/du)(du/dw)(dw/dx)$$

$$d/dx[f(x)]^n = n[f(x)]^{n-1} f'(x)$$

$$d/dx[1/f(x)] = -f'(x)/[f(x)]^2$$

EXAMPLE 5(I) (FROM THE UNIT)

$$y = 3u, u = v^2, v = 4x^2 + 5.$$

$$dy/du = 3, du/dv = 2v, dv/dx = 8x.$$

$$dy/dx = 3 \cdot 2v \cdot 8x = 48xv = 48x(4x^2 + 5).$$

EXAMPLE 5(II) (FROM THE UNIT)

$$y = u^2, u = 3v, v = x/(x + 1).$$

$$dy/du = 2u, du/dv = 3, dv/dx = [(x + 1) \cdot 1 - x \cdot 1]/(x + 1)^2 = 1/(x + 1)^2.$$

$$dy/dx = 2u \cdot 3 \cdot 1/(x + 1)^2 = 6u/(x + 1)^2 = 18v/(x + 1)^2 = 18x/(x + 1)^3.$$

EXAMPLE 3(XVII), 3(XVIII) (FROM THE UNIT)

$$(xvii) y = (x^2 + 1/x^2)^2.$$

$$dy/dx = 2(x^2 + 1/x^2)(2x - 2/x^3) = 4(x^3 - 1/x + 1/x - 1/x^5) = 4(x^3 - 1/x^5).$$

$$(xviii) y = (x + 1/x)^{100}.$$

$$dy/dx = 100(x + 1/x)^{99}(1 - 1/x^2).$$

EXAMPLE (CHECK YOUR PROGRESS 1, Q3(V))

$$y = (x^3 + 1/x^3)^2.$$

$$dy/dx = 2(x^3 + 1/x^3)(3x^2 - 3/x^4) = 6(x^5 - 1/x + 1/x - 1/x^7) = 6(x^5 - 1/x^7).$$

$$\text{Check by expanding: } y = x^6 + 2 + x^{-6} \Rightarrow dy/dx = 6x^5 - 6x^{-7}. \checkmark$$

EXAMPLE (EXTRA)

$$(a) y = (x^2 + 1)^5 \Rightarrow 5(x^2 + 1)^4 \cdot 2x = 10x(x^2 + 1)^4.$$

$$(b) y = \sqrt{1 + x^3} \Rightarrow (1/2)(1 + x^3)^{-1/2} \cdot 3x^2 = 3x^2/(2\sqrt{1 + x^3}).$$

$$(c) y = 1/(x^2 - 3x) \Rightarrow -(2x - 3)/(x^2 - 3x)^2.$$

Exam tip: The unit's answer key for Check Your Progress 1, Q3(v) ends with $4(x^5 - 1/x^7)$. The correct answer is $6(x^5 - 1/x^7)$: the factor $2 \times 3 = 6$ is taken out, as the expansion check above confirms.

Exponential and logarithmic functions

Definition: $d/dx(e^x) = e^x$, $d/dx(a^x) = a^x \log a$ and $d/dx(\log x) = 1/x$, where $\log$ is the natural logarithm (base e).

$d/dx(e^x) = e^x$, $d/dx(a^x) = a^x \log a$, $d/dx(\log x) = 1/x$; ഇവിടെ $\log$ എന്നത് natural logarithm (base e) ആണ്.

e^x is its own derivative, a^x picks up a factor $\log a$, and $\log x$ differentiates to $1/x$. With a function inside, multiply by its derivative (chain rule). Use the laws of logarithms and $a^{\log_a f(x)} = f(x)$ to simplify before differentiating; a change of base, $\log_b m = \log m / \log b$, turns any base into natural log.

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e^x -നെ differentiate ചെയ്താൽ അത് മാറില്ല, അതാണ് അതിന്റെ പ്രത്യേകത. a^x -ന് ഒരു $\log a$ കൂടെ വരും. $\log f(x)$ -ന്റെ derivative, "അകത്തെ derivative $\div$ അകത്തെ function". Differentiate ചെയ്യും മുമ്പ് log rules വെച്ച് ചുരുക്കിയാൽ പകുതി പണി കുറയും.

$$d/dx(a^x) = a^x \log a; d/dx(a^{bx}) = b a^{bx} \log a$$

$$d/dx(e^{f(x)}) = e^{f(x)} f'(x)$$

$$d/dx(\log f(x)) = f'(x)/f(x)$$

$$d/dx(\log_a x) = (1/x) \log_a e$$

$$\log m^n = n \log m; \log_b m = \log m / \log b; a^{\log_a f(x)} = f(x)$$

EXAMPLE 6(I)-(IV) (FROM THE UNIT)

$$(i) y = 2^x \Rightarrow 2^x \log 2.$$

$$(ii) y = 5^{ax} \Rightarrow 5^{ax} \log 5 \cdot d/dx(ax) = a \cdot 5^{ax} \log 5.$$

$$(iii) y = e^{3x} \Rightarrow 3e^{3x}.$$

$$(iv) y = e^{9x^2} \Rightarrow e^{9x^2} \cdot 18x = 18x e^{9x^2}.$$

EXAMPLE 6(V)-(IX) (FROM THE UNIT)

$$(v) y = \log(1 + x^2) \Rightarrow 2x/(1 + x^2).$$

$$(vi) y = \log_x x^2 = \log x^2 / \log x = 2 \log x / \log x = 2 \Rightarrow dy/dx = 0.$$

$$(vii) y = \log_2 x \Rightarrow (1/x) \log_2 e.$$

$$(viii) y = \log_x 2 = \log 2 / \log x = (\log 2)(\log x)^{-1}.$$

$$dy/dx = -(\log 2)(\log x)^{-2} \cdot (1/x) = -\log 2 / (x(\log x)^2).$$

$$(ix) y = 1/\log(1 + x) = [\log(1 + x)]^{-1}.$$

$$dy/dx = -[\log(1 + x)]^{-2} \cdot 1/(1 + x) = -1/((1 + x)[\log(1 + x)]^2).$$

EXAMPLE (CHECK YOUR PROGRESS 2, Q1)

$$(i) y = a^{\log_a 2^x} = 2^x \Rightarrow dy/dx = 2^x \log 2.$$

$$(ii) y = 3^{\log_3 x^2} = x^2 \Rightarrow dy/dx = 2x.$$

EXAMPLE (EXTRA)

(a) $y = e^{x^2 + 3x} \Rightarrow (2x + 3)e^{x^2 + 3x}$.

(b) $y = \log(x^3 + 5x) \Rightarrow (3x^2 + 5)/(x^3 + 5x)$.

(c) $y = x^2 e^x$ (product) $\Rightarrow x^2 e^x + 2x e^x = x e^x(x + 2)$.

(d) $y = \log x/x$ (quotient) $\Rightarrow [x \cdot (1/x) - \log x \cdot 1]/x^2 = (1 - \log x)/x^2$.

Exam tip: In IGNOU papers "log" without a base means $\log_e$. Do not confuse $d/dx(a^x) = a^x \log a$ with $d/dx(x^n) = nx^{n-1}$: the variable is in the exponent in the first, in the base in the second.

Parametric functions

Definition: Parametric functions give x and y in terms of a third variable t (the parameter), $x = f(t)$, $y = g(t)$; then $dy/dx = (dy/dt)/(dx/dt)$, $dx/dt \neq 0$.

x -ഉം y -ഉം ഒരു മൂന്നാം variable t (parameter) വഴി $x = f(t)$, $y = g(t)$ എന്ന് തരുന്നവയാണ് parametric functions; അപ്പോൾ $dy/dx = (dy/dt)/(dx/dt)$, $dx/dt \neq 0$.

Sometimes x and y are both given in terms of a third variable t (the parameter): $x = f(t)$, $y = g(t)$. Differentiate each with respect to t and divide: $dy/dx = (dy/dt)/(dx/dt)$, provided $dx/dt \neq 0$.

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ഇവിടെ x -ഉം y -ഉം നേരിട്ട് ബന്ധിച്ചിട്ടില്ല; രണ്ടും t എന്ന ഒരു മൂന്നാമന്റെ (സമയം പോലെ) function ആണ്. ഒരു ബസ് സമയം t -ൽ എവിടെ എത്തി എന്ന് x -ഉം y -ഉം പറയുന്നതുപോലെ. രണ്ടിനെയും t വെച്ച് differentiate ചെയ്ത്, dy/dt -നെ dx/dt കൊണ്ട് ഹരിച്ചാൽ dy/dx കിട്ടും.

$$x = f(t), y = g(t)$$

$$dy/dx = (dy/dt)/(dx/dt), dx/dt \neq 0$$

EXAMPLE 7(I) (FROM THE UNIT)

$$x = 1 + t, y = 2 + t^2.$$

$$dx/dt = 1, dy/dt = 2t \Rightarrow dy/dx = 2t/1 = 2t.$$

EXAMPLE 7(II) (FROM THE UNIT)

$$x = (1 + 3t^2)/(1 - t), y = (1 + t^2)/(1 - t).$$

$$dx/dt = [(1 - t)(6t) - (1 + 3t^2)(-1)]/(1 - t)^2 = (6t - 6t^2 + 1 + 3t^2)/(1 - t)^2 = (1 + 6t - 3t^2)/(1 - t)^2.$$

$$dy/dt = [(1 - t)(2t) - (1 + t^2)(-1)]/(1 - t)^2 = (2t - 2t^2 + 1 + t^2)/(1 - t)^2 = (1 + 2t - t^2)/(1 - t)^2.$$

$$dy/dx = (1 + 2t - t^2)/(1 + 6t - 3t^2) \text{ (the } (1 - t)^2 \text{ cancels).}$$

EXAMPLE (CHECK YOUR PROGRESS 2, Q2)

$$x = 2 + 4t^2, y = 9t^2 + 3t + 1.$$

$$dx/dt = 8t, dy/dt = 18t + 3 \Rightarrow dy/dx = (18t + 3)/(8t).$$

EXAMPLE (EXTRA)

$$x = at^2, y = 2at. \text{ Find } dy/dx \text{ and its value at } t = 2.$$

$$dx/dt = 2at, dy/dt = 2a \Rightarrow dy/dx = 2a/(2at) = 1/t. \text{ At } t = 2, dy/dx = 1/2.$$

Exam tip: Keep the answer in terms of t unless the question asks you to eliminate t . Put dy/dt on top, not dx/dt .

Implicit functions

Explicit function: A function in which y is expressed directly in terms of x , $y = f(x)$.

y -യെ x -ന്റെ രൂപത്തിൽ നേരിട്ട് $y = f(x)$ എന്ന് എഴുതിയ function.

Implicit function: A function given by a relation $f(x, y) = c$ in which y is not expressed directly in terms of x .

y -യെ x -ന്റെ രൂപത്തിൽ നേരിട്ട് എഴുതാതെ $f(x, y) = c$ എന്ന relation ആയി തരുന്ന function.

$y = f(x)$ is an explicit function. When the relation is $f(x, y) = c$ and y cannot easily be made the subject, it is an implicit function. Differentiate every term with respect to x , treating y as a function of x (so y^2 gives $2y dy/dx$ and xy needs the product rule), collect the dy/dx terms on the left, and divide.

മലയാളത്തിൽ

ചിലപ്പോൾ y -നെ ഒറ്റയ്ക്ക് ഒരു വശത്ത് കൊണ്ടുവരാൻ പറ്റില്ല, x -ഉം y -ഉം കെട്ടിപ്പിണഞ്ഞു കിടക്കും. അപ്പോൾ അതുപോലെ തന്നെ x വെച്ച് differentiate ചെയ്യുക; y വരുന്നിടത്തെല്ലാം ഒരു dy/dx കൂടെ ചേർക്കണം (y , x -നെ ആശ്രയിക്കുന്നതുകൊണ്ട്). പിന്നെ dy/dx ഉള്ള terms ഒരു വശത്തേക്ക് മാറ്റി ഹരിച്ചാൽ ഉത്തരം.

$$d/dx(y^n) = n y^{n-1} dy/dx$$

$$d/dx(xy) = x dy/dx + y$$

$$d/dx(e^{-y}) = -e^{-y} dy/dx$$

EXAMPLE 8 (FROM THE UNIT)

$$(i) x^2 + y^2 = c^2 \Rightarrow 2x + 2y dy/dx = 0 \Rightarrow dy/dx = -x/y.$$

$$(ii) (x - a)^2 + (y - b)^2 = r^2 \Rightarrow 2(x - a) + 2(y - b) dy/dx = 0 \Rightarrow dy/dx = -(x - a)/(y - b).$$

$$(iii) x^3 + y^3 + xy = 5 \Rightarrow 3x^2 + 3y^2 dy/dx + x dy/dx + y = 0 \\ \Rightarrow (3y^2 + x) dy/dx = -(3x^2 + y) \Rightarrow dy/dx = -(3x^2 + y)/(x + 3y^2).$$

EXAMPLE (CHECK YOUR PROGRESS 3, Q1)

$$xy^3 + xe^x + xe^{-y} = 3.$$

Term by term (product rule on each):

$$d/dx(xy^3) = 3xy^2 dy/dx + y^3$$

$$d/dx(xe^x) = xe^x + e^x$$

$$d/dx(xe^{-y}) = -xe^{-y} dy/dx + e^{-y}$$

$$\text{Sum} = 0 \Rightarrow (3xy^2 - xe^{-y}) dy/dx = -(y^3 + xe^x + e^x + e^{-y})$$

$$\Rightarrow dy/dx = (y^3 + xe^x + e^x + e^{-y})/(xe^{-y} - 3xy^2).$$

EXAMPLE (EXTRA)

$$x^2 + xy + y^2 = 7. \text{ Find } dy/dx \text{ at } (1, 2).$$

$$2x + (x dy/dx + y) + 2y dy/dx = 0 \Rightarrow (x + 2y) dy/dx = -(2x + y) \Rightarrow dy/dx = -(2x + y)/(x + 2y).$$

$$\text{At } (1, 2): -(2 + 2)/(1 + 4) = -4/5.$$

Exam tip: The most common slip is writing $d/dx(y^2) = 2y$ and forgetting dy/dx , or differentiating xy as just dy/dx . Every y -term carries a dy/dx ; every product of x and y needs the product rule.

Derivatives of higher orders

Definition: The second derivative $d^2y/dx^2 = d/dx(dy/dx)$ is the derivative of dy/dx ; differentiating y n times gives the n th order derivative d^ny/dx^n .

dy/dx -ന്റെ derivative ആണ് second derivative $d^2y/dx^2 = d/dx(dy/dx)$; y -യെ n തവണ differentiate ചെയ്താൽ n th order derivative d^ny/dx^n കിട്ടും.

Differentiating dy/dx again gives the second derivative d^2y/dx^2 (also $f''(x)$); again gives d^3y/dx^3 , and n times gives d^ny/dx^n . A polynomial of degree n has all derivatives beyond the n -th equal to zero.

മലയാളത്തിൽ

ഒരിക്കൽ differentiate ചെയ്ത ഉത്തരം വീണ്ടും differentiate ചെയ്യുന്നതേ ഉള്ളൂ. Position-ന്റെ derivative velocity, velocity-യുടെ derivative acceleration എന്നതുപോലെ. ഓരോ step-ലും ഉത്തരം simplify ചെയ്ത് വെച്ചാൽ അടുത്ത step എളുപ്പമാകും.

$$d^2y/dx^2 = d/dx(dy/dx), d^3y/dx^3 = d/dx(d^2y/dx^2), \dots$$

$$y = e^{ax} \Rightarrow d^ny/dx^n = a^n e^{ax}$$

EXAMPLE 9 (FROM THE UNIT)

(i) $y = 1/x$: $dy/dx = -1/x^2$, $d^2y/dx^2 = 2/x^3$, $d^3y/dx^3 = -6/x^4$, $d^4y/dx^4 = 24/x^5$.

(ii) $y = e^{ax}$: $dy/dx = ae^{ax}$, $d^2y/dx^2 = a^2e^{ax}$, $d^3y/dx^3 = a^3e^{ax}$.

(iii) $y = (ax + b)^7$: $dy/dx = 7a(ax + b)^6$, $d^2y/dx^2 = 42a^2(ax + b)^5$, $d^3y/dx^3 = 210a^3(ax + b)^4$.

(iv) $y = x^2 + x + 1$: $dy/dx = 2x + 1$, $d^2y/dx^2 = 2$, $d^3y/dx^3 = 0$, $d^4y/dx^4 = 0$.

EXAMPLE (EXTRA)

$y = x^3 \log x$. Find d^2y/dx^2 .

$$dy/dx = 3x^2 \log x + x^3 \cdot (1/x) = 3x^2 \log x + x^2.$$

$$d^2y/dx^2 = 6x \log x + 3x^2 \cdot (1/x) + 2x = 6x \log x + 5x.$$

Exam tip: Spot the pattern for $1/x$: the n -th derivative is $(-1)^n n!/x^{n+1}$. Check: $n = 4$ gives $24/x^5$.

Maxima and minima: the idea and the tests

Local maxima: x_1 is a point of local maxima of f if $f(x_1) \geq f(x)$ for all x sufficiently close to x_1 .

x_1 -നോട് വളരെ അടുത്തുള്ള എല്ലാ x -നും $f(x_1) \geq f(x)$ ആണെങ്കിൽ x_1 , f -ന്റെ local maxima point ആണ്.

Local minima: x_2 is a point of local minima of f if $f(x_2) \leq f(x)$ for all x sufficiently close to x_2 .

x_2 -നോട് വളരെ അടുത്തുള്ള എല്ലാ x -നും $f(x_2) \leq f(x)$ ആണെങ്കിൽ x_2 , f -ന്റെ local minima point ആണ്.

Stationary point: A point c where $f'(c) = 0$; the tangent there is parallel to the x -axis.

$f'(c) = 0$ ആകുന്ന point c ; അവിടെ tangent x -axis-ന് parallel ആണ്.

x_1 is a point of **local maxima** if $f(x_1) \geq f(x)$ for all x very close to x_1 ; x_2 is a point of **local minima** if $f(x_2) \leq f(x)$ for all x close to x_2 . At such a peak or valley the tangent is parallel to the x -axis, so $f' = 0$ there (if f' exists). The converse is false: $f(x) = -x^3$ has $f'(0) = 0$ but no max or min at 0. And a minimum can occur where f' does not exist, e.g. $|x|$ at 0 (a global minimum).

മലയാളത്തിൽ

മല കയറി ഇറങ്ങുന്ന ഒരു റോഡ് ഓർക്കുക. മലമുകളിലും (maxima) താഴ്വരയിലും (minima) ഒരു നിമിഷം റോഡ് നിർപ്പാണ്, അതായത് slope = 0. അതുകൊണ്ട് $f'(x) = 0$ ആക്കി candidate points കണ്ടുപിടിക്കുന്നു. പക്ഷേ നിർപ്പായ എല്ലാ സ്ഥലവും മുകളോ താഴോ ആകണമെന്നില്ല ($-x^3$ പോലെ); അത് തീരുമാനിക്കാൻ second derivative test ഉപയോഗിക്കുന്നു.

Step 1: solve $f'(x) = 0$ (stationary points)

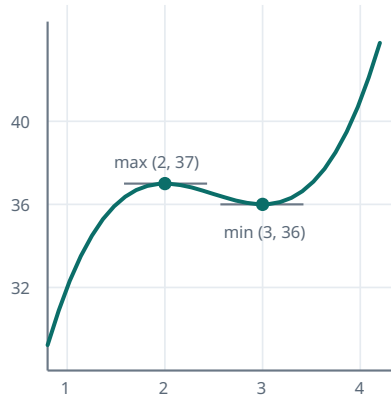
Second derivative test at c with $f'(c) = 0$:

$f''(c) > 0 \Rightarrow$ local minima at c

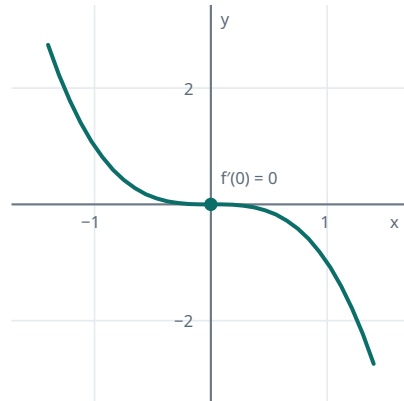
$f''(c) < 0 \Rightarrow$ local maxima at c

$f''(c) = 0 \Rightarrow$ test fails; use the first derivative test

First derivative test: f' changes + to - across $c \Rightarrow$ maxima; - to + $\Rightarrow$ minima; no sign change $\Rightarrow$ neither (point of inflection)



$f(x) = 2x^3 - 15x^2 + 36x + 9$ (y-axis starts at 28). The tangents at $x = 2$ and $x = 3$ are horizontal: $f' = 0$ at both.



$y = -x^3$. $f'(0) = 0$, yet the curve keeps falling on both sides: $x = 0$ is a point of inflection, not a max or min.

In machine learning, training a model means minimising a loss $L(w)$; gradient descent repeats $w \leftarrow w - \eta \cdot dL/dw$, stepping downhill until the derivative is (nearly) zero, exactly the $f' = 0$ condition of this topic.

Exam tip: Always write the two parts separately: the point of maxima/minima (the x value) and the maximum/minimum value (f at that x). The normal curve has its points of inflection at $x = \mu \pm \sigma$, found by setting the second derivative of the normal density to zero.

Maxima and minima: worked problems

The exam procedure: find $f'(x)$, solve $f'(x) = 0$, find $f''(x)$, check its sign at each root, then substitute each root into the original $f(x)$ to get the maximum or minimum value.

മലയാളത്തിൽ

അഞ്ച് steps: f' കണ്ടുപിടിക്കുക, $f' = 0$ solve ചെയ്യുക, f'' കണ്ടുപിടിക്കുക, ഓരോ root-ലും f'' -ന്റെ sign നോക്കുക (negative എങ്കിൽ max, positive എങ്കിൽ min), പിന്നെ ആ x -നെ original $f(x)$ -ൽ ഇട്ട് value കണക്കാക്കുക. അവസാനത്തെ arithmetic-ൽ ആണ് കൂടുതൽ പേർക്കും mark പോകുന്നത്, അത് ശ്രദ്ധിക്കണം.

$f'(x) = 0 \rightarrow$ roots c

$f''(c) < 0 \rightarrow$ maximum value $f(c)$

$f''(c) > 0 \rightarrow$ minimum value $f(c)$

EXAMPLE 10 (FROM THE UNIT)

Find the local maximum and minimum values of $f(x) = 2x^3 - 15x^2 + 36x + 9$.

$$f'(x) = 6x^2 - 30x + 36 = 6(x^2 - 5x + 6) = 6(x - 2)(x - 3). f'(x) = 0 \Rightarrow x = 2, 3.$$

$$f''(x) = 12x - 30.$$

$$x = 2: f''(2) = 24 - 30 = -6 < 0 \Rightarrow \text{maxima. } f(2) = 16 - 60 + 72 + 9 = 37.$$

$$x = 3: f''(3) = 36 - 30 = 6 > 0 \Rightarrow \text{minima. } f(3) = 54 - 135 + 108 + 9 = 36.$$

Local maximum value 37 (at $x = 2$); local minimum value 36 (at $x = 3$).

Exam tip: The unit's solution to Example 10 says $f(2) = 16 - 60 + 72 + 9 = 27$. The correct value is 37 ($16 - 60 = -44$, $-44 + 72 = 28$, $28 + 9 = 37$). The graph confirms the local max 37 sits just above the local min 36.

EXAMPLE 11 (FROM THE UNIT)

Find the local maximum and minimum values of $f(x) = x^4/4 + x^3/3 - 2x^2 - 4x + 5$.

$$f'(x) = x^3 + x^2 - 4x - 4. x = -1 \text{ is a root by inspection, so}$$

$$f'(x) = (x + 1)(x^2 - 4) = (x + 1)(x - 2)(x + 2). \text{ Roots } x = -1, 2, -2.$$

$$f''(x) = 3x^2 + 2x - 4.$$

$$f''(-1) = 3 - 2 - 4 = -3 < 0 \Rightarrow \text{maxima.}$$

$$f''(2) = 12 + 4 - 4 = 12 > 0 \Rightarrow \text{minima.}$$

$$f''(-2) = 12 - 4 - 4 = 4 > 0 \Rightarrow \text{minima.}$$

$$f(2) = 4 + 8/3 - 8 - 8 + 5 = -13/3.$$

$$f(-2) = 4 - 8/3 - 8 + 8 + 5 = 19/3.$$

$$f(-1) = 1/4 - 1/3 - 2 + 4 + 5 = 83/12.$$

Local minimum values $-13/3$ ($x = 2$) and $19/3$ ($x = -2$); local maximum value $83/12$ ($x = -1$).

EXAMPLE (CHECK YOUR PROGRESS 3, Q2)

$$f(x) = 4x^3 - 21x^2 + 18x + 9.$$

$$f'(x) = 12x^2 - 42x + 18 = 6(2x^2 - 7x + 3) = 6(x - 3)(2x - 1) \Rightarrow x = 3, 1/2.$$

$$f''(x) = 24x - 42. f''(3) = 30 > 0 \Rightarrow \text{minima}; f''(1/2) = -30 < 0 \Rightarrow \text{maxima.}$$

$$f(3) = 108 - 189 + 54 + 9 = -18 \text{ (local minimum value).}$$

$$f(1/2) = 1/2 - 21/4 + 9 + 9 = (2 - 21 + 72)/4 = 53/4 \text{ (local maximum value).}$$

EXAMPLE (EXTRA, TEST FAILS)

$f(x) = x^4$. $f'(x) = 4x^3 = 0 \Rightarrow x = 0$; $f''(0) = 12 \cdot 0 = 0$, so the second derivative test fails.

First derivative test: for $x < 0$, $f' < 0$; for $x > 0$, $f' > 0$. Sign goes $-$ to $+$ $\Rightarrow$ minima at $x = 0$, minimum value 0.

Compare $f(x) = -x^3$: $f'(x) = -3x^2 \leq 0$ on both sides, no sign change $\Rightarrow$ neither max nor min (inflection).

Exam tip: Factorise $f'(x)$ fully and take out the common number first ($6x^2 - 30x + 36 = 6(x^2 - 5x + 6)$). For a cubic f' , try $x = \pm 1, \pm 2$ to find the first root.

Quick recap

- $f'(a) = \lim_{h \rightarrow 0} (f(a+h) - f(a))/h$; geometrically it is the slope of the tangent, $\tan \psi$.
- $f'(a)$ exists only if L.H.D. = R.H.D.; at a corner (like $|x|$ at 0) it does not exist.
- First principle: increment δx , find δy , divide by δx , take the limit.
- $d/dx(k) = 0$, $d/dx(x^n) = nx^{n-1}$, $d/dx(ax+b)^n = na(ax+b)^{n-1}$.
- $d/dx(e^{ax}) = ae^{ax}$, $d/dx(a^x) = a^x \log a$, $d/dx(\log x) = 1/x$, $d/dx(\log_a x) = (1/x) \log_a e$.
- Product rule: $(uv)' = uv' + vu'$; for three factors each takes a turn.
- Quotient rule: $(u/v)' = (vu' - uv')/v^2$; if the numerator is constant, use the power -1 instead.
- Chain rule: $dy/dx = (dy/du)(du/dw)(dw/dx)$; $[f(x)]^n \rightarrow n[f(x)]^{n-1}f'(x)$.
- $d/dx e^{f(x)} = e^{f(x)}f'(x)$; $d/dx \log f(x) = f'(x)/f(x)$.
- Parametric: $dy/dx = (dy/dt)/(dx/dt)$.
- Implicit: differentiate each term, attach dy/dx to every y -term, collect and divide.
- Higher order: differentiate again; $d^n/dx^n(e^{ax}) = a^n e^{ax}$.
- At a local max/min $f' = 0$ (if f' exists), but $f' = 0$ does not guarantee one ($-x^3$).
- Second derivative test: $f'' < 0 \Rightarrow$ max, $f'' > 0 \Rightarrow$ min, $f'' = 0 \Rightarrow$ use the sign change of f' .

Check yourself

Q. 1. Using the definition, find $f'(2)$ for $f(x) = x^2 - 3x$.

$$f(2) = 4 - 6 = -2. \quad f(2+h) = 4 + 4h + h^2 - 6 - 3h = -2 + h + h^2.$$

$$f'(2) = \lim_{h \rightarrow 0} (h + h^2)/h = \lim_{h \rightarrow 0} (1 + h) = 1.$$

Q. 2. Is $f(x) = |x - 2|$ differentiable at $x = 2$?

$$(f(2 + h) - f(2))/h = |h|/h.$$

L.H.D. ($h < 0$) = $-h/h = -1$; R.H.D. ($h > 0$) = $h/h = 1$. Since $-1 \neq 1$, f is not differentiable at $x = 2$ (a corner).

Q. 3. Use the product rule to differentiate $y = (x^2 + 2)(x^3 - 1)$.

$$dy/dx = (x^2 + 2)(3x^2) + (x^3 - 1)(2x) = 3x^4 + 6x^2 + 2x^4 - 2x = 5x^4 + 6x^2 - 2x.$$

Q. 4. Differentiate $y = (2x + 3)/(x - 4)$.

$$dy/dx = [(x - 4)(2) - (2x + 3)(1)]/(x - 4)^2 = (2x - 8 - 2x - 3)/(x - 4)^2 = -11/(x - 4)^2.$$

Q. 5. Find dy/dx if $y = u^3$, $u = 2v + 1$, $v = x^2$.

$$dy/du = 3u^2, du/dv = 2, dv/dx = 2x.$$

$$dy/dx = 3u^2 \cdot 2 \cdot 2x = 12xu^2 = 12x(2x^2 + 1)^2.$$

Q. 6. Differentiate (a) $e^{5x^2 - x}$, (b) $\log(x^2 + 5)$, (c) 3^{2x} .

$$(a) (10x - 1)e^{5x^2 - x}. (b) 2x/(x^2 + 5). (c) 2 \cdot 3^{2x} \log 3.$$

Q. 7. If $x = t^2$, $y = t^3$, find dy/dx and its value at $t = 2$.

$$dx/dt = 2t, dy/dt = 3t^2 \Rightarrow dy/dx = 3t^2/(2t) = 3t/2. \text{ At } t = 2, dy/dx = 3.$$

Q. 8. Find dy/dx if $x^2 + y^2 + 3xy = 10$.

$$2x + 2y dy/dx + 3(x dy/dx + y) = 0 \Rightarrow (2y + 3x) dy/dx = -(2x + 3y).$$

$$dy/dx = -(2x + 3y)/(3x + 2y).$$

Q. 9. For $y = x^4 - 3x^2 + 2$ find d^2y/dx^2 and d^3y/dx^3 .

$$dy/dx = 4x^3 - 6x; d^2y/dx^2 = 12x^2 - 6; d^3y/dx^3 = 24x.$$

Q. 10. Find the local maximum and minimum values of $f(x) = x^3 - 3x + 2$.

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \Rightarrow x = 1, -1. f''(x) = 6x.$$

$$x = 1: f''(1) = 6 > 0 \Rightarrow \text{minima}, f(1) = 1 - 3 + 2 = 0.$$

$$x = -1: f''(-1) = -6 < 0 \Rightarrow \text{maxima}, f(-1) = -1 + 3 + 2 = 4.$$

Local maximum value 4, local minimum value 0.

UNIT 13

Indefinite Integration

Differentiation-ന്റെ നേരെ തിരിച്ചുള്ള പ്രക്രിയയാണ് integration. ഈ unit-ൽ standard formulas ഉപയോഗിച്ച് നേരിട്ട് integrate ചെയ്യുന്നതും, പിന്നെ substitution, partial fractions, integration by parts എന്നീ മൂന്ന് പ്രധാന methods-ഉം പഠിക്കുന്നു.

Meaning and terminology

Table of standard integrals and rules

Powers and polynomials

Linear forms: $(ax + b)^n$ and $1/(ax + b)$

Exponential forms and $a^{\log_a f(x)}$

Integration by substitution

More substitution patterns

Partial fractions: Type 1 (distinct linear factors)

Partial fractions: Type 2 (repeated linear factors)

Improper fractions: divide first

Integration by parts and the ILATE order

Repeated integration by parts

Meaning and terminology

Antiderivative: $F(x)$ is an antiderivative of $f(x)$ if $F'(x) = f(x)$; then the indefinite integral is $\int f(x) dx = F(x) + c$.

$F'(x) = f(x)$ ആണെങ്കിൽ $F(x)$, $f(x)$ -ന്റെ antiderivative ആണ്; അപ്പോൾ indefinite integral $\int f(x) dx = F(x) + c$.

Constant of integration: The arbitrary constant c added to every indefinite integral, since the derivative of a constant is zero.

ഒരു constant-ന്റെ derivative zero ആയതിനാൽ എല്ലാ indefinite integral-ലും കൂട്ടുന്ന arbitrary constant c .

Integrand: The function $f(x)$ being integrated in $\int f(x) dx$.

$\int f(x) dx$ -ൽ integrate ചെയ്യപ്പെടുന്ന function $f(x)$.

Integration is the reverse of differentiation: we are given a derivative and must find the function it came from. That function is called an

antiderivative. Because the derivative of a constant is zero, if $F'(x) = f(x)$ then $(F(x) + c)' = f(x)$ too, so we always add an arbitrary **constant of integration** c . This is why the result is called an indefinite integral.

In $\int f(x) dx$, $f(x)$ is the **integrand**, dx says "integrate with respect to x ", and the whole process is **integration**. The sign $\int$ is a stretched letter S, for "summation": integration grew out of adding up a continuous quantity.

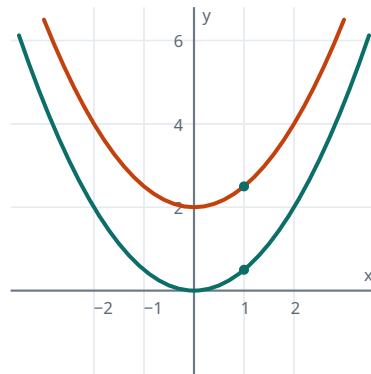
മലയാളത്തിൽ

Differentiation ഒരു function-ൽ നിന്ന് അതിന്റെ derivative കണ്ടുപിടിക്കുന്നു; integration അതിന്റെ നേരെ തിരിച്ച്, derivative തന്നിട്ട് "ഇത് ഏത് function-ന്റെ derivative ആണ്?" എന്ന് ചോദിക്കുന്നു. ഒരു ചായക്കടയിലെ ബാക്കി പൈസ കണ്ടിട്ട് ആദ്യം എത്ര കൊടുത്തു എന്ന് ഊഹിക്കുന്നത് പോലെ. Constant-ന്റെ derivative zero ആയതുകൊണ്ട് ആ constant എത്രയാണെന്ന് അറിയാൻ പറ്റില്ല, അതിനാൽ എപ്പോഴും "+ c" ചേർക്കണം.

If $d/dx [F(x)] = f(x)$, then $\int f(x) dx = F(x) + c$

c = constant of integration

Check any answer: d/dx [answer] must give back the integrand



$y = x^2/2$ and $y = x^2/2 + 2$ are both antiderivatives of x . At $x = 1$ both have slope 1; they differ only by the constant c .

EXAMPLE (FROM THE UNIT)

Find $\int x dx$.

We know $d/dx (x^2/2) = 2x/2 = x$. So $x^2/2$ is an antiderivative of x .

$$\int x dx = x^2/2 + c.$$

EXAMPLE (REMARK 2 OF THE UNIT)

Show that $\int (x^3 + x + 5) dx = x^4/4 + x^2/2 + 5x + c$ by differentiating.

Let $F(x) = x^4/4 + x^2/2 + 5x + c$.

$F'(x) = 4x^3/4 + 2x/2 + 5 + 0 = x^3 + x + 5$, which is the integrand. So the answer is correct.

Exam tip: Never drop "+ c" in an indefinite integral; IGNOU examiners deduct for it. In this unit "log" means the natural log (base e), and $\int (1/x) dx = \log |x| + c$ needs the modulus because log is defined only for positive numbers.

Table of standard integrals and rules

Standard integrals: Standard derivatives read backwards, e.g. $\int (1/x) dx = \log |x| + c$ and $\int e^x dx = e^x + c$.

Standard derivatives തിരിച്ചു വായിച്ചതാണ് standard integrals; ഉദാ. $\int (1/x) dx = \log |x| + c$,
 $\int e^x dx = e^x + c$.

Linearity: $\int [a f(x) \pm b g(x)] dx = a \int f(x) dx \pm b \int g(x) dx$, for constants a, b.

a, b constants ആണെങ്കിൽ $\int [a f(x) \pm b g(x)] dx = a \int f(x) dx \pm b \int g(x) dx$.

Each formula below is just a derivative read backwards. Learn this table by heart: every method in the unit (substitution, partial fractions, by parts) ends by reducing the problem to one of these rows.

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ഇത് integration-ന്റെ "ഗുണനപ്പട്ടിക" ആണ്. ഓരോ row-ഉം ഒരു derivative തിരിച്ചു വായിച്ചതാണ്. Substitution ആയാലും partial fractions ആയാലും by parts ആയാലും അവസാനം ഈ table-ലെ ഏതെങ്കിലും ഒരു row-ലേക്ക് എത്തിക്കുക എന്നതാണ് ലക്ഷ്യം, അതിനാൽ ഇത് മനഃപാഠമാക്കുക.

No.	f(x)	$\int f(x) dx$	Condition / note
1	k (constant)	$kx + c$	$\int 0 dx = c$
2	x^n	$x^{n+1}/(n+1) + c$	$n \neq -1$
3	$1/x$	$\log x + c$	the $n = -1$ case
4	$(ax + b)^n$	$(ax + b)^{n+1}/[a(n+1)] + c$	$n \neq -1, a \neq 0$
5	$1/(ax + b)$	$(1/a) \log ax + b + c$	$a \neq 0$
6(i)	a^{mx}	$a^{mx}/(m \log a) + c$	$a > 0, a \neq 1$
6(ii)	a^{mx+n}	$a^{mx+n}/(m \log a) + c$	special case: $\int a^x dx = a^x/\log a + c$
6(iii)	e^{ax}	$e^{ax}/a + c$	$\int e^x dx = e^x + c$
6(iv)	e^{ax+b}	$e^{ax+b}/a + c$	
Results derived in this unit (use them directly)			
7	$f'(x)/f(x)$	$\log f(x) + c$	substitution $t = f(x)$
8	$[f(x)]^n f'(x)$	$[f(x)]^{n+1}/(n+1) + c$	$n \neq -1$
9	$\log x$	$x \log x - x + c$	by parts
10	$x e^x$	$x e^x - e^x + c$	by parts

Remark 1 (linearity), a, b real constants:

$$\int a f(x) dx = a \int f(x) dx$$

$$\int [a f(x) \pm b g(x)] dx = a \int f(x) dx \pm b \int g(x) dx$$

Useful identity: $a^{\log_a f(x)} = f(x)$, in particular $e^{\log f(x)} = f(x)$

EXAMPLE

Use the table to write $\int (4x^3 - 6/x + 2e^{3x}) dx$.

By linearity: $4 \int x^3 dx - 6 \int (1/x) dx + 2 \int e^{3x} dx$.

$$= 4 \cdot x^4/4 - 6 \log |x| + 2 \cdot e^{3x}/3 + c = x^4 - 6 \log |x| + (2/3) e^{3x} + c.$$

Exam tip: Rows 4, 5, 6(ii), 6(iv) all work only when the inside is **linear** ($ax + b$). For $(x^2 + 1)^5$ you cannot divide by " $a(n+1)$ "; expand it or use substitution. There is no product rule or quotient rule for integrals: $\int f g dx \neq \int f dx \cdot \int g dx$.

Powers and polynomials

Definition: Power rule: $\int x^n dx = x^{n+1}/(n+1) + c$, $n \neq -1$; raise the power by one and divide by the new power.

Power rule: $\int x^n dx = x^{n+1}/(n+1) + c$, $n \neq -1$; power ഒന്ന് കൂട്ടി, പുതിയ power കൊണ്ട് ഹരിക്കുക.

To integrate a polynomial, integrate term by term with the power rule: raise the power by one and divide by the new power. Roots and reciprocals must first be written as powers ($\sqrt{x} = x^{1/2}$, $1/x^5 = x^{-5}$). Products and quotients are expanded or split into separate powers before integrating.

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Power rule ലളിതമാണ്: power ഒന്ന് കൂട്ടുക, പുതിയ power കൊണ്ട് ഹരിക്കുക. പക്ഷേ ആദ്യം എല്ലാം x^n രൂപത്തിലാക്കണം; $\sqrt{x}$ എന്നത് $x^{1/2}$, $1/x^3$ എന്നത് x^{-3} . ഗുണനമോ ഹരണമോ ഉണ്ടെങ്കിൽ, പച്ചക്കുറി അരിയുന്നതിന് മുമ്പ് കഴുകുന്നത് പോലെ, ആദ്യം expand ചെയ്ത് വേർതിരിക്കുക, എന്നിട്ട് ഓരോ term-ഉം integrate ചെയ്യുക.

$$\int x^n dx = x^{n+1}/(n+1) + c, n \neq -1$$
$$\sqrt{x} = x^{1/2}, \sqrt[3]{x} = x^{1/3}, 1/x^n = x^{-n}, x^m/x^n = x^{m-n}$$

EXAMPLE 1 (I)-(IV) OF THE UNIT

(i) $\int 5 dx = 5x + c$. (ii) $\int 0 dx = c$. (iii) $\int \pi dx = \pi x + c$ (π is a constant).

(iv) $\int x^3 dx = x^{3+1}/(3+1) + c = x^4/4 + c$.

EXAMPLE 1 (V)-(VIII): FRACTIONAL AND NEGATIVE POWERS

(v) $\int x^{7/2} dx = x^{9/2}/(9/2) + c = (2/9) x^{9/2} + c$.

(vi) $\int (1/x^5) dx = \int x^{-5} dx = x^{-4}/(-4) + c = -1/(4x^4) + c$.

(vii) $\int (1/x^{7/2}) dx = \int x^{-7/2} dx = x^{-5/2}/(-5/2) + c = -(2/5) x^{-5/2} + c$.

(viii) $\int (1/\sqrt[3]{x}) dx = \int x^{-1/3} dx = x^{2/3}/(2/3) + c = (3/2) x^{2/3} + c$.

Exam tip: In Example 1(vii) the unit prints the final answer as $-(2/5) x^{-7/2} + c$. That is a misprint; the correct answer is $-(2/5) x^{-5/2} + c$, since the power goes up from $-7/2$ to $-7/2 + 1 = -5/2$. Check: $d/dx [-(2/5) x^{-5/2}] = -(2/5)(-5/2) x^{-7/2} = x^{-7/2}$.

EXAMPLE 1 (X)-(XI): EXPAND OR SPLIT FIRST

$$(x) \int (x^2 + 1)(x - 1) dx = \int (x^3 - x^2 + x - 1) dx = x^4/4 - x^3/3 + x^2/2 - x + c.$$

$$(xi) \int (x^6 - x^4 + 1)/x^2 dx = \int (x^4 - x^2 + x^{-2}) dx = x^5/5 - x^3/3 + x^{-1}/(-1) + c = x^5/5 - x^3/3 - 1/x + c.$$

EXAMPLE 1 (XII)-(XIV): ROOTS IN THE DENOMINATOR, SQUARES

$$(xii) \int (x^4 + x^3 + 3)/\sqrt{x} dx. \text{ Divide each term by } x^{1/2}: \int (x^{7/2} + x^{5/2} + 3x^{-1/2}) dx.$$

$$= x^{9/2}/(9/2) + x^{7/2}/(7/2) + 3x^{1/2}/(1/2) + c = (2/9)x^{9/2} + (2/7)x^{7/2} + 6\sqrt{x} + c.$$

$$(xiii) \int (\sqrt{x} + 1/\sqrt{x}) dx = \int (x^{1/2} + x^{-1/2}) dx = (2/3)x^{3/2} + 2\sqrt{x} + c.$$

$$(xiv) \int (x + 1/x)^2 dx = \int (x^2 + 2 + x^{-2}) dx = x^3/3 + 2x - 1/x + c.$$

EXAMPLE 1 (XV)

$$\int (8x^3 + 2x - 3 + 1/x^2 + 2/x^3) dx = \int (8x^3 + 2x - 3 + x^{-2} + 2x^{-3}) dx$$

$$= 8x^4/4 + 2x^2/2 - 3x + x^{-1}/(-1) + 2x^{-2}/(-2) + c = 2x^4 + x^2 - 3x - 1/x - 1/x^2 + c.$$

Exam tip: The commonest slip is $\int x^{-1} dx = x^0/0$. The power rule fails at $n = -1$; use $\int (1/x) dx = \log |x| + c$ instead.

Linear forms: $(ax + b)^n$ and $1/(ax + b)$

Definition: $\int (ax + b)^n dx = (ax + b)^{n+1}/[a(n + 1)] + c$ ($n \neq -1$) and $\int dx/(ax + b) = (1/a) \log |ax + b| + c$: integrate as if $ax + b$ were x , then divide by a .

$\int (ax + b)^n dx = (ax + b)^{n+1}/[a(n + 1)] + c$ ($n \neq -1$), $\int dx/(ax + b) = (1/a) \log |ax + b| + c$: $ax + b$ -യെ x ആയി കണ്ട് integrate ചെയ്ത്, a കൊണ്ട് ഹരിക്കുക.

When the base is a linear expression $ax + b$, integrate as if $ax + b$ were x , then divide by a (the derivative of $ax + b$). This gives rows 4 and 5 of the table. The same "divide by a " idea works for e^{ax+b} and a^{mx+n} .

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ഉള്ളിൽ $ax + b$ പോലെ ഒരു linear expression ആണെങ്കിൽ, അതിനെ ഒരു x പോലെ കരുതി power rule ചെയ്യുക, എന്നിട്ട് ഉള്ളിലെ x -ന്റെ coefficient a കൊണ്ട് ഹരിക്കുക. Differentiate ചെയ്യുമ്പോൾ chain rule ഒരു a പുറത്ത് കൊണ്ടുവരും; അത് cancel ചെയ്യാനാണ് ഈ ഹരണം. a negative ആണെങ്കിൽ sign ശ്രദ്ധിക്കുക.

$$\int (ax + b)^n dx = (ax + b)^{n+1}/[a(n + 1)] + c, n \neq -1$$

$$\int dx/(ax + b) = (1/a) \log |ax + b| + c$$

EXAMPLE 2 (I)-(IV) OF THE UNIT

$$(i) \int (2x + 3)^5 dx: a = 2, n = 5, \text{ so } = (2x + 3)^6/(2 \cdot 6) + c = (2x + 3)^6/12 + c.$$

$$(ii) \int (5 - 9x)^6 dx: a = -9, n = 6, \text{ so } = (5 - 9x)^7/(7 \cdot (-9)) + c = -(1/63)(5 - 9x)^7 + c.$$

$$(iii) \int (9x + 5)^{3/2} dx: a = 9, n = 3/2, \text{ so } = (9x + 5)^{5/2}/(9 \cdot 5/2) + c = (2/45)(9x + 5)^{5/2} + c.$$

$$(iv) \int \sqrt[5]{(8 - 3x)} dx = \int (8 - 3x)^{1/5} dx: a = -3, n = 1/5, \text{ so } = (8 - 3x)^{6/5}/((6/5)(-3)) + c = -(5/18)(8 - 3x)^{6/5} + c.$$

EXAMPLE 2 (V)-(VII) OF THE UNIT

$$(v) \int (3 + 2x)^{7/2}/\sqrt{(3 + 2x)} dx = \int (3 + 2x)^{7/2 - 1/2} dx = \int (3 + 2x)^3 dx = (3 + 2x)^4/(4 \cdot 2) + c = (1/8)(3 + 2x)^4 + c.$$

$$(vi) \int dx/(7x + 2)^3 = \int (7x + 2)^{-3} dx = (7x + 2)^{-2}/(-2 \cdot 7) + c = -1/[14(7x + 2)^2] + c.$$

$$(vii) \int dx/\sqrt{(3x + 5)} = \int (3x + 5)^{-1/2} dx = (3x + 5)^{1/2}/(3 \cdot 1/2) + c = (2/3)\sqrt{(3x + 5)} + c.$$

EXAMPLE 3 OF THE UNIT: THE LOG FORM

$$(i) \int (1/x) dx = \log |x| + c. (ii) \int (3/x) dx = 3 \log |x| + c.$$

$$(iii) \int 5/(x + 1) dx = 5 \log |x + 1| + c.$$

$$(iv) \int 7/(5x + 2) dx = 7 \cdot (1/5) \log |5x + 2| + c = (7/5) \log |5x + 2| + c.$$

$$(v) \int 3/(9 - 2x) dx = 3 \cdot (1/(-2)) \log |9 - 2x| + c = -(3/2) \log |9 - 2x| + c.$$

EXAMPLE

Evaluate $\int (2x + 1)^2 dx$ in two ways and compare.

$$\text{Linear form: } (2x + 1)^3/(2 \cdot 3) + c = (2x + 1)^3/6 + c.$$

Expanding: $\int (4x^2 + 4x + 1) dx = (4/3)x^3 + 2x^2 + x + c$. Since $(2x + 1)^3/6 = (4/3)x^3 + 2x^2 + x + 1/6$, the two answers differ only by the constant $1/6$. Both are correct.

Exam tip: With a negative a , as in $(5 - 9x)$ or $(9 - 2x)$, the answer picks up a minus sign. Students often forget it. Differentiate your answer once to catch this.

Exponential forms and $a^{\log_a f(x)}$

Definition: $\int a^x dx = a^x / \log a + c$ ($a > 0$, $a \neq 1$) and $\int e^{ax+b} dx = e^{ax+b} / a + c$; also $a^{\log_a f(x)} = f(x)$.

$\int a^x dx = a^x / \log a + c$ ($a > 0$, $a \neq 1$), $\int e^{ax+b} dx = e^{ax+b} / a + c$; കൂടാതെ $a^{\log_a f(x)} = f(x)$.

For a^x we divide by $\log a$; for e^{ax} we divide by a (since $\log e = 1$). Many exam integrands look complicated but simplify at once using $a^{\log_a f(x)} = f(x)$, or by combining powers with the same exponent, such as $5^x 2^x = 10^x$. Terms like a^a , e^a , a^e are constants.

മലയാളത്തിൽ

a^x -ന്റെ integral എടുക്കുമ്പോൾ $\log a$ കൊണ്ട് ഹരിക്കണം; e -ക്ക് $\log e = 1$

ആയതുകൊണ്ട് $\int e^x dx = e^x$ തന്നെ. പല ചോദ്യങ്ങളും പേടിപ്പിക്കുന്ന

രൂപത്തിലായിരിക്കും, പക്ഷേ $a^{\log_a f(x)} = f(x)$ ഉപയോഗിച്ചാൽ അത് ഒരു സാധാരണ

power ആയി മാറും. x ഇല്ലാത്ത a^a , e^a പോലുള്ളവ വെറും constants ആണ്; അവയുടെ integral kx .

$$\int a^x dx = a^x / \log a + c; \int a^{mx+n} dx = a^{mx+n} / (m \log a) + c$$

$$\int e^{ax+b} dx = e^{ax+b} / a + c$$

$$a^{\log_a f(x)} = f(x); e^{k \log f(x)} = [f(x)]^k; a^x b^x = (ab)^x; a^x / b^x = (a/b)^x$$

EXAMPLE 2 (X)-(XIV) OF THE UNIT

$$(x) \int a^{3x} dx = a^{3x} / (3 \log a) + c. (xi) \int e^{3x} dx = e^{3x} / 3 + c.$$

$$(xii) \int e^{5x+7} dx = e^{5x+7} / 5 + c. (xiii) \int a^{3-2x} dx: m = -2, \text{ so } = -a^{3-2x} / (2 \log a) + c.$$

$$(xiv) \int (5e^{7x} + x^2) dx = (5/7) e^{7x} + x^3 / 3 + c.$$

EXAMPLE 2 (VIII)-(IX): THE LOG IDENTITY

$$(viii) \int e^{5 \log \sqrt{3x-5}} dx. \text{ Since } 5 \log \sqrt{3x-5} = \log (3x-5)^{5/2}, \text{ the integrand is } (3x-5)^{5/2}.$$

$$\int (3x-5)^{5/2} dx = (3x-5)^{7/2} / ((7/2) \cdot 3) + c = (2/21)(3x-5)^{7/2} + c.$$

$$(ix) \int a^{\log_a \sqrt{4x+5}} dx = \int (4x+5)^{1/2} dx = (4x+5)^{3/2} / ((3/2) \cdot 4) + c = (1/6)(4x+5)^{3/2} + c.$$

EXAMPLE 2 (XV)-(XVII): CONSTANTS AND COMBINED BASES

$$(xv) \int (a^x + e^x + a^a + e^a + a^e) dx = a^x / \log a + e^x + a^a x + e^a x + a^e x + c.$$

$$(xvi) \int 5^x 2^x dx = \int 10^x dx = 10^x / \log 10 + c.$$

$$(xvii) \int 2^x / 3^x dx = \int (2/3)^x dx = (2/3)^x / \log (2/3) + c = (2/3)^x / (\log 2 - \log 3) + c.$$

EXAMPLE 2 (XVIII)

$$\int (a^x - b^x)^2 / (a^x b^x) dx.$$

Expand the top: $a^{2x} + b^{2x} - 2a^x b^x$. Divide each term by $a^x b^x$: $a^x / b^x + b^x / a^x - 2 = (a/b)^x + (b/a)^x - 2$.

$$\text{Integral} = (a/b)^x / \log (a/b) + (b/a)^x / \log (b/a) - 2x + c.$$

EXAMPLE 2 (XIX)-(XX): MIXED

$$(xix) \int (e^{a \log x} + a^{x \log_a a} + a^{m \log_a a}) dx = \int (x^a + a^x + a^m) dx = x^{a+1} / (a+1) + a^x / \log a + a^m x + c.$$

$$(xx) \int [x/3 + (5x-3)^3 + x\sqrt{x} + 1/\sqrt{(5+2x)} + a^x] dx = \int [x/3 + (5x-3)^3 + x^{3/2} + (5+2x)^{-1/2} + a^x] dx \\ = x^2/6 + (5x-3)^4/20 + (2/5)x^{5/2} + (5+2x)^{1/2} + a^x x + c.$$

Exam tip: $\int a^x dx$ is **not** $a^{x+1} / (x+1)$. The power rule is for a variable base with a constant power (x^n); a^x has a constant base with a variable power, so it uses the exponential row.

Integration by substitution

Definition: Integration by substitution replaces part $f(x)$ of the integrand by a new variable t , with $f'(x) dx = dt$, so that the integral becomes a standard form in t .

Integrand-ലെ ഒരു ഭാഗം $f(x)$ -നെ പുതിയ variable t ആക്കി ($f'(x) dx = dt$), integral-നെ t -യിലുള്ള ഒരു standard form ആക്കുന്ന രീതിയാണ് integration by substitution.

When no formula fits directly, we replace part of the integrand by a new variable t so that the integral becomes a standard form in t . The method works well when the integrand contains a function $f(x)$ together with its derivative $f'(x)$, as in $[f(x)]^n f'(x)$ or $f'(x)/[f(x)]^n$. This is the chain rule used in reverse.

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ഒരു function-ഉം അതിന്റെ derivative-ഉം ഒരുമിച്ച് integrand-ൽ കണ്ടാൽ substitution ഉപയോഗിക്കുക. ആ function-നെ t എന്ന് വിളിക്കുക, dx -ന് പകരം dt കൊണ്ടുവരിക; അപ്പോൾ ചോദ്യം table-ലെ ഒരു ലളിത രൂപമാകും. ഒരു വലിയ ചാക്ക് അരി ചെറിയ പാത്രങ്ങളിലാക്കി കൊണ്ടുപോകുന്നത് പോലെ: രൂപം മാറുന്നു, സാധനം അതുതന്നെ. അവസാനം t -ന് പകരം വീണ്ടും x -ലേക്ക് മാറ്റാൻ മറക്കരുത്.

Step I: Put $f(x) = t$

Step II: Differentiate: $f'(x) dx = dt$

Step III: Rewrite the whole integral in t (no x left)

Step IV: Integrate using a standard form

Step V: Replace t by $f(x)$

EXAMPLE 4 (I) OF THE UNIT

Evaluate $\int x^9/(x^{10} + 1) dx$.

Put $x^{10} + 1 = t$. Then $10x^9 dx = dt$, so $x^9 dx = dt/10$.

$I = \int (1/t)(dt/10) = (1/10) \log |t| + c = (1/10) \log (x^{10} + 1) + c$. ($x^{10} + 1 > 0$, so the modulus can be dropped.)

Alternatively put $x^{10} = t$: $I = (1/10) \int dt/(t + 1) = (1/10) \log |t + 1| + c$, the same answer.

EXAMPLE 4 (II)-(IV) OF THE UNIT

(ii) $\int x^{n-1}/(x^n + a) dx$. Put $x^n + a = t$, $n x^{n-1} dx = dt$. $I = (1/n) \int dt/t = (1/n) \log |x^n + a| + c$.

(iii) $\int e^x/(e^x + 5) dx$. Put $e^x + 5 = t$, $e^x dx = dt$. $I = \int dt/t = \log (e^x + 5) + c$.

(iv) $\int dx/(x \log x)$. Put $\log x = t$, $(1/x) dx = dt$. $I = \int dt/t = \log |\log x| + c$.

EXAMPLE 4 (V)-(VI) OF THE UNIT

(v) $\int (2ax + b)/(ax^2 + bx + c) dx$. Put $ax^2 + bx + c = t$, $(2ax + b) dx = dt$. $I = \int dt/t = \log |ax^2 + bx + c| + k$ (k is the constant, since c is already used).

(vi) $\int (8x^3 + 4x)/(x^4 + x^2 + 1)^6 dx = 2 \int (4x^3 + 2x)/(x^4 + x^2 + 1)^6 dx$. Put $x^4 + x^2 + 1 = t$, $(4x^3 + 2x) dx = dt$.

$I = 2 \int t^{-6} dt = 2 \cdot t^{-5}/(-5) + c = -(2/5)(x^4 + x^2 + 1)^{-5} + c$.

EXAMPLE 4 (VII)-(VIII) OF THE UNIT

(vii) $\int (e^{2x} + e^{-2x})/(e^{2x} - e^{-2x}) dx$. Put $e^{2x} - e^{-2x} = t$. Then $(2e^{2x} + 2e^{-2x}) dx = dt$, so $(e^{2x} + e^{-2x}) dx = dt/2$.

$$I = (1/2) \int dt/t = (1/2) \log |e^{2x} - e^{-2x}| + c.$$

(viii) $\int dx/(x + \sqrt{x}) = \int dx/[\sqrt{x}(\sqrt{x} + 1)]$. Put $\sqrt{x} + 1 = t$, $dx/(2\sqrt{x}) = dt$, so $dx/\sqrt{x} = 2 dt$.

$$I = 2 \int dt/t = 2 \log |\sqrt{x} + 1| + c.$$

EXAMPLE 4 (IX)-(X) OF THE UNIT

(ix) $\int (2ax + b)\sqrt{ax^2 + bx + c} dx$. Put $ax^2 + bx + c = t$, $(2ax + b) dx = dt$. $I = \int t^{1/2} dt = (2/3) t^{3/2} + k = (2/3)(ax^2 + bx + c)^{3/2} + k$.

(x) $\int 2x/[(1 + x^2) \log(1 + x^2)] dx$. Put $\log(1 + x^2) = t$, $2x dx/(1 + x^2) = dt$. $I = \int dt/t = \log |\log(1 + x^2)| + c$.

Exam tip: How to choose t : look for the "inner" function whose derivative is also sitting in the integrand (possibly up to a constant factor). If after substituting any x is still left over, your choice of t is wrong or you need to express that x in terms of t .

More substitution patterns

Numerator = derivative of denominator: $\int f'(x)/f(x) dx = \log |f(x)| + c$.

Numerator, denominator-ന്റെ derivative ആണെങ്കിൽ $\int f'(x)/f(x) dx = \log |f(x)| + c$.

Power of $f(x)$ times $f'(x)$: $\int [f(x)]^n f'(x) dx = [f(x)]^{n+1}/(n+1) + c$, $n \neq -1$.

$f(x)$ -ന്റെ power-നെ $f'(x)$ കൊണ്ട് ഗുണിച്ചതാണെങ്കിൽ $\int [f(x)]^n f'(x) dx = [f(x)]^{n+1}/(n+1) + c$, $n \neq -1$.

Two shortcuts come out of the substitution method and save time in the exam. If the numerator is exactly the derivative of the denominator, the answer is $\log |\text{denominator}|$. If the integrand is a power of $f(x)$ times $f'(x)$, the answer is that power raised by one. A third pattern, $\sqrt{x+a}$ with an extra x outside, is handled by putting $x+a = t^2$, so that the root disappears.

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Numerator കൃത്യമായി denominator-ന്റെ derivative ആണെങ്കിൽ ഉത്തരം നേരെ $\log |\text{denominator}|$. ഒരു function-ന്റെ power-ഉം അതിന്റെ derivative-ഉം

ഒന്നിച്ചാണെങ്കിൽ power rule തന്നെ. $\sqrt{x+a}$ -ഉം പുറത്ത് ഒരു x -ഉം ഉണ്ടെങ്കിൽ $x+a = t^2$ എന്ന് വെക്കുക; അപ്പോൾ root പോയി, t -യിൽ ഒരു polynomial മാത്രം ബാക്കിയാകും.

$$\int f'(x)/f(x) dx = \log |f(x)| + c$$

$$\int [f(x)]^n f'(x) dx = [f(x)]^{n+1}/(n+1) + c, n \neq -1$$

For $\sqrt{x+a}$: put $x+a = t^2$, $dx = 2t dt$, $x = t^2 - a$

EXAMPLE

(a) $\int (3x^2 + 2)/(x^3 + 2x + 5) dx$. The numerator is $d/dx (x^3 + 2x + 5)$. Answer: $\log |x^3 + 2x + 5| + c$.

(b) $\int x(x^2 + 1)^4 dx$. Put $x^2 + 1 = t$, $2x dx = dt$, $x dx = dt/2$. $I = (1/2) \int t^4 dt = t^5/10 + c = (x^2 + 1)^5/10 + c$.

(c) $\int (\log x)^2/x dx$. Put $\log x = t$, $dx/x = dt$. $I = \int t^2 dt = t^3/3 + c = (\log x)^3/3 + c$.

EXAMPLE

(a) $\int x e^{x^2} dx$. Put $x^2 = t$, $x dx = dt/2$. $I = (1/2) \int e^t dt = (1/2) e^{x^2} + c$.

(b) $\int e^{\sqrt{x}}/\sqrt{x} dx$. Put $\sqrt{x} = t$, $dx/(2\sqrt{x}) = dt$. $I = 2 \int e^t dt = 2e^{\sqrt{x}} + c$.

EXAMPLE (CHECK YOUR PROGRESS 1, Q3(II))

Evaluate $\int x\sqrt{x+a} dx$.

Put $x+a = t^2$, so $dx = 2t dt$ and $x = t^2 - a$.

$$I = \int (t^2 - a) \cdot t \cdot 2t dt = 2 \int (t^4 - at^2) dt = 2 [t^5/5 - at^3/3] + c.$$

$$\text{Replace } t = \sqrt{x+a}: I = 2 [(x+a)^{5/2}/5 - a(x+a)^{3/2}/3] + c.$$

EXAMPLE (CHECK YOUR PROGRESS 1, Q3(III))

Evaluate $\int x/\sqrt{x+a} dx$.

$$\text{Put } x+a = t^2, dx = 2t dt: I = \int (t^2 - a)/t \cdot 2t dt = 2 \int (t^2 - a) dt = 2 [t^3/3 - at] + c.$$

$$= 2 [(x+a)^{3/2}/3 - a\sqrt{x+a}] + c.$$

Exam tip: For $\int x\sqrt{x+a} \, dx$, putting $t = x + a$ (without the square) also works: $x = t - a$, and $I = \int (t - a) t^{1/2} dt = \int (t^{3/2} - a t^{1/2}) dt$ gives the same answer. Use whichever you find easier.

Partial fractions: Type 1 (distinct linear factors)

Proper fraction: A rational function $P(x)/Q(x)$ in which the degree of P is less than the degree of Q .

P-യുടെ degree, Q-യുടെ degree-യെക്കാൾ കുറവായ rational function $P(x)/Q(x)$.

Partial fractions (Type 1): When $Q(x)$ has distinct linear factors, $P(x)/Q(x) = A/(x - a) + B/(x - b) + \dots$, and each fraction integrates to a log.

$Q(x)$ -ന് വ്യത്യസ്തമായ linear factors ആണെങ്കിൽ $P(x)/Q(x) = A/(x - a) + B/(x - b) + \dots$; ഓരോ fraction-ഉം integrate ചെയ്യുമ്പോൾ ഒരു log കിട്ടും.

A rational function $P(x)/Q(x)$ whose denominator factorises can be split into simple fractions $A/(x - a)$, each of which integrates to a log. First check degrees: if the degree of the numerator is less than that of the denominator (a proper fraction), go ahead; otherwise divide first (topic 10). In Type 1 every factor of the denominator is linear and appears only once.

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ഒരു വലിയ fraction-നെ ചെറിയ ചെറിയ fractions ആക്കി പൊളിക്കുന്നതാണ് partial fractions. ഒരു കല്യാണ സന്ധ്യയുടെ bill ഓരോ വീട്ടുകാർക്കും പങ്കിടുന്നത് പോലെ, ഓരോ linear factor-നും ഒരു പങ്ക് (A, B, C). ഓരോ constant-ഉം കണ്ടുപിടിക്കാൻ ആ factor zero ആക്കുന്ന x-ന്റെ value, ആ factor ഒഴികെ ബാക്കി എല്ലായിടത്തും ഇടുക. ഇതിനെ "cover-up" method എന്നും പറയും.

$$P(x)/[(x - a)(x - b)(x - c)] = A/(x - a) + B/(x - b) + C/(x - c)$$

A = value of $P(x)/[(x - b)(x - c)]$ at $x = a$ (cover up $x - a$), and so on

$$\int A/(x - a) \, dx = A \log |x - a| + c$$

WORKED RESOLUTION FROM THE UNIT (TYPE 1)

Resolve $(x + 5)/[(x - 1)(x - 2)(x - 3)]$ and integrate.

Factors vanish at $x = 1, 2, 3$.

$$A (x = 1): (1 + 5)/[(1 - 2)(1 - 3)] = 6/2 = 3.$$

$$B (x = 2): (2 + 5)/[(2 - 1)(2 - 3)] = 7/(-1) = -7.$$

$$C (x = 3): (3 + 5)/[(3 - 1)(3 - 2)] = 8/2 = 4.$$

So the integrand $= 3/(x - 1) - 7/(x - 2) + 4/(x - 3)$, and the integral $= 3 \log |x - 1| - 7 \log |x - 2| + 4 \log |x - 3| + c$.

EXAMPLE 5 (I) OF THE UNIT

Evaluate $\int (4x + 1)/[(x - 1)(x - 2)] dx$.

$$A (x = 1): (4 + 1)/(1 - 2) = -5. \quad B (x = 2): (8 + 1)/(2 - 1) = 9.$$

$$I = \int [-5/(x - 1) + 9/(x - 2)] dx = -5 \log |x - 1| + 9 \log |x - 2| + c.$$

EXAMPLE 5 (II) OF THE UNIT

Evaluate $\int (3x + 4)/(x^2 - x - 12) dx$.

Factorise: $x^2 - x - 12 = (x - 4)(x + 3)$.

$$A (x = 4): (12 + 4)/(4 + 3) = 16/7. \quad B (x = -3): (-9 + 4)/(-3 - 4) = 5/7.$$

$$I = (16/7) \log |x - 4| + (5/7) \log |x + 3| + c.$$

EXAMPLE

(a) $\int dx/(x^2 - 1)$. Here $1/[(x - 1)(x + 1)]$: $A (x = 1) = 1/2$, $B (x = -1) = 1/(-2) = -1/2$.

$$I = (1/2) \log |x - 1| - (1/2) \log |x + 1| + c = (1/2) \log |(x - 1)/(x + 1)| + c.$$

(b) $\int (2x + 3)/(x^2 + 3x + 2) dx$. The numerator is the derivative of the denominator, so the answer is $\log |x^2 + 3x + 2| + c$ directly. (Partial fractions give $1/(x + 1) + 1/(x + 2)$, i.e. $\log |x + 1| + \log |x + 2|$, the same thing.)

Exam tip: Always check whether the numerator is the derivative of the denominator before doing partial fractions; it saves several steps. After finding A, B, C, test at one extra x (say $x = 0$) that both sides agree.

Partial fractions: Type 2 (repeated linear factors)

Definition: A repeated linear factor $(x - b)^k$ in the denominator needs one partial fraction for each power: $A_1/(x - b) + A_2/(x - b)^2 + \dots + A_k/(x - b)^k$.

Denominator-ലെ repeated linear factor $(x - b)^k$ -ന് ഓരോ power-നും ഓരോ partial fraction വേണം: $A_1/(x - b) + A_2/(x - b)^2 + \dots + A_k/(x - b)^k$.

If a linear factor is repeated, say $(x + 2)^3$, it needs one fraction for each power: $B/(x + 2) + C/(x + 2)^2 + D/(x + 2)^3$. Multiply both sides by the full denominator. Putting the roots gives the constants over the non-repeated factors and the highest power; the rest come from comparing coefficients (usually of the highest power of x and of the constant term).

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ഒരു factor പല തവണ (square, cube) വന്നാൽ, ഓരോ power-നും ഓരോ fraction വേണം. Root ഇട്ടാൽ എല്ലാ constants-ഉം കിട്ടില്ല; ബാക്കിയുള്ളവ കണ്ടുപിടിക്കാൻ ഇരുവശത്തെയും x^3 -ന്റെ coefficient-ഉം constant term-ഉം തുല്യമാക്കുക. $\int 1/(x + 2)^2 dx$ log അല്ല, power rule ആണ് എന്ന് ഓർക്കുക.

$$P(x)/[(x - a)(x - b)^3] = A/(x - a) + B/(x - b) + C/(x - b)^2 + D/(x - b)^3$$
$$\int dx/(x - b)^k = (x - b)^{1-k}/(1 - k) + c, k \geq 2 \text{ (not a log)}$$

WORKED RESOLUTION FROM THE UNIT (TYPE 2), CORRECTED

Resolve $(x^2 + x + 5)/[(x + 5)(x + 2)^3]$.

Let it equal $A/(x + 5) + B/(x + 2) + C/(x + 2)^2 + D/(x + 2)^3$. Multiply out:

$$x^2 + x + 5 = A(x + 2)^3 + B(x + 5)(x + 2)^2 + C(x + 5)(x + 2) + D(x + 5). \dots (1)$$

$$x = -5: 25 - 5 + 5 = A(-3)^3 \Rightarrow 25 = -27A \Rightarrow A = -25/27.$$

$$x = -2: 4 - 2 + 5 = 3D \Rightarrow D = 7/3.$$

$$\text{Coefficient of } x^3: 0 = A + B \Rightarrow B = 25/27.$$

$$\text{Constant term: } 5 = 8A + 20B + 10C + 5D = -200/27 + 500/27 + 10C + 35/3 = 100/9 + 105/9 + 10C.$$

$$10C = 5 - 205/9 = -160/9 \Rightarrow C = -16/9.$$

$$\text{So } (x^2 + x + 5)/[(x + 5)(x + 2)^3] = (-25/27)/(x + 5) + (25/27)/(x + 2) - (16/9)/(x + 2)^2 + (7/3)/(x + 2)^3.$$

$$\text{Integrating: } -(25/27) \log |x + 5| + (25/27) \log |x + 2| + 16/[9(x + 2)] - 7/[6(x + 2)^2] + c.$$

Exam tip: The unit's Type 2 illustration writes $C = -48$, which is wrong. From its own equation $10C = 5 + 200/27 - 20(25/27) - 35/3 = -160/9$, so $C = -16/9$. Check at $x = 0$: left side $5/40 = 1/8$; right side $(-25/27)/5 + (25/27)/2 + (-16/9)/4 + (7/3)/8 = -5/27 + 25/54 - 4/9 + 7/24 = 1/8$. ✓

EXAMPLE 5 (III) OF THE UNIT

Evaluate $\int 8x/[(x+1)(x-3)^2] dx$.

$$8x = A(x-3)^2 + B(x+1)(x-3) + C(x+1). \dots (2)$$

$$x = -1: -8 = 16A \Rightarrow A = -1/2. \quad x = 3: 24 = 4C \Rightarrow C = 6. \quad \text{Coefficient of } x^2: 0 = A + B \Rightarrow B = 1/2.$$

$$I = \int [(-1/2)/(x+1) + (1/2)/(x-3) + 6(x-3)^{-2}] dx = -(1/2) \log|x+1| + (1/2) \log|x-3| - 6/(x-3) + c.$$

EXAMPLE 5 (IV) OF THE UNIT, CORRECTED

Evaluate $\int (x^2 + x + 2)/[(x+2)(x+1)^3] dx$.

$$x^2 + x + 2 = A(x+1)^3 + B(x+2)(x+1)^2 + C(x+2)(x+1) + D(x+2). \dots (2)$$

$$x = -2: 4 - 2 + 2 = A(-1)^3 \Rightarrow 4 = -A \Rightarrow A = -4.$$

$$x = -1: 1 - 1 + 2 = D(1) \Rightarrow D = 2.$$

$$\text{Coefficient of } x^3: 0 = A + B \Rightarrow B = 4.$$

$$\text{Constant term: } 2 = A + 2B + 2C + 2D = -4 + 8 + 2C + 4 \Rightarrow 2C = -6 \Rightarrow C = -3.$$

$$I = \int [-4/(x+2) + 4/(x+1) - 3/(x+1)^2 + 2/(x+1)^3] dx \\ = -4 \log|x+2| + 4 \log|x+1| + 3/(x+1) - 1/(x+1)^2 + c.$$

Exam tip: In Example 5(iv) the unit substitutes $x = -2$ into $A(x+1)^3$ but writes $A(-2+1)^2$, getting $A = 4$ instead of -4 . That error carries through ($B = -4$, $C = 1$ in the unit). The correct values are $A = -4$, $B = 4$, $C = -3$, $D = 2$, giving the answer above. Check at $x = 1$: left side $4/(3 \cdot 8) = 1/6$; correct values give $-4/3 + 2 - 3/4 + 1/4 = 1/6$ ✓, while the unit's values give $4/3 - 2 + 1/4 + 1/4 = -1/6$ ✗.

EXAMPLE

Evaluate $\int dx/[x(x-1)^2]$.

$$1 = A(x-1)^2 + Bx(x-1) + Cx. \quad x = 0: A = 1. \quad x = 1: C = 1. \quad \text{Coefficient of } x^2: 0 = A + B \Rightarrow B = -1.$$

$$I = \int [1/x - 1/(x-1) + 1/(x-1)^2] dx = \log|x| - \log|x-1| - 1/(x-1) + c = \log|x/(x-1)| - 1/(x-1) + c.$$

Exam tip: The unit states that Type 3 (quadratic factors) and Type 4 (repeated quadratic factors) are beyond its scope, so expect only linear factors, distinct or repeated.

Improper fractions: divide first

Definition: An improper fraction $P(x)/Q(x)$ has $\deg P \geq \deg Q$; divide first to write it as quotient + remainder/ $Q(x)$, where the remainder fraction is proper.

$\deg P \geq \deg Q$ ആയ $P(x)/Q(x)$ ആണ് improper fraction; ആദ്യം divide ചെയ്ത് quotient + remainder/ $Q(x)$ എന്ന് എഴുതുക, ഇവിടെ remainder fraction proper ആയിരിക്കും.

If the degree of the numerator is greater than or equal to the degree of the denominator, first divide (long division) to get quotient + remainder/denominator. The quotient is a polynomial, integrated by the power rule; the remainder fraction is proper and is then resolved into partial fractions.

മലയാളത്തിൽ

Numerator-ന്റെ degree denominator-ന്റേതിന് തുല്യമോ അതിൽ കൂടുതലോ ആണെങ്കിൽ നേരെ partial fractions ചെയ്യാൻ പറ്റില്ല. ആദ്യം സ്കൂളിൽ പഠിച്ച long division ചെയ്യുക: quotient-ഉം ഒരു ചെറിയ remainder fraction-ഉം കിട്ടും. Quotient-നെ power rule കൊണ്ടും, remainder fraction-നെ partial fractions കൊണ്ടും integrate ചെയ്യുക.

$P(x)/Q(x) = \text{quotient} + \text{remainder}/Q(x)$, when $\deg P \geq \deg Q$

EXAMPLE 5 (V) OF THE UNIT

Evaluate $\int (x^2 + 1)/(x^2 - 1) dx$.

Divide: $(x^2 + 1)/(x^2 - 1) = 1 + 2/(x^2 - 1) = 1 + 2/[(x - 1)(x + 1)]$.

$2/[(x - 1)(x + 1)]$: $A(x = 1) = 2/2 = 1$, $B(x = -1) = 2/(-2) = -1$.

$I = \int [1 + 1/(x - 1) - 1/(x + 1)] dx = x + \log|x - 1| - \log|x + 1| + c = x + \log|(x - 1)/(x + 1)| + c$.

EXAMPLE (CHECK YOUR PROGRESS 2, Q1(II))

Evaluate $\int (x^3 + 5x + 1)/(x^2 - 4) dx$.

Divide: $x^3 + 5x + 1 = x(x^2 - 4) + 9x + 1$, so the integrand $= x + (9x + 1)/[(x - 2)(x + 2)]$.

$A(x = 2): (18 + 1)/4 = 19/4$. $B(x = -2): (-18 + 1)/(-4) = 17/4$.

$I = x^2/2 + (19/4) \log|x - 2| + (17/4) \log|x + 2| + c = x^2/2 + (1/4) \log|(x - 2)^{19}(x + 2)^{17}| + c$.

EXAMPLE

Evaluate $\int x^2/(x+1) dx$.

Divide: $x^2 = (x-1)(x+1) + 1$, so $x^2/(x+1) = x-1 + 1/(x+1)$.

$I = x^2/2 - x + \log|x+1| + c$.

Integration by parts and the ILATE order

Integration by parts: $\int I \cdot II dx = I \int II dx - \int [(dI/dx) \cdot \int II dx] dx$, where I is the first function and II the second function.

$\int I \cdot II dx = I \int II dx - \int [(dI/dx) \cdot \int II dx] dx$; ഇവിടെ I first function-ഉം II second function-ഉം ആണ്.

ILATE: The order for choosing the first function: Inverse trigonometric, Logarithmic, Algebraic, Trigonometric, Exponential.

First function തിരഞ്ഞെടുക്കാനുള്ള order: Inverse trigonometric, Logarithmic, Algebraic, Trigonometric, Exponential.

Integration by parts handles a product of two functions. It comes from integrating the product rule $d(uv)/dx = u dv/dx + v du/dx$. We call one factor the first function (I) and the other the second function (II). The formula is: $\text{first} \times (\text{integral of second}) - \int [\text{derivative of first} \times (\text{integral of second})] dx$. The second function must be one we can integrate, and the first should become simpler when differentiated.

മലയാളത്തിൽ

രണ്ട് functions-ന്റെ ഗുണനഫലം integrate ചെയ്യാനുള്ള വഴിയാണ് by parts.

ഒന്നിനെ first (I), മറ്റേതിനെ second (II) ആയി എടുക്കുന്നു. "ആദ്യത്തേത് $\times$

രണ്ടാമത്തേതിന്റെ integral - $\int$ (ആദ്യത്തേതിന്റെ derivative $\times$ രണ്ടാമത്തേതിന്റെ integral)" എന്നത് ഒരു മന്ത്രം പോലെ ഓർക്കുക. ഏതിനെ first ആക്കണം

എന്നറിയാൻ ILATE എന്ന ക്രമം ഉപയോഗിക്കുക: ഈ list-ൽ ആദ്യം വരുന്നത് first function.

$$\int u (dv/dx) dx = uv - \int v (du/dx) dx$$

$$\int I \cdot II dx = I \int II dx - \int [(dI/dx) \cdot \int II dx] dx$$

ILATE order for the first function: Inverse trig $\rightarrow$ Logarithmic $\rightarrow$ Algebraic (polynomial) $\rightarrow$ Trigonometric $\rightarrow$ Exponential

The unit's Remark 3 says: the choice matters; if one factor is a polynomial, take the polynomial as the first function; and by parts is only a technique, it does not promise that every product can be integrated. ILATE extends this: a log beats a polynomial (log x is hard to integrate but easy to differentiate), and a polynomial beats an exponential.

EXAMPLE 6 (I) OF THE UNIT: WHY THE CHOICE MATTERS

Evaluate $\int x e^x dx$.

ILATE: x is Algebraic, e^x is Exponential, so $I = x$, $II = e^x$.

$$I = x \int e^x dx - \int [1 \cdot e^x] dx = x e^x - e^x + c.$$

Wrong choice ($I = e^x$, $II = x$): $I = e^x \cdot x^2/2 - \int e^x \cdot x^2/2 dx$. The new integral has x^2 in it, which is harder than what we started with. So the choice of first function is important.

EXAMPLE 6 (IV) OF THE UNIT: $\int \log x dx$

Write $\log x = \log x \cdot 1$. Take $I = \log x$ (L comes before A), $II = 1$.

$$I = (\log x)(x) - \int (1/x)(x) dx = x \log x - \int 1 dx = x \log x - x + c.$$

EXAMPLE

(a) $\int x \log x dx$. $I = \log x$, $II = x$.

$$= \log x \cdot x^2/2 - \int (1/x)(x^2/2) dx = (x^2/2) \log x - \int (x/2) dx = (x^2/2) \log x - x^2/4 + c.$$

(b) $\int x^2 \log x dx$. $I = \log x$, $II = x^2$.

$$= \log x \cdot x^3/3 - \int (1/x)(x^3/3) dx = (x^3/3) \log x - x^3/9 + c.$$

EXAMPLE

(a) $\int \log x/x^2 dx$. $I = \log x$, $II = x^{-2}$ ($\int x^{-2} dx = -1/x$).

$$= \log x \cdot (-1/x) - \int (1/x)(-1/x) dx = -(\log x)/x + \int x^{-2} dx = -(\log x)/x - 1/x + c.$$

(b) $\int x \cdot 2^x dx$. $I = x$, $II = 2^x$ ($\int 2^x dx = 2^x/\log 2$).

$$= x \cdot 2^x/\log 2 - \int 1 \cdot 2^x/\log 2 dx = x 2^x/\log 2 - 2^x/(\log 2)^2 + c.$$

EXAMPLE

$\int x e^{-x} dx$. $I = x$, $II = e^{-x}$ ($\int e^{-x} dx = -e^{-x}$).

$$= x(-e^{-x}) - \int 1 \cdot (-e^{-x}) dx = -x e^{-x} + \int e^{-x} dx = -x e^{-x} - e^{-x} + c.$$

Exam tip: Write "I" and "II" under the two factors in your answer, as IGNOU's model answers do; it shows the examiner your choice. If $\log x$ or an inverse trig function appears alone, multiply by 1 and take 1 as the second function.

Repeated integration by parts

Definition: When the first function is a polynomial of degree n , integration by parts is applied n times, keeping the polynomial as the first function each time, until it disappears.

First function degree n ഉള്ള ഒരു polynomial ആണെങ്കിൽ, ഓരോ തവണയും polynomial-നെ first function ആയി വെച്ച്, അത് ഇല്ലാതാകുന്നതുവരെ integration by parts n തവണ ചെയ്യുന്നു.

When the polynomial has degree 2 or more, one round of by parts only lowers its degree by one. Apply by parts again to the new integral, keeping the polynomial as the first function each time, until the polynomial is gone. Sometimes a substitution is needed first to bring the integral into a by-parts form.

മലയാളത്തിൽ

$x^2 e^{3x}$ പോലെയുള്ളതിൽ ഒരു തവണ by parts ചെയ്താൽ $x^2 \rightarrow x$ ആകുന്നു; ഒരിക്കൽ കൂടി ചെയ്താൽ $x \rightarrow 1$. ഒരു തേങ്ങ പൊളിക്കുന്നത് പോലെ, ഓരോ അടിയിലും polynomial-ന്റെ degree ഒന്ന് കുറയും. Signs ശ്രദ്ധിക്കാൻ ഓരോ round-ഉം brackets-ൽ എഴുതുക.

$\int x^n e^{ax} dx$ or $\int x^n a^x dx$: apply by parts n times, polynomial always first

Signs alternate: $+ - + - \dots$

EXAMPLE 6 (II) OF THE UNIT

Evaluate $\int x^2 e^{3x} dx$.

Round 1 ($I = x^2$, $II = e^{3x}$): $= x^2 \cdot e^{3x}/3 - \int 2x \cdot e^{3x}/3 dx = (x^2 e^{3x})/3 - (2/3) \int x e^{3x} dx$.

Round 2 ($I = x$, $II = e^{3x}$): $\int x e^{3x} dx = x e^{3x}/3 - \int e^{3x}/3 dx = x e^{3x}/3 - e^{3x}/9$.

$I = (x^2 e^{3x})/3 - (2/3)[x e^{3x}/3 - e^{3x}/9] + c = (1/3) x^2 e^{3x} - (2/9) x e^{3x} + (2/27) e^{3x} + c$.

EXAMPLE 6 (III) OF THE UNIT

Evaluate $\int x^3 a^x dx$. Write $L = \log a$, so $\int a^x dx = a^x/L$.

Round 1: $= x^3 a^x/L - (3/L) \int x^2 a^x dx$.

Round 2: $\int x^2 a^x dx = x^2 a^x/L - (2/L) \int x a^x dx$.

Round 3: $\int x a^x dx = x a^x/L - a^x/L^2$.

Put back: $I = x^3 a^x/\log a - 3x^2 a^x/(\log a)^2 + 6x a^x/(\log a)^3 - 6a^x/(\log a)^4 + c$.

EXAMPLE (CHECK YOUR PROGRESS 2, Q2(I))

Evaluate $\int x^2 e^{-x} dx$.

Round 1 ($I = x^2, \int e^{-x} dx = -e^{-x}$): $= -x^2 e^{-x} + 2 \int x e^{-x} dx$.

Round 2: $\int x e^{-x} dx = -x e^{-x} - e^{-x}$ (from topic 11).

$$I = -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} + c = -e^{-x}(x^2 + 2x + 2) + c.$$

EXAMPLE (CHECK YOUR PROGRESS 2, Q2(II)): SUBSTITUTION, THEN BY PARTS

Evaluate $\int x^3 e^{x^2} dx$.

Write $x^3 e^{x^2} = x^2 \cdot e^{x^2} \cdot x$. Put $x^2 = t, x dx = dt/2$: $I = (1/2) \int t e^t dt$.

By parts ($I = t, II = e^t$): $\int t e^t dt = t e^t - e^t$.

$$I = (1/2)(x^2 e^{x^2} - e^{x^2}) + c = (1/2)(x^2 - 1) e^{x^2} + c.$$

EXAMPLE

Evaluate $\int (\log x)^2 dx$.

$$I = (\log x)^2, II = 1: = x(\log x)^2 - \int 2 \log x \cdot (1/x) \cdot x dx = x(\log x)^2 - 2 \int \log x dx.$$

$$\text{Using } \int \log x dx = x \log x - x: I = x(\log x)^2 - 2x \log x + 2x + c.$$

Exam tip: Check a repeated by-parts answer by differentiating: for $\int x^2 e^{-x} dx$, $d/dx [-e^{-x}(x^2 + 2x + 2)] = e^{-x}(x^2 + 2x + 2) - e^{-x}(2x + 2) = x^2 e^{-x} \checkmark$. It takes thirty seconds and catches sign errors.

Quick recap

- Integration is anti-differentiation: $\int f(x) dx = F(x) + c$ exactly when $F'(x) = f(x)$.
- Always add the constant of integration c ; check any answer by differentiating it.
- $\int x^n dx = x^{n+1}/(n+1) + c$ for $n \neq -1$; $\int (1/x) dx = \log |x| + c$.
- For a linear inside $(ax + b)$, integrate as usual and divide by a .
- $\int a^x dx = a^x/\log a + c$; $\int e^{ax+b} dx = e^{ax+b}/a + c$.
- Simplify first: $a^{\log_a f(x)} = f(x)$, $a^x b^x = (ab)^x$, split fractions, expand products.
- Integral of a sum is the sum of integrals; constants come out of the integral.
- Substitution: put the inner function $f(x) = t$ when $f'(x)$ also appears; convert fully to t , integrate, return to x .
- $\int f'(x)/f(x) dx = \log |f(x)| + c$; $\int [f(x)]^n f'(x) dx = [f(x)]^{n+1}/(n+1) + c$.
- For $\sqrt{x+a}$ with an x outside, put $x+a = t^2$.
- Partial fractions: proper fraction only (else divide first); Type 1 by cover-up, Type 2 needs a fraction for each power plus comparing coefficients.

- $\int dx/(x - b)^k$ for $k \geq 2$ uses the power rule, not log.
- By parts: $\int I \cdot II \, dx = I \int II \, dx - \int [I' \int II \, dx] \, dx$; choose I by ILATE.
- x^n times an exponential needs n rounds of by parts; $\int \log x \, dx = x \log x - x + c$.

Check yourself

Q. 1. Evaluate $\int (x^2 + 1/x^2)^2 \, dx$.

Expand: $x^4 + 2 + x^{-4}$. Integrate: $x^5/5 + 2x + x^{-3}/(-3) + c = x^5/5 + 2x - 1/(3x^3) + c$.

Q. 2. Evaluate $\int (\sqrt{x} + 1/\sqrt{x})^3 \, dx$.

$(p + q)^3 = p^3 + q^3 + 3pq(p + q)$ with $p = \sqrt{x}$, $q = 1/\sqrt{x}$, $pq = 1$: $x^{3/2} + x^{-3/2} + 3x^{1/2} + 3x^{-1/2}$.

Integral = $(2/5)x^{5/2} - 2x^{-1/2} + 2x^{3/2} + 6x^{1/2} + c = (2/5)x^{5/2} - 2/\sqrt{x} + 2x^{3/2} + 6\sqrt{x} + c$.

Q. 3. Evaluate $\int (x + 1/x)(x^3 + 1/x^3) \, dx$ and $\int (x^a + 1)(x^b + 1) \, dx$.

(a) Product = $x^4 + x^{-2} + x^2 + x^{-4}$. Integral = $x^5/5 - 1/x + x^3/3 - 1/(3x^3) + c$.

(b) Product = $x^{a+b} + x^a + x^b + 1$. Integral = $x^{a+b+1}/(a+b+1) + x^{a+1}/(a+1) + x^{b+1}/(b+1) + x + c$, provided a, b, $a+b \neq -1$.

Q. 4. Evaluate $\int (a^x + e^x a^x + x/a) \, dx$.

$e^x a^x = (ea)^x$, and $\log(ea) = \log e + \log a = 1 + \log a$.

Integral = $a^x/\log a + (ea)^x/(1 + \log a) + x^2/(2a) + c$.

Q. 5. Evaluate $\int (3^{2 \log_3 x} + 3^{x \log_3 a} + a^{a \log_a x} + a^{a \log_a a}) \, dx$.

Use $k \log m = \log m^k$ and $a^{\log_a f} = f$: the integrand is $x^2 + a^x + x^a + a^a$.

Integral = $x^3/3 + a^x/\log a + x^{a+1}/(a+1) + a^a x + c$.

Q. 6. Evaluate $\int (2x + 1)/(x^2 + x + 7)^5 \, dx$.

Put $x^2 + x + 7 = t$, $(2x + 1) \, dx = dt$. $I = \int t^{-5} \, dt = t^{-4}/(-4) + c = -1/[4(x^2 + x + 7)^4] + c$.

Q. 7. Evaluate $\int dx/[(1+x) \log(1+x)]$.

Put $\log(1+x) = t$, $dx/(1+x) = dt$. $I = \int dt/t = \log|t| + c = \log|\log(1+x)| + c$.

Q. 8. Evaluate $\int (3x+2)/[(x-1)(x-2)(x-3)] dx$.

Cover-up: $A(x=1) = 5/[(1)(-2)] = 5/2$; $B(x=2) = 8/[(1)(-1)] = -8$; $C(x=3) = 11/[(2)(1)] = 11/2$.

$I = (5/2) \log|x-1| - 8 \log|x-2| + (11/2) \log|x-3| + c$.

Q. 9. Evaluate $\int (x+5)/[(x-1)(x-2)(x-3)] dx$ and check one constant by substituting $x=0$.

From topic 8: $A=3$, $B=-7$, $C=4$, so $I = 3 \log|x-1| - 7 \log|x-2| + 4 \log|x-3| + c$.

Check at $x=0$: left side $5/[(1)(2)(3)] = 5/6$; right side $3/(-1) - 7/(-2) + 4/(-3) = -3 + 7/2 - 4/3 = -5/6$ ✓.

Q. 10. Evaluate $\int x^2 \log x dx$ and $\int x^3 e^x dx$ (exam style).

(a) ILATE: $I = \log x$, $II = x^2$. $= (x^3/3) \log x - \int (1/x)(x^3/3) dx = (x^3/3) \log x - x^3/9 + c$.

(b) $I = x^3$, $II = e^x$, three rounds: $x^3 e^x - 3 \int x^2 e^x dx$; $\int x^2 e^x dx = x^2 e^x - 2(x e^x - e^x)$.

So $I = e^x(x^3 - 3x^2 + 6x - 6) + c$. Check: derivative $= e^x(x^3 - 3x^2 + 6x - 6) + e^x(3x^2 - 6x + 6) = x^3 e^x$ ✓.

UNIT 14

Definite Integration

Indefinite integral-ന് ഒരു "+ c" ഉണ്ട്, അതിനാൽ ഒരു fixed value ഇല്ല. ഈ unit-ൽ limits a, b കൊടുത്ത് integral-ന് ഒരു കൃത്യമായ number കിട്ടുന്ന definite integral, അതിന്റെ area അർത്ഥം, substitution/partial fractions ഉപയോഗിച്ചുള്ള evaluation, calculation ചുരുക്കുന്ന properties എന്നിവ പഠിക്കുന്നു.

Meaning of the definite integral

Geometrical interpretation: area

Definite integrals of commonly used functions

Substitution: change the limits too

Partial fractions with limits

Properties P1 to P3

Property P4: even and odd functions

Properties P5 to P7

Using P3: modulus and piecewise functions

Using P4: odd functions give zero

Using P5: the "add I to itself" trick

Meaning of the definite integral

Definite integral: If $F'(x) = f(x)$ on $[a, b]$, then $\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$ (fundamental theorem of integral calculus).

$[a, b]$ -ൽ $F'(x) = f(x)$ ആണെങ്കിൽ $\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$ (fundamental theorem of integral calculus).

Limits of integration: a is the lower limit, b the upper limit, and $[a, b]$ the interval of integration.

a ആണ് lower limit, b ആണ് upper limit, $[a, b]$ ആണ് interval of integration.

If $d/dx (F(x)) = f(x)$, then $\int f(x) dx = F(x) + c$, which has no fixed value because c is arbitrary. The **definite integral** of $f(x)$ from $x = a$ to $x = b$ is written $\int_a^b f(x) dx$ and is defined by the **fundamental theorem of integral calculus**: find an antiderivative F, then compute $F(b) - F(a)$. Here a is the **lower limit**, b the **upper limit**, and $[a, b]$ the **interval of integration**. The answer is a single number that depends only on f, a and b.

മലയാളത്തിൽ

Indefinite integral ഒരു family of functions ആണ്, "+ c" കാരണം value ഉറപ്പില്ല.

Definite integral-ൽ നമ്മൾ ആദ്യം $F(x)$ കണ്ടുപിടിക്കും, പിന്നെ upper limit b ഇട്ട value-ൽ നിന്ന് lower limit a ഇട്ട value കുറയ്ക്കും. Bus ticket-ലെ fare പോലെ: എവിടെ കയറി (a), എവിടെ ഇറങ്ങി (b) എന്നത് മാത്രം മതി, വഴിയിലെ കാര്യങ്ങൾ ഒന്നും വേണ്ട. c ഇട്ടാലും $F(b) + c - (F(a) + c)$ എന്നതിൽ c cancel ആകും, അതിനാൽ definite integral-ൽ c എഴുതാറില്ല.

If $d/dx (F(x)) = f(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

$[F(x)]_a^b$ means: evaluate F at the top limit, evaluate at the bottom limit, subtract.

EXAMPLE (FROM THE UNIT)

Evaluate $\int_2^5 (x + 3) dx$.

Step 1: antiderivative $F(x) = x^2/2 + 3x$.

Step 2: $F(5) = 25/2 + 15 = 55/2$ and $F(2) = 4/2 + 6 = 8$.

Step 3: $\int_2^5 (x + 3) dx = 55/2 - 8 = 39/2$.

EXAMPLE (FROM THE UNIT): WHY C IS DROPPED

Evaluate $\int_1^2 (x^2 + x + 6) dx$.

With c : $[x^3/3 + x^2/2 + 6x + c]_1^2 = (8/3 + 2 + 12 + c) - (1/3 + 1/2 + 6 + c)$. The two c 's cancel.

So we simply write $[x^3/3 + x^2/2 + 6x]_1^2 = (8/3 + 2 + 12) - (1/3 + 1/2 + 6) = 50/3 - 41/6 = 100/6 - 41/6 = 59/6$.

Exam tip: Always write "top value minus bottom value", never the other way round. The unit mentions that definite integrals are classed as proper or improper (it says first-kind improper integrals are used in later courses); every integral in this unit is proper: finite limits and f defined on all of $[a, b]$.

Geometrical interpretation: area

Definition: If $f(x) \geq 0$ on $[a, b]$, $\int_a^b f(x) dx$ is the area bounded by $y = f(x)$, the x-axis and the lines $x = a$, $x = b$; in general it is net signed area.

$[a, b]$ -ൽ $f(x) \geq 0$ ആണെങ്കിൽ, $y = f(x)$, x-axis, $x = a$, $x = b$ എന്നീ lines എന്നിവയാൽ ചുറ്റപ്പെട്ട area ആണ് $\int_a^b f(x) dx$; പൊതുവെ ഇത് net signed area ആണ്.

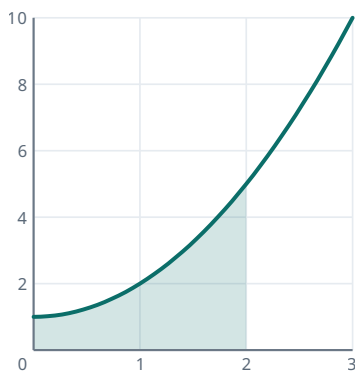
When $f(x) \geq 0$ on $[a, b]$, $\int_a^b f(x) dx$ is the **area of the region bounded by the curve $y = f(x)$, the x-axis and the lines $x = a$ and $x = b$** . When the curve lies below the x-axis the integral comes out negative, so in general the definite integral is a net signed area: area above the axis counts plus, area below counts minus.

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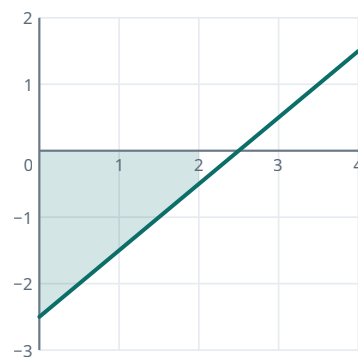
Curve $y = f(x)$, x-axis, $x = a$, $x = b$ എന്നീ നാല് അതിരുകൾക്കുള്ളിലെ ഭാഗത്തിന്റെ area ആണ് definite integral. ഒരു പഠനത്തിന്റെ അളവ് എടുക്കുന്നത് പോലെ: ഒരു വശത്ത് വേലി (curve), താഴെ റോഡ് (x-axis), രണ്ട് വശത്ത് അതിർത്തികൾ ($x = a$, $x = b$). Curve x-axis-ന് താഴെ ആണെങ്കിൽ ആ area negative ആയി കണക്കാക്കും, അതുകൊണ്ട് answer negative ആകും.

$f(x) \geq 0$ on $[a, b] \Rightarrow \int_a^b f(x) dx = \text{area under the curve}$

$f(x) \leq 0$ on $[a, b] \Rightarrow \int_a^b f(x) dx = -(\text{area between curve and x-axis})$



Shaded area under $y = x^2 + 1$ from $x = 0$ to $x = 2$ equals $\int_0^2 (x^2 + 1) dx = 14/3$.



$y = x - 5/2$ lies below the x-axis on $[0, 2]$, so $\int_0^2 (x - 5/2) dx = -3$ (area 3, counted negative).

EXAMPLE

Find the area under $y = x^2 + 1$ between $x = 0$ and $x = 2$ (first graph).

The curve is above the x -axis, so area = $\int_0^2 (x^2 + 1) dx = [x^3/3 + x]_0^2 = (8/3 + 2) - 0 = 14/3$ square units (≈ 4.67). This agrees with the picture: the region is a bit bigger than a 2×2 square.

EXAMPLE (FROM CHECK YOUR PROGRESS 1)

Evaluate $\int_0^2 (x - 5/2) dx$ and explain the sign (second graph).

$$[x^2/2 - 5x/2]_0^2 = (2 - 5) - 0 = -3.$$

On $[0, 2]$, $y = x - 5/2$ is negative, so the region (a trapezium with parallel sides $5/2$ and $1/2$, width 2, area $(5/2 + 1/2)/2 \times 2 = 3$) lies below the axis and counts as -3 .

Exam tip: If a question asks for an area, the answer must be positive; take the absolute value of each part that lies below the x -axis. If it asks only to evaluate the integral, a negative answer is fine.

Definite integrals of commonly used functions

Definition: Take the standard indefinite integral $F(x)$ of $f(x)$ (without c) and compute $F(b) - F(a)$, e.g. $\int_a^b x^n dx = [x^{n+1}/(n+1)]_a^b$, $n \neq -1$.

$f(x)$ -ന്റെ standard indefinite integral $F(x)$ (c ഇല്ലാതെ) എടുത്ത് $F(b) - F(a)$ കണക്കാക്കുക; ഉദാ. $\int_a^b x^n dx = [x^{n+1}/(n+1)]_a^b$, $n \neq -1$.

Use the standard indefinite-integral formulas from Unit 13, then put in the limits. Here $\log$ means natural $\log$ (base e), and $\log 1 = 0$, $e^0 = 1$ are used often.

മലയാളത്തിൽ

ഇവിടെ പുതിയ formula ഒന്നുമില്ല. Unit 13-ലെ formula ഉപയോഗിച്ച് $F(x)$ എഴുതുക, പിന്നെ b ഇട്ട് a ഇട്ട് കുറയ്ക്കുക. Mistakes മിക്കവാറും വരുന്നത് arithmetic-ൽ ആണ്, അതുകൊണ്ട് $F(b)$, $F(a)$ വേറെ വേറെ bracket-ൽ എഴുതുന്നത് നല്ല ശീലമാണ്.

$$\int_a^b x^n dx = [x^{n+1}/(n+1)]_a^b, n \neq -1$$

$$\int_a^b (px + q)^n dx = [(px + q)^{n+1}/(p(n+1))]_a^b$$

$$\int_a^b 1/x dx = [\log |x|]_a^b, \int_a^b 1/(px + q) dx = [(1/p) \log |px + q|]_a^b$$

$$\int_a^b e^{px + q} dx = [e^{px + q}/p]_a^b, \int_a^b k^x dx = [k^x/\log k]_a^b$$

EXAMPLE 1 (FROM THE UNIT): POWERS AND POLYNOMIALS

$$(i) \int_2^6 8 \, dx = [8x]_2^6 = 48 - 16 = 32.$$

$$(ii) \int_0^2 (x^2 + 1) \, dx = [x^3/3 + x]_0^2 = 8/3 + 2 - 0 = 14/3.$$

$$(iii) \int_2^3 (3x + 4) \, dx = [3x^2/2 + 4x]_2^3 = (27/2 + 12) - (6 + 8) = 27/2 - 2 = 23/2.$$

$$(iv) \int_2^5 4x^3 \, dx = [x^4]_2^5 = 625 - 16 = 609.$$

$$(v) \int_2^3 x^a \, dx = [x^{a+1}/(a+1)]_2^3 = (3^{a+1} - 2^{a+1})/(a+1), a \neq -1.$$

$$(vii) \int_1^2 (x+1)(x^2-1) \, dx = \int_1^2 (x^3 + x^2 - x - 1) \, dx = [x^4/4 + x^3/3 - x^2/2 - x]_1^2$$

$$\text{At } 2: 4 + 8/3 - 2 - 2 = 8/3. \text{ At } 1: 1/4 + 1/3 - 1/2 - 1 = (3 + 4 - 6 - 12)/12 = -11/12.$$

$$\text{Answer: } 8/3 - (-11/12) = 32/12 + 11/12 = 43/12.$$

$$(viii) \int_2^3 (x^3 + 5)/x^2 \, dx = \int_2^3 (x + 5x^{-2}) \, dx = [x^2/2 - 5/x]_2^3 = (9/2 - 5/3) - (2 - 5/2) = 17/6 + 1/2 = 10/3.$$

EXAMPLE 1 (FROM THE UNIT): (PX + Q)^N, LOG AND EXPONENTIAL FORMS

$$(vi) \int_2^4 \sqrt{(18x-24)} \, dx = [(18x-24)^{3/2}/(18 \times 3/2)]_2^4 = (1/27)[48^{3/2} - 12^{3/2}]$$

$$48^{3/2} = 48\sqrt{48} = 48 \times 4\sqrt{3} = 192\sqrt{3} \text{ and } 12^{3/2} = 12 \times 2\sqrt{3} = 24\sqrt{3}, \text{ so the answer is } (1/27)(168\sqrt{3}) = 56\sqrt{3}/9.$$

$$(ix) \int_2^5 1/(2x-3) \, dx = [(1/2) \log |2x-3|]_2^5 = (1/2)(\log 7 - \log 1) = (1/2) \log 7.$$

$$(x) \int_2^3 4/x \, dx = [4 \log |x|]_2^3 = 4(\log 3 - \log 2) = 4 \log (3/2).$$

$$(xi) \int_1^4 2^x \, dx = [2^x/\log 2]_1^4 = (16 - 2)/\log 2 = 14/\log 2.$$

$$(xii) \int_0^2 e^{3x} \, dx = [e^{3x}/3]_0^2 = (e^6 - 1)/3.$$

$$(xiii) \int_0^2 3^{4x+1} \, dx = [3^{4x+1}/(4 \log 3)]_0^2 = (3^9 - 3)/(4 \log 3) = 19680/(4 \log 3) = 4920/\log 3.$$

$$(xiv) \int_1^3 e^{2x+5} \, dx = [e^{2x+5}/2]_1^3 = (e^{11} - e^7)/2.$$

Exam tip: In $\int (px + q)^n \, dx$ and $\int e^{px+q} \, dx$ do not forget to divide by p . In (vi) the divisor is $18 \times 3/2 = 27$, not just $3/2$.

Substitution: change the limits too

Definition: On substituting $x = g(t)$, the limits must also change to t -values: $\int_a^b f(x) \, dx = \int_\alpha^\beta f(g(t)) g'(t) \, dt$, where $x = a \Rightarrow t = \alpha$ and $x = b \Rightarrow t = \beta$.

$x = g(t)$ എന്നു substitute ചെയ്യുമ്പോൾ limits-ഉം t -values ആക്കണം: $\int_a^b f(x) \, dx = \int_\alpha^\beta f(g(t)) g'(t) \, dt$; ഇവിടെ $x = a \Rightarrow t = \alpha$, $x = b \Rightarrow t = \beta$.

When you substitute $x \rightarrow t$, the limits a and b are values of x , so they must be converted to values of t . After that you never need to go back to x : just evaluate in t .

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Substitution ചെയ്യുമ്പോൾ dx മാത്രമല്ല, limits-ഉം മാറ്റണം. $x = 1$ ആകുമ്പോൾ t എത്ര, $x = 6$ ആകുമ്പോൾ t എത്ര എന്ന് ആദ്യം കണ്ടുപിടിച്ച് എഴുതുക. Rupees-ൽ നിന്ന് dollar-ലേക്ക് മാറ്റിയാൽ എല്ലാ തുകയും dollar-ൽ തന്നെ എഴുതണം, പക്ഷെ rupees പക്ഷെ dollar എഴുതരുത്, അതുപോലെ തന്നെ.

Put $x = g(t)$: $dx = g'(t) dt$; $x = a \Rightarrow t = \alpha$, $x = b \Rightarrow t = \beta$

$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f(g(t)) g'(t) dt$$

EXAMPLE 2(I) (FROM THE UNIT)

Evaluate $I = \int_1^6 x\sqrt{x+3} dx$.

Put $\sqrt{x+3} = t$, so $x+3 = t^2$, $x = t^2 - 3$ and $dx = 2t dt$.

Limits: $x = 1 \Rightarrow t = 2$; $x = 6 \Rightarrow t = 3$.

$$\begin{aligned} I &= \int_2^3 (t^2 - 3) \cdot t \cdot 2t dt = 2 \int_2^3 (t^4 - 3t^2) dt = 2[t^5/5 - t^3]_2^3 \\ &= 2[(243/5 - 27) - (32/5 - 8)] = 2[211/5 - 19] = 2 \times 116/5 = \mathbf{232/5}. \end{aligned}$$

EXAMPLE 2(II) (FROM THE UNIT)

Evaluate $I = \int_0^1 (2x+3)/(x^2+3x+5) dx$.

Put $x^2+3x+5 = t$, so $(2x+3) dx = dt$. Limits: $x = 0 \Rightarrow t = 5$; $x = 1 \Rightarrow t = 9$.

$$I = \int_5^9 dt/t = [\log t]_5^9 = \log 9 - \log 5 = \mathbf{\log (9/5)}.$$

EXAMPLE (FROM CHECK YOUR PROGRESS 1)

Evaluate $I = \int_2^5 x/(x^2+3) dx$.

Put $x^2 = t$, $2x dx = dt$, so $x dx = dt/2$. Limits: $x = 2 \Rightarrow t = 4$; $x = 5 \Rightarrow t = 25$.

$$I = (1/2) \int_4^{25} dt/(t+3) = (1/2)[\log(t+3)]_4^{25} = (1/2)(\log 28 - \log 7) = (1/2) \log 4 = (1/2)(2 \log 2) = \mathbf{\log 2}.$$

Exam tip: The most common mistake is changing the variable but keeping the old limits (e.g. writing $\int_1^6 \dots dt$). Write the "when $x = \dots$, $t = \dots$ " line every time; IGNOU gives marks for it.

Partial fractions with limits

Definition: Split the rational integrand into partial fractions, integrate each piece to a log or a power, then substitute the limits and combine the logs.

Rational integrand-നെ partial fractions ആക്കി, ഓരോ ഭാഗവും integrate ചെയ്ത് log അല്ലെങ്കിൽ power ആക്കി, തുടർന്ന് limits substitute ചെയ്ത് logs ഒന്നിപ്പിക്കുക.

For a rational function whose denominator factorises, split it into partial fractions (as in Unit 13), integrate each piece to a log or a power, and then put in the limits. Combine logs at the end using $\log m - \log n = \log (m/n)$ and $\log m^n = n \log m$.

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വലിയ fraction-നെ ചെറിയ കഷ്ണങ്ങളാക്കി (partial fractions) ഓരോന്നും വേറെ integrate ചെയ്യുക. $A/(x - k)$ എന്ന കഷ്ണം $A \log |x - k|$ ആകും, $A/(x - k)^2$ എന്നത് $-A/(x - k)$ ആകും. Limits ഇടുമ്പോൾ $|x - k|$ എന്ന modulus ശ്രദ്ധിക്കുക, log-ന്റെ ഉള്ളിൽ negative number വരരുത്.

$$\int A/(x - k) dx = A \log |x - k|$$

$$\int A/(x - k)^2 dx = -A/(x - k)$$

$$\int A/(x - k)^3 dx = -A/(2(x - k)^2)$$

EXAMPLE 2(III) (FROM THE UNIT)

Evaluate $I = \int_0^2 (2x + 7)/((x - 3)(x + 1)(x - 4)) dx$.

Write $(2x + 7)/((x - 3)(x + 1)(x - 4)) = A/(x - 3) + B/(x + 1) + C/(x - 4)$. Cover-up rule:

$$A = (6 + 7)/((4)(-1)) = -13/4, B = (-2 + 7)/((-4)(-5)) = 1/4, C = (8 + 7)/((1)(5)) = 3.$$

$$I = -(13/4)[\log |x - 3|]_0^2 + (1/4)[\log |x + 1|]_0^2 + 3[\log |x - 4|]_0^2$$

$$= -(13/4)(\log 1 - \log 3) + (1/4)(\log 3 - \log 1) + 3(\log 2 - \log 4)$$

$$= (13/4) \log 3 + (1/4) \log 3 + 3(\log 2 - 2 \log 2) = (14/4) \log 3 - 3 \log 2 = (7/2) \log 3 - 3 \log 2 (\approx 1.766).$$

EXAMPLE 2(IV) (FROM THE UNIT)

Evaluate $I = \int_0^3 (x - 5)/((x + 1)(x + 2)^2) dx$.

Let $(x - 5)/((x + 1)(x + 2)^2) = A/(x + 1) + B/(x + 2) + C/(x + 2)^2$, so $x - 5 = A(x + 2)^2 + B(x + 1)(x + 2) + C(x + 1)$.

$x = -1$: $-6 = A$. $x = -2$: $-7 = -C$, so $C = 7$. Coefficient of x^2 : $0 = A + B$, so $B = 6$.

$$\begin{aligned} I &= -6[\log |x + 1|]_0^3 + 6[\log |x + 2|]_0^3 + 7[-1/(x + 2)]_0^3 \\ &= -6 \log 4 + 6(\log 5 - \log 2) + 7(-1/5 + 1/2) = -12 \log 2 + 6 \log 5 - 6 \log 2 + 21/10 \\ &= 6 \log 5 - 18 \log 2 + 21/10 (\approx -0.720). \end{aligned}$$

Exam tip: Check each constant by putting a value like $x = 0$ into both sides of the partial-fraction identity before integrating. One wrong constant spoils the whole answer (see Q5 of "Check yourself" for a slip in the unit's own answer key).

Properties P1 to P3

P1: $\int_a^b f(x) dx = \int_a^b f(t) dt$ (the variable's letter does not matter).

$\int_a^b f(x) dx = \int_a^b f(t) dt$ (variable-ന് ഏത് അക്ഷരം ഉപയോഗിച്ചാലും value മാറില്ല).

P2: $\int_a^b f(x) dx = -\int_b^a f(x) dx$ (interchanging the limits changes the sign).

$\int_a^b f(x) dx = -\int_b^a f(x) dx$ (limits പരസ്പരം മാറ്റിയാൽ sign മാറും).

P3: $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, $a < c < b$.

$a < c < b$ ആണെങ്കിൽ $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$.

P1 says the letter used for the variable does not matter. **P2** says swapping the limits changes the sign. **P3** says you can break the interval at any point c (or several points) and add the pieces. All three follow directly from $\int_a^b f = F(b) - F(a)$.

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P1: variable-ന്റെ പേര് x ആയാലും t ആയാലും answer ഒന്നുതന്നെ, പേര് മാറ്റിയാൽ ആള് മാറില്ല. **P2:** limits തിരിച്ചിട്ടാൽ sign മാറും, തൃശ്ശൂർ മുതൽ എറണാകുളം വരെ +75 km എങ്കിൽ തിരിച്ച് -75 km. **P3:** ഒരു യാത്രയെ രണ്ട് stage ആയി മുറിച്ച് കൂട്ടിയാലും ആകെ ദൂരം അതുതന്നെ; modulus, piecewise functions ഒക്കെ ചെയ്യാൻ ഇതാണ് ഉപയോഗിക്കുന്നത്.

P1: $\int_a^b f(x) dx = \int_a^b f(t) dt$ (change of variable)

P2: $\int_a^b f(x) dx = -\int_b^a f(x) dx$ (interchange of limits)

P3: $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, $a < c < b$

In general: $\int_a^b f(x) dx = \int_a^{c_1} f(x) dx + \int_{c_1}^{c_2} f(x) dx + \dots + \int_{c_n}^b f(x) dx$, $a < c_1 < c_2 < \dots < c_n < b$

PROOFS (AS IN THE UNIT)

Let $\int f(x) dx = F(x)$.

P1: both sides equal $F(b) - F(a)$; only the name of the variable differs.

P2: $\int_b^a f(x) dx = F(a) - F(b) = -(F(b) - F(a)) = -\int_a^b f(x) dx$.

P3: $\int_a^c f + \int_c^b f = (F(c) - F(a)) + (F(b) - F(c)) = F(b) - F(a) = \int_a^b f$.

EXAMPLE

Check P2 and P3 with $f(x) = 2x$.

P2: $\int_1^3 2x dx = [x^2]_1^3 = 8$, and $\int_3^1 2x dx = 1 - 9 = -8$. ✓

P3 with $c = 2$: $\int_1^2 2x dx + \int_2^3 2x dx = (4 - 1) + (9 - 4) = 3 + 5 = 8$. ✓

Property P4: even and odd functions

Even function: $f(-x) = f(x)$ for all x ; its graph is symmetric about the y-axis.

എല്ലാ x -നും $f(-x) = f(x)$; graph y-axis-നെ ആധാരമാക്കി symmetric ആണ്.

Odd function: $f(-x) = -f(x)$ for all x ; its graph is symmetric about the origin.

എല്ലാ x -നും $f(-x) = -f(x)$; graph origin-നെ ആധാരമാക്കി symmetric ആണ്.

P4: $\int_{-a}^a f(x) dx = 2\int_0^a f(x) dx$ if f is even, and $= 0$ if f is odd.

f even ആണെങ്കിൽ $\int_{-a}^a f(x) dx = 2\int_0^a f(x) dx$, f odd ആണെങ്കിൽ ഇത് 0.

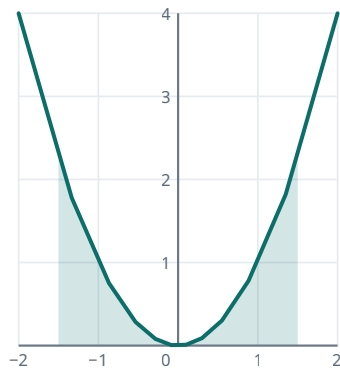
A function is **even** if $f(-x) = f(x)$ (graph symmetric about the y-axis, e.g. x^2 , $|x|$, $e^{2|x|}$) and **odd** if $f(-x) = -f(x)$ (graph symmetric about the origin, e.g. x^3 , x , $x|x|$). On a symmetric interval $[-a, a]$, an even function gives twice the integral over $[0, a]$, and an odd function gives 0 because the area below the axis exactly cancels the area above.

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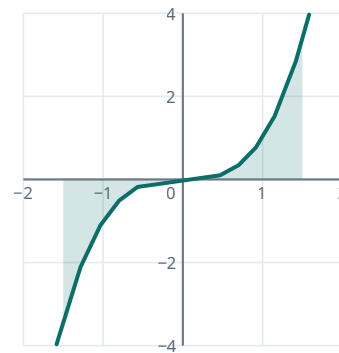
Limits $-a$ മുതൽ a വരെ ആണെങ്കിൽ ആദ്യം $f(-x)$ കണ്ടുപിടിക്കുക. $f(-x) = f(x)$ ആണെങ്കിൽ even, ഇടത് പകുതിയും വലത് പകുതിയും കണ്ണാടിയിലെ പോലെ ഒരുപോലെ, അതിനാൽ $2 \times$ (0 മുതൽ a വരെ). $f(-x) = -f(x)$ ആണെങ്കിൽ odd, ഒരു വശത്തെ + area മറുവശത്തെ - area-യുമായി cancel ആകും, answer നേരെ 0. ഒരു calculation-ഉം വേണ്ട!

P4: $\int_{-a}^a f(x) dx = 2\int_0^a f(x) dx$, if f is even ($f(-x) = f(x)$)

$\int_{-a}^a f(x) dx = 0$, if f is odd ($f(-x) = -f(x)$)



Even: $y = x^2$. The two halves are mirror images, so $\int_{-a}^a x^2 dx = 2\int_0^a x^2 dx$.



Odd: $y = x^3$. The shaded piece below the axis cancels the piece above, so $\int_{-a}^a x^3 dx = 0$.

PROOF (AS IN THE UNIT)

By P3, $\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$.

In the first integral put $x = -t$, $dx = -dt$; $x = -a \Rightarrow t = a$, $x = 0 \Rightarrow t = 0$. It becomes $-\int_a^0 f(-t) dt = \int_0^a f(-x) dx$ (by P2 and P1).

So $\int_{-a}^a f(x) dx = \int_0^a f(-x) dx + \int_0^a f(x) dx$. If f is even this is $2\int_0^a f(x) dx$; if f is odd it is $-\int_0^a f + \int_0^a f = 0$.

EXAMPLE

Evaluate $\int_{-2}^2 (x^2 + x^3) dx$.

Split: x^2 is even and x^3 is odd.

$\int_{-2}^2 x^2 dx = 2\int_0^2 x^2 dx = 2 \times 8/3 = 16/3$, and $\int_{-2}^2 x^3 dx = 0$.

Answer: **16/3**.

Exam tip: P4 works only when the limits are $-a$ and a . Always show the line " $f(-x) = \dots = -f(x)$, so f is odd" to earn the marks.

Properties P5 to P7

P5: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$; in particular $\int_0^a f(x) dx = \int_0^a f(a - x) dx$.

$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$; പ്രത്യേകിച്ച് $\int_0^a f(x) dx = \int_0^a f(a - x) dx$.

P6: $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx$.

$\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx$.

P7: $\int_0^{2a} f(x) dx = 2\int_0^a f(x) dx$ if $f(2a - x) = f(x)$, and $= 0$ if $f(2a - x) = -f(x)$.

$f(2a - x) = f(x)$ ആണെങ്കിൽ $\int_0^{2a} f(x) dx = 2\int_0^a f(x) dx$; $f(2a - x) = -f(x)$ ആണെങ്കിൽ ഇത് 0.

P5 lets you replace x by $a + b - x$ inside the integral without changing its value (for limits 0 and a this is $a - x$). **P6** splits $[0, 2a]$ into two halves and folds the second half onto the first. **P7** follows from P6: if $f(2a - x) = f(x)$ the two halves are equal, and if $f(2a - x) = -f(x)$ they cancel.

മലയാളത്തിൽ

P5 ഒരു "മറിച്ചിടൽ" ആണ്: interval-നെ തലകീഴായി വായിച്ചാലും (a -യിൽ നിന്ന് b -യിലേക്ക് പോകുന്നതിന് പകരം b -യിൽ നിന്ന് a -യിലേക്ക്) area മാറില്ല. x -ന് പകരം $a + b - x$ ഇട്ടാൽ integrand ചിലപ്പോൾ വളരെ ലളിതമാകും. P6, P7 എന്നിവ 0 മുതൽ $2a$ വരെയുള്ള integral-നെ രണ്ട് പകുതിയായി മടക്കി നോക്കുന്നു, ഒരു paper നട്ടുക്ക് മടക്കുന്നത് പോലെ.

P5: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$; in particular $\int_0^a f(x) dx = \int_0^a f(a - x) dx$

P6: $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx$

P7: $\int_0^{2a} f(x) dx = 2\int_0^a f(x) dx$, if $f(2a - x) = f(x)$

$\int_0^{2a} f(x) dx = 0$, if $f(2a - x) = -f(x)$

PROOFS (AS IN THE UNIT)

P5: in $I = \int_a^b f(x) dx$ put $x = a + b - t$, $dx = -dt$; $x = a \Rightarrow t = b$, $x = b \Rightarrow t = a$. Then $I = -\int_b^a f(a + b - t) dt = \int_a^b f(a + b - t) dt = \int_a^b f(a + b - x) dx$ (P2, P1).

P6: by P3, $\int_0^{2a} f = \int_0^a f + \int_a^{2a} f$. In the second, put $x = 2a - t$: $x = a \Rightarrow t = a$, $x = 2a \Rightarrow t = 0$, $dx = -dt$, so it becomes $\int_0^a f(2a - x) dx$.

P7: put $f(2a - x) = f(x)$ or $-f(x)$ into P6.

EXAMPLE (FROM CHECK YOUR PROGRESS 2)

Evaluate $I = \int_0^2 x(2 - x)^{11/2} dx$.

By P5 with $a = 0$, $b = 2$, replace x by $2 - x$: $I = \int_0^2 (2 - x) x^{11/2} dx = \int_0^2 (2x^{11/2} - x^{13/2}) dx$.

$$= [2x^{13/2}/(13/2) - x^{15/2}/(15/2)]_0^2 = (4/13) 2^{13/2} - (2/15) 2^{15/2}.$$

$$2^{13/2} = 64\sqrt{2} \text{ and } 2^{15/2} = 128\sqrt{2}, \text{ so } I = 256\sqrt{2}/13 - 256\sqrt{2}/15 = 256\sqrt{2} \times 2/195 = \mathbf{512\sqrt{2}/195} (\approx 3.713).$$

EXAMPLE (P7)

Evaluate $\int_0^2 (x - 1)^3 dx$ without expanding.

$$\text{Here } 2a = 2, a = 1. f(2 - x) = (2 - x - 1)^3 = (1 - x)^3 = -(x - 1)^3 = -f(x).$$

By P7 the integral is **0**. (Check: $[(x - 1)^4/4]_0^2 = 1/4 - 1/4 = 0$.)

Exam tip: Use P5 when the integrand has a factor like $(2 - x)^n$ with an awkward power: moving the power onto x makes it easy to integrate.

Using P3: modulus and piecewise functions

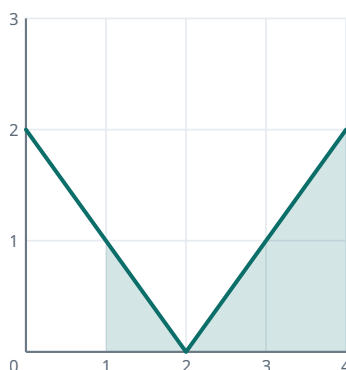
To integrate $|g(x)|$, find where $g(x) = 0$, split the interval there using P3, and on each piece replace $|g(x)|$ by $g(x)$ (where $g \geq 0$) or $-g(x)$ (where $g \leq 0$). A piecewise (split) function is handled the same way: split at the points where the formula changes.

മലയാളത്തിൽ

Modulus കണ്ടാൽ ഉടനെ "ഇത് എവിടെ zero ആകുന്നു?" എന്ന് ചോദിക്കുക. $|x - 2|$ ആണെങ്കിൽ $x = 2$ -ൽ interval മുറിക്കുക. 2-ന് മുമ്പ് $x - 2$ negative, അതിനാൽ $|x - 2| = -(x - 2)$; 2-ന് ശേഷം positive, അതിനാൽ $|x - 2| = x - 2$. Graph നോക്കിയാൽ രണ്ട് triangle-കൾ, അവയുടെ area കൂട്ടിയാൽ മതി.

$|g(x)| = g(x)$ where $g(x) \geq 0$; $|g(x)| = -g(x)$ where $g(x) \leq 0$

Split points = zeros of $g(x)$ (or the break points of a piecewise f) inside $[a, b]$



$y = |x - 2|$ on $[1, 4]$: two triangles of area $1/2$ and 2 , total $5/2$.

EXAMPLE 3(I) (FROM THE UNIT)

Evaluate $\int_1^4 |x - 2| dx$.

$x - 2 = 0$ at $x = 2$. For $1 \leq x \leq 2$, $|x - 2| = -(x - 2)$; for $2 \leq x \leq 4$, $|x - 2| = x - 2$.

$$I = -[x^2/2 - 2x]_1^2 + [x^2/2 - 2x]_2^4 = -[(2 - 4) - (1/2 - 2)] + [(8 - 8) - (2 - 4)] = -[-2 + 3/2] + [2] = 1/2 + 2 = 5/2.$$

The graph above confirms this: triangles of area $(1 \times 1)/2 = 1/2$ and $(2 \times 2)/2 = 2$.

EXAMPLE 3(II) (FROM THE UNIT)

Evaluate $\int_0^2 |2x - 3| dx$.

$$2x - 3 = 0 \text{ at } x = 3/2. I = \int_0^{3/2} -(2x - 3) dx + \int_{3/2}^2 (2x - 3) dx = -[x^2 - 3x]_0^{3/2} + [x^2 - 3x]_{3/2}^2 \\ = -(9/4 - 9/2) + [(4 - 6) - (9/4 - 9/2)] = 9/4 + (-2 + 9/4) = 9/4 + 1/4 = 5/2.$$

EXAMPLE 3(III) (FROM THE UNIT): PIECEWISE FUNCTION

Evaluate $\int_0^3 f(x) dx$, where $f(x) = x + 1$ for $0 \leq x \leq 1$ and $f(x) = 2x + 3$ for $1 \leq x \leq 3$.

$$I = \int_0^1 (x + 1) dx + \int_1^3 (2x + 3) dx = [x^2/2 + x]_0^1 + [x^2 + 3x]_1^3 = 3/2 + (18 - 4) = 31/2.$$

EXAMPLE 3(VI) (FROM THE UNIT)

Evaluate $\int_{-3}^3 e^{2|x|} dx$.

For $-3 \leq x \leq 0$, $|x| = -x$; for $0 \leq x \leq 3$, $|x| = x$.

$$I = \int_{-3}^0 e^{-2x} dx + \int_0^3 e^{2x} dx = [e^{-2x}/(-2)]_{-3}^0 + [e^{2x}/2]_0^3 = -(1/2)(1 - e^6) + (1/2)(e^6 - 1) = e^6 - 1.$$

Shortcut: $e^{2|x|}$ is even, so by P4, $I = 2\int_0^3 e^{2x} dx = 2 \times (e^6 - 1)/2 = e^6 - 1$.

Exam tip: The integral of a modulus can never be negative. If your answer for $\int |g(x)| dx$ comes out negative or zero (with g not identically 0), a sign on one piece is wrong.

Using P4: odd functions give zero

When the limits are $-a$ and a , test $f(-x)$ first. If the integrand is odd, the answer is 0 at once, even when the integrand looks impossible to integrate directly.

മലയാളത്തിൽ

Question-ൽ -99 മുതൽ 99 വരെ, -2 മുതൽ 2 വരെ എന്നിങ്ങനെ symmetric limits കണ്ടാൽ അത് ഒരു സൂചനയാണ്: "odd function ആണോ എന്ന് നോക്കൂ" എന്ന്. Odd ആണെന്ന് തെളിയിച്ചാൽ answer 0, വലിയ calculation ഒന്നും ചെയ്യേണ്ട.

Test: replace x by $-x$; if $f(-x) = -f(x)$, then $\int_{-a}^a f(x) dx = 0$

Useful facts: odd $\times$ even = odd; odd + odd = odd; $\log(1/m) = -\log m$

EXAMPLE 3(IV) (FROM THE UNIT)

Evaluate $\int_{-99}^{99} (x^3 + x + e^x - e^{-x}) dx$.

$f(-x) = -x^3 - x + e^{-x} - e^x = -(x^3 + x + e^x - e^{-x}) = -f(x)$, so f is odd.

By P4, the integral = 0.

EXAMPLE 3(V) (FROM THE UNIT)

Evaluate $\int_{-2}^2 \log((5-x)/(5+x)) dx$.

$f(-x) = \log((5+x)/(5-x)) = \log[((5-x)/(5+x))^{-1}] = -\log((5-x)/(5+x)) = -f(x)$, so f is odd.

By P4, the integral = 0.

EXAMPLE (FROM CHECK YOUR PROGRESS 2)

Evaluate $\int_{-3}^3 4x|x| dx$.

$f(-x) = 4(-x)|-x| = -4x|x| = -f(x)$ (since $|-x| = |x|$), so f is odd and the integral is 0.

Exam tip: In Example 3(iv) the unit's last line prints the integrand as $x^3 + x + e^x + e^{-x}$; that is a typo. It must be $e^x - e^{-x}$ (with a plus sign the function would not be odd, and the integral would be $2(e^{99} - e^{-99})$, not 0).

Using P5: the "add I to itself" trick

For integrals of the form $\int_a^b f(x)/(f(x) + f(a + b - x)) dx$: call the integral I , apply P5 to get a second form of I , then add the two. The numerators add up to the denominator, so $2I = \int_a^b 1 dx = b - a$.

മലയാളത്തിൽ

ഈ type question IGNOU exam-ൽ വീണ്ടും വീണ്ടും വരുന്നതാണ്. എഴുതുക, P5

ഉപയോഗിച്ച് x -ന് പകരം $a + b - x$ ഇട്ട് രണ്ടാമത്തെ I എഴുതുക, രണ്ടും കൂട്ടുക.

Numerator-ഉം denominator-ഉം ഒന്നാകും, $2I = b - a$, അതിനാൽ $I = (b - a)/2$. ഒരു നോട്ടത്തിൽ answer: limits-ന്റെ വ്യത്യാസത്തിന്റെ പകുതി.

$$\int_a^b f(x)/(f(x) + f(a + b - x)) dx = (b - a)/2$$

EXAMPLE 3(VII) (FROM THE UNIT): THE GENERAL RESULT

Show that $I = \int_a^b f(x)/(f(x) + f(a + b - x)) dx = (b - a)/2$.

By P5, $I = \int_a^b f(a + b - x)/(f(a + b - x) + f(x)) dx$ (because $a + b - (a + b - x) = x$).

Adding the two forms: $2I = \int_a^b (f(x) + f(a + b - x))/(f(x) + f(a + b - x)) dx = \int_a^b 1 dx = b - a$. So $I = (b - a)/2$.

EXAMPLE 3(VIII) (FROM THE UNIT)

Evaluate $I = \int_2^3 \sqrt{x}/(\sqrt{x} + \sqrt{5 - x}) dx$.

Here $a + b = 5$. By P5, $I = \int_2^3 \sqrt{5 - x}/(\sqrt{5 - x} + \sqrt{x}) dx$.

Adding: $2I = \int_2^3 1 dx = 3 - 2 = 1$, so $I = 1/2$.

EXAMPLE 3(IX) (FROM THE UNIT)

Evaluate $I = \int_2^7 \sqrt[5]{x + 1}/(\sqrt[5]{x + 1} + \sqrt[5]{10 - x}) dx$ ($\sqrt[5]{} =$ fifth root).

$a + b = 9$. Replacing x by $9 - x$: $x + 1 \rightarrow 10 - x$ and $10 - x \rightarrow x + 1$, so $I = \int_2^7 \sqrt[5]{10 - x}/(\sqrt[5]{10 - x} + \sqrt[5]{x + 1}) dx$.

Adding: $2I = \int_2^7 1 dx = 5$, so $I = 5/2$.

Exam tip: Before using the trick, check that replacing x by $a + b - x$ really swaps the two terms of the denominator. In 3(ix), $9 - x + 1 = 10 - x$ and $10 - (9 - x) = x + 1$, so it works.

Quick recap

- $\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$, where $F' = f$ (fundamental theorem of integral calculus).

- The constant c cancels, so it is not written in definite integrals.
- For $f \geq 0$, the definite integral is the area bounded by $y = f(x)$, the x -axis, $x = a$ and $x = b$; parts below the x -axis count negative.
- In substitution, change dx and the limits; then finish in the new variable.
- Partial fractions: $A/(x - k) \rightarrow A \log |x - k|$, $A/(x - k)^2 \rightarrow -A/(x - k)$.
- P1: the variable's name does not matter.
- P2: $\int_a^b = -\int_b^a$.
- P3: $\int_a^b = \int_a^c + \int_c^b$; use it for modulus and piecewise functions.
- P4: on $[-a, a]$, even f gives $2\int_0^a f$, odd f gives 0.
- P5: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$; $\int_0^a f(x) dx = \int_0^a f(a - x) dx$.
- P6: $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx$.
- P7: $\int_0^{2a} f = 2\int_0^a f$ if $f(2a - x) = f(x)$, and 0 if $f(2a - x) = -f(x)$.
- $\int_a^b f(x)/(f(x) + f(a + b - x)) dx = (b - a)/2$.

Check yourself

Q. 1. Evaluate $\int_2^4 x^2 dx$.

$$[x^3/3]_2^4 = (64 - 8)/3 = 56/3.$$

Q. 2. Evaluate $\int_0^2 (x - 5/2) dx$. Why is the answer negative?

$[x^2/2 - 5x/2]_0^2 = 2 - 5 = -3$. The line $y = x - 5/2$ lies below the x -axis on $[0, 2]$, so the area counts as negative.

Q. 3. Evaluate $\int_2^5 x/(x^2 + 3) dx$.

Put $x^2 = t$, $x dx = dt/2$; $x = 2 \Rightarrow t = 4$, $x = 5 \Rightarrow t = 25$.

$$(1/2)[\log(t + 3)]_4^{25} = (1/2) \log(28/7) = (1/2) \log 4 = \log 2.$$

Q. 4. Evaluate $\int_0^2 e^{2x}/(3 + e^{2x}) dx$.

Put $e^{2x} = t$, $e^{2x} dx = dt/2$; $x = 0 \Rightarrow t = 1$, $x = 2 \Rightarrow t = e^4$.

$$(1/2) \int_1^{e^4} dt/(3 + t) = (1/2)[\log(3 + t)]_1^{e^4} = (1/2) \log((3 + e^4)/4) (\approx 1.334).$$

Note: the question in the unit appears to print the numerator as e^x , but its worked answer uses e^{2x} , which is what is solved here. (With e^x on top, the substitution $e^x = t$ gives $(1/\sqrt{3})(\tan^{-1}(e^2/\sqrt{3}) - \pi/6) \approx 0.472$ instead.)

Q. 5. Evaluate $\int_0^1 (2x^2 + 1)/((x - 4)(x - 2)^3) dx$.

Let $(2x^2 + 1)/((x - 4)(x - 2)^3) = A/(x - 4) + B/(x - 2) + C/(x - 2)^2 + D/(x - 2)^3$, so

$$2x^2 + 1 = A(x - 2)^3 + B(x - 4)(x - 2)^2 + C(x - 4)(x - 2) + D(x - 4).$$

$x = 4$: $33 = 8A$, $A = 33/8$. $x = 2$: $9 = -2D$, $D = -9/2$. Coefficient of x^3 : $0 = A + B$, $B = -33/8$. Constant term: $1 = -8A - 16B + 8C - 4D = -33 + 66 + 8C + 18$, so $C = -25/4$.

$$\begin{aligned} I &= (33/8)[\log|x - 4|]_0^1 - (33/8)[\log|x - 2|]_0^1 - (25/4)[-1/(x - 2)]_0^1 - (9/2)[-1/(2(x - 2)^2)]_0^1 \\ &= (33/8)(\log 3 - \log 4) - (33/8)(0 - \log 2) - (25/4)(1 - 1/2) - (9/2)(-1/2 + 1/8) \\ &= (33/8) \log 3 - (33/8) \log 2 - 25/8 + 27/16 = \mathbf{(33/8) \log (3/2) - 23/16} \approx 0.235. \end{aligned}$$

Exam tip: The unit's answer key puts $x = 4$ as $33 = A(4 - 2)^2 = 4A$ and gets $A = 33/4$, and its final answer is $(33/4) \log 3 - (33/4) \log 2 - 56/16 \approx -0.155$. The factor is $(x - 2)^3$, so $33 = A(2)^3 = 8A$ and $A = 33/8$; the correct answer is $(33/8) \log (3/2) - 23/16$ (checked numerically).

Q. 6. Evaluate $\int_0^5 |x^2 - 3x + 2| dx$.

$x^2 - 3x + 2 = (x - 1)(x - 2)$: positive on $[0, 1]$, negative on $[1, 2]$, positive on $[2, 5]$. Let $G(x) = x^3/3 - 3x^2/2 + 2x$.

$$G(0) = 0, G(1) = 5/6, G(2) = 2/3, G(5) = 85/6.$$

$$I = (G(1) - G(0)) - (G(2) - G(1)) + (G(5) - G(2)) = 5/6 + 1/6 + 81/6 = 87/6 = \mathbf{29/2}.$$

Q. 7. Evaluate $\int_1^5 (|x - 1| + |x + 2| + |x - 3|) dx$.

On $[1, 5]$, $|x - 1| = x - 1$ and $|x + 2| = x + 2$. Split $|x - 3|$ at $x = 3$.

$$\int_1^5 (x - 1) dx = [x^2/2 - x]_1^5 = 15/2 + 1/2 = 8; \int_1^5 (x + 2) dx = [x^2/2 + 2x]_1^5 = 45/2 - 5/2 = 20.$$

$$\int_1^5 |x - 3| dx = \int_1^3 (3 - x) dx + \int_3^5 (x - 3) dx = 2 + 2 = 4.$$

$$\text{Total: } 8 + 20 + 4 = \mathbf{32}.$$

Q. 8. Evaluate $\int_{-1}^2 f(x) dx$, where $f(x) = 4 - 3x$ for $-1 \leq x \leq 1$ and $f(x) = 2x + 1$ for $1 \leq x \leq 2$.

$$\text{By P3: } \int_{-1}^1 (4 - 3x) dx + \int_1^2 (2x + 1) dx = [4x - 3x^2/2]_{-1}^1 + [x^2 + x]_1^2 = (5/2 + 11/2) + (6 - 2) = 8 + 4 = \mathbf{12}.$$

(The unit's key cites "property 2" here; the splitting uses property 3.)

Q. 9. Evaluate (a) $\int_{-77}^{77} x^5 \sqrt{a^2 - x^2} \, dx$ ($a \geq 77$) and (b) $\int_{-1}^1 (x^5 + x^7)/(4 - x^4) \, dx$.

(a) $f(-x) = (-x)^5 \sqrt{a^2 - x^2} = -f(x)$: odd, so the integral is **0**.

(b) $f(-x) = (-x^5 - x^7)/(4 - x^4) = -f(x)$: odd, so the integral is **0** (by P4).

Q. 10. Evaluate $\int_1^5 f(x) \, dx$, where $f(x) = 1$ for $1 \leq x < 2$, 2 for $2 \leq x < 3$, 3 for $3 \leq x < 4$, 4 for $4 \leq x < 5$, and **5** for $x = 5$.

By P3: $\int_1^2 1 \, dx + \int_2^3 2 \, dx + \int_3^4 3 \, dx + \int_4^5 4 \, dx = 1 + 2 + 3 + 4 = \mathbf{10}$.

The single value $f(5) = 5$ does not change the area (a single point has no width).

Based on the IGNOU unit texts. Check definitions against the PDFs before the exam.